

Risolvere tramite metodo dello sviluppo l'equazione

$$T(n) = 5T\left(\frac{n}{2}\right) + n$$

$$T(1) = 1$$

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Sviluppo:

$$T\left(\frac{n}{2}\right) = 5T\left(\frac{n}{4}\right) + \frac{n}{2}$$

$$T\left(\frac{n}{4}\right) = 5T\left(\frac{n}{8}\right) + \frac{n}{4}$$

Risolvere tramite metodo dello sviluppo l'equazione

$$T(n) = 5T\left(\frac{n}{2}\right) + n$$

$$T(1) = 1$$

Sostituzione:

$$\begin{aligned} T(n) &= n + \frac{5}{2}n + \left(\frac{5}{2}\right)^2 n + 5^3 T\left(\frac{n}{8}\right) \\ &= n \sum_{0 \leq i \leq s} \left(\frac{5}{2}\right)^i \end{aligned}$$

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Terminazione per $n = 1$, ogni volta dimezzo:

$$\frac{n}{2^s} = 1$$

$$2^s = n$$

$$s = \log_2(n)$$

Risolvere tramite metodo dello sviluppo l'equazione

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$$T(1) = 1$$

Sostituzione:

$$\begin{aligned} T(n) &= n \sum_{0 \leq i \leq \log_2(n)} \left(\frac{5}{2}\right)^i \\ &= n \left(\frac{\left(\frac{5}{2}\right)^{\log_2(n)+1} - 1}{\frac{5}{2} - 1} \right) \end{aligned}$$

Equazioni alle ricorrenze

Risolvere tramite metodo dello sviluppo l'equazione

$$T(n) = 5T\left(\frac{n}{2}\right) + n$$

$$T(1) = 1$$

Sostituzione:

$$\begin{aligned} n \left(\frac{\left(\frac{5}{2}\right)^{\log_2(n)+1} - 1}{\frac{5}{2} - 1} \right) &= n \frac{2}{3} \left(\frac{5}{2} \cdot \frac{5^{\log_2(n)}}{2^{\log_2(n)}} - 1 \right) \\ &= n \frac{1}{3} \left(5 \cdot \frac{5^{\log_2(n)}}{n} - 2 \right) \\ &= n \frac{1}{3} \left(5 \cdot \frac{n^{\log_2 5}}{n} - 2 \right) \\ &= \frac{1}{3} (5 \cdot n^{\log_2 5} - 2n) \end{aligned}$$

Equazioni alle ricorrenze

Risolvere tramite metodo dello sviluppo l'equazione

$$T(n) = 5T\left(\frac{n}{2}\right) + n$$

$$T(1) = 1$$

Sostituzione:

$$T(n) = \frac{1}{3} (5 \cdot n^{\log_2 5} - 2n)$$

Quindi:

$$T(n) = O(n^{\log_2 5})$$

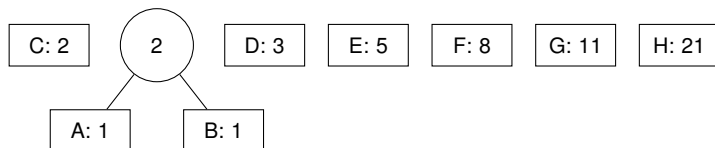
Determinare il codice di Huffman ottimo per i seguenti caratteri con le frequenze specificate:

A: 1 B: 1 C: 2 D: 3 E: 5 F: 8 G: 11 H: 21

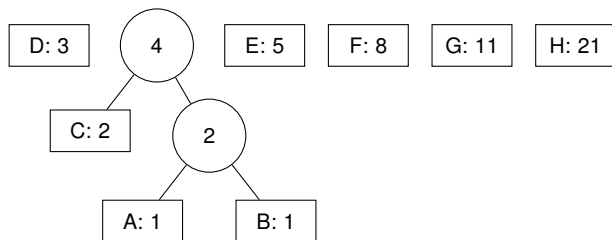
Memorizziamo le informazioni in una coda a priorità



Estraiamo i due elementi a frequenza minore, A e B, e creiamo un nuovo albero che fonde i due nodi. Il nuovo albero ha frequenza 2, e va reinserito nella coda a priorità

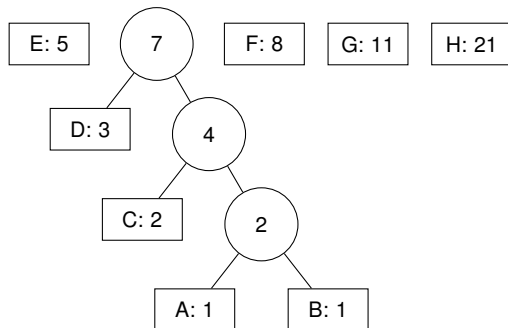


Estraiamo nuovamente i due elementi a frequenza minore, C e l'albero che contiene A, B, e li fondiamo in un nuovo albero con frequenza 4. Reinseriamo il nuovo albero nella coda

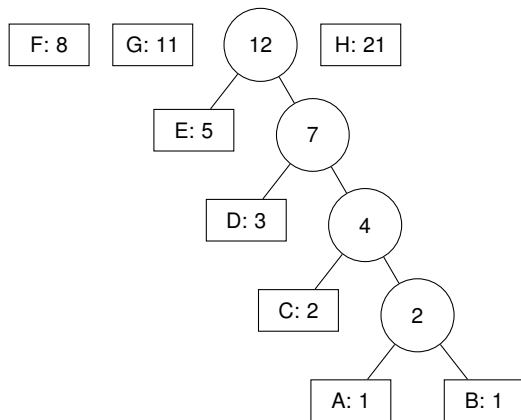


Codice Huffman

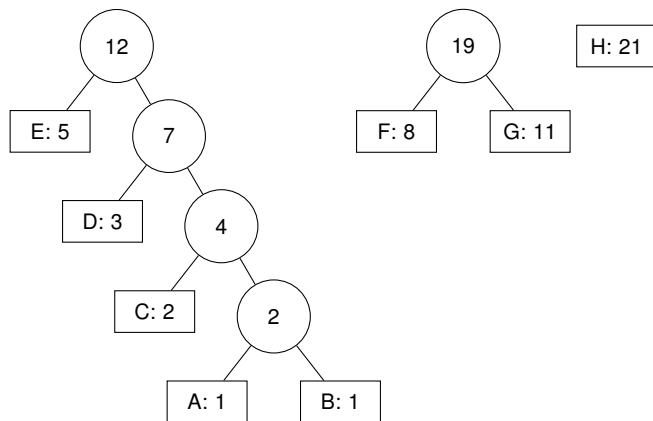
Iterando, si ottiene:



Codice Huffman

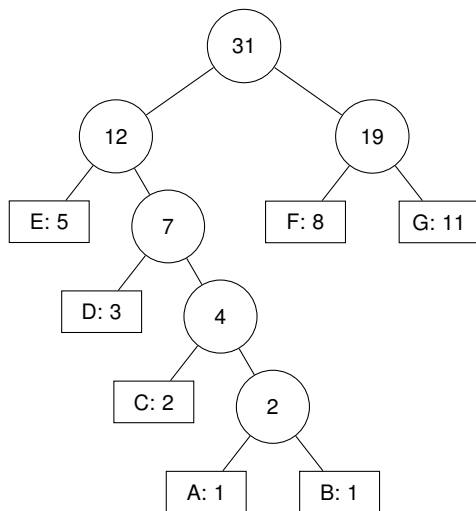


Codice Huffman

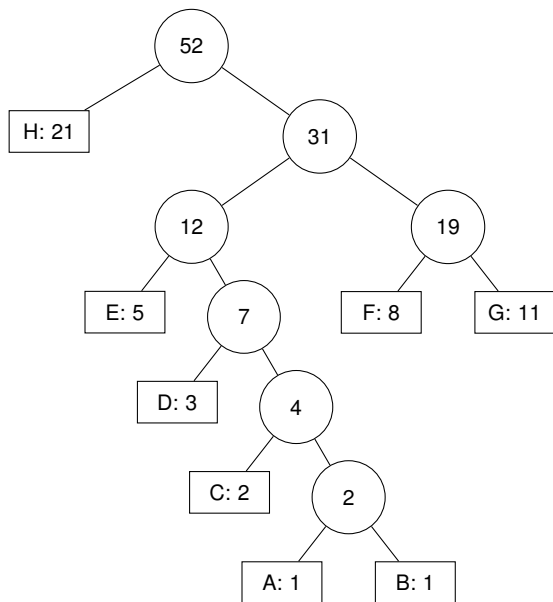


Codice Huffman

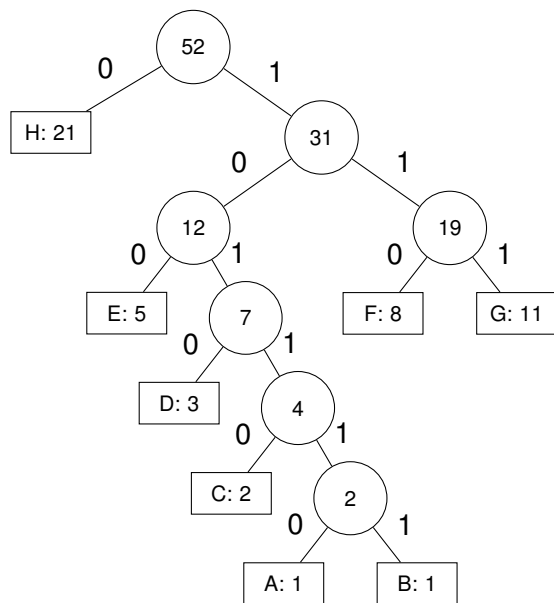
H: 21



Codice Huffman

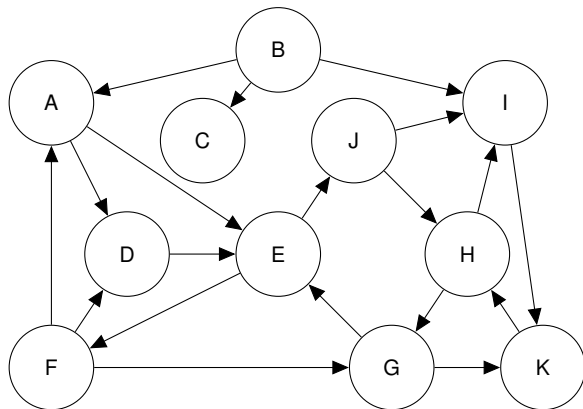


Codice Huffman



A	101110
B	101111
C	10110
D	1010
E	100
F	110
G	111
H	0

Consideriamo il seguente grafo



Considerando B come nodo iniziale:

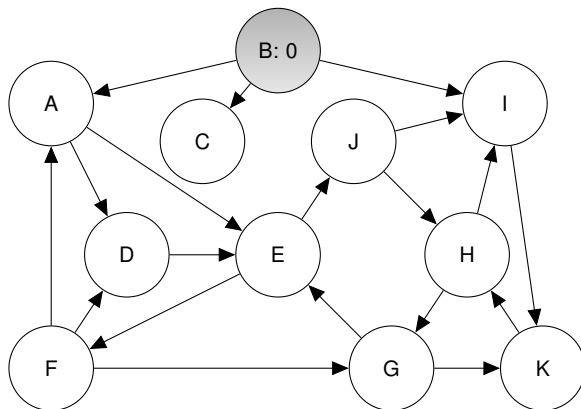
- Si effettui una visita in ampiezza
- Si effettui una visita in profondità, considerando i vertici in ordine alfabetico e etichettando i vertici con tempo di inizio/fine elaborazione
- Si etichettino gli archi come T (tree), B (back), F (forward), C (cross)

Usiamo una coda per contenere i nodi scoperti e non ancora visitati

Partiamo dal nodo B, lo inseriamo nella coda e lo etichettiamo come nodo a distanza 0 da B stesso

Proseguiamo estraendo dalla coda il primo nodo X, andando quindi a inserire tutti i nodi adiacenti a X non ancora scoperti, che saranno etichettati come a distanza $D + 1$, dove D è la distanza di X da B

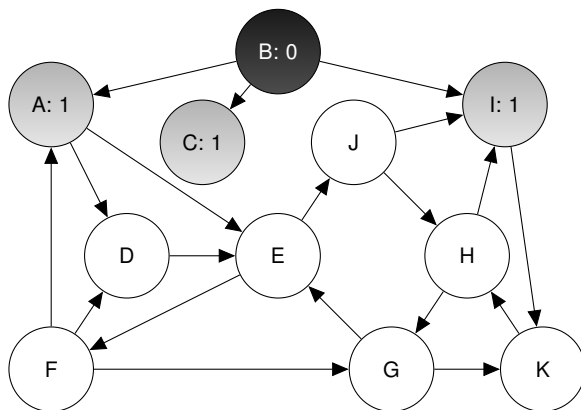
Visita in ampiezza



Coda:

B: 0

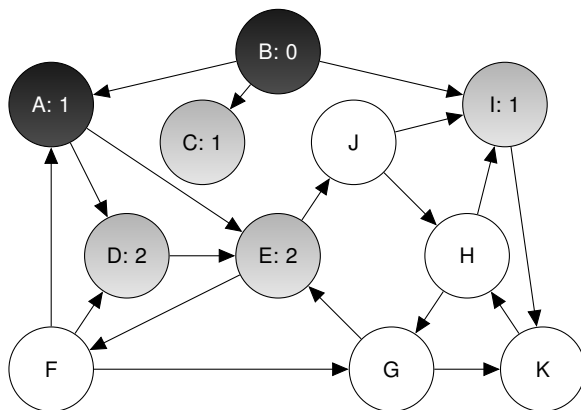
Visita in ampiezza



Coda:

A: 1
C: 1
I: 1

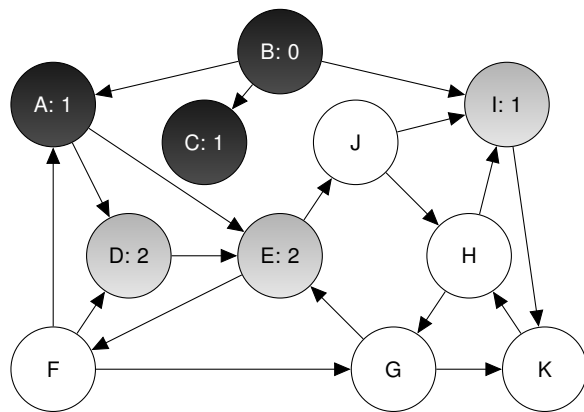
Visita in ampiezza



Coda:

C: 1
I: 1
D: 2
E: 2

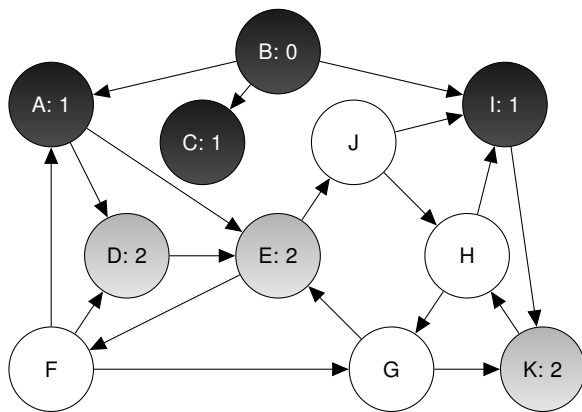
Visita in ampiezza



Coda:

I: 1
D: 2
E: 2

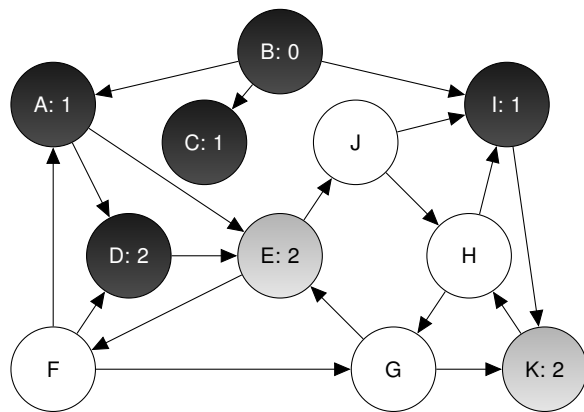
Visita in ampiezza



Coda:

D: 2
E: 2
K: 2

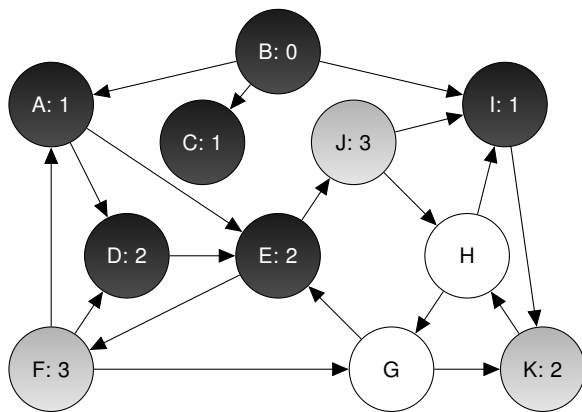
Visita in ampiezza



Coda:

E: 2
K: 2

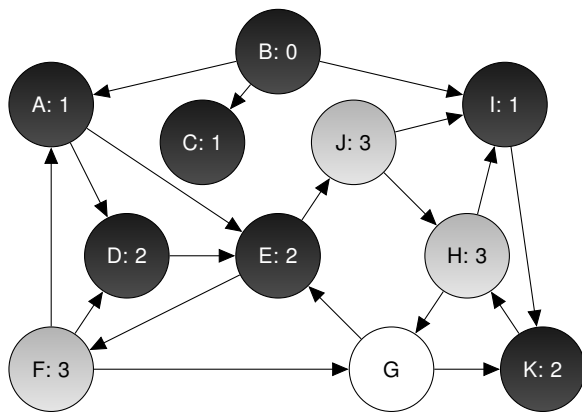
Visita in ampiezza



Coda:

K: 2
F: 3
J: 3

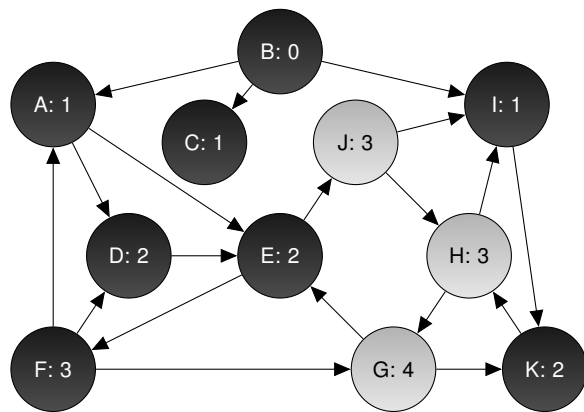
Visita in ampiezza



Coda:

F: 3
J: 3
H: 3

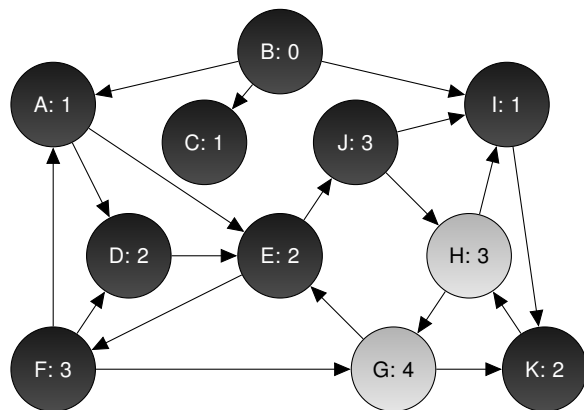
Visita in ampiezza



Coda:

J: 3
H: 3
G: 4

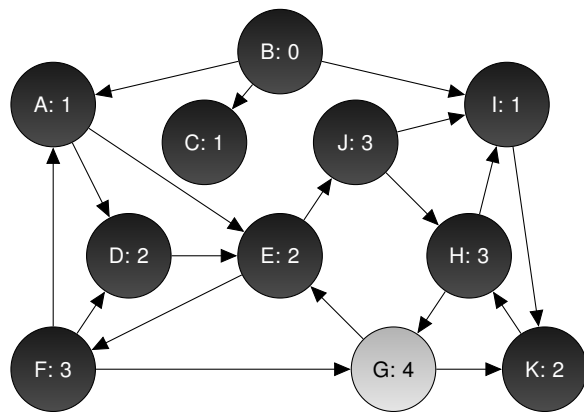
Visita in ampiezza



Coda:

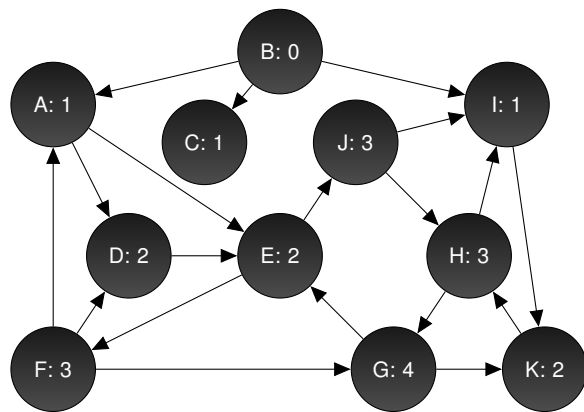
H: 3
G: 4

Visita in ampiezza



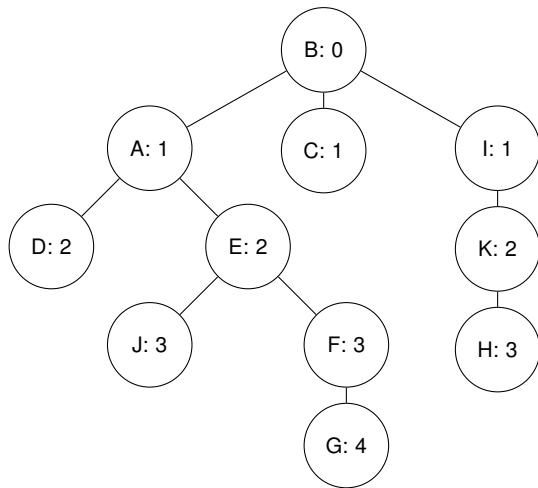
Coda:

G: 4



Coda:

Albero della visita in ampiezza:



Visita in profondità

Partiamo dal nodo B e iniziamo a percorrere cammini finchè possibile senza ripassare da nodi già visitati.

Quando ciò non è possibile, torniamo indietro fino al primo nodo da cui partono ancora cammini inesplorati e proseguiamo la visita

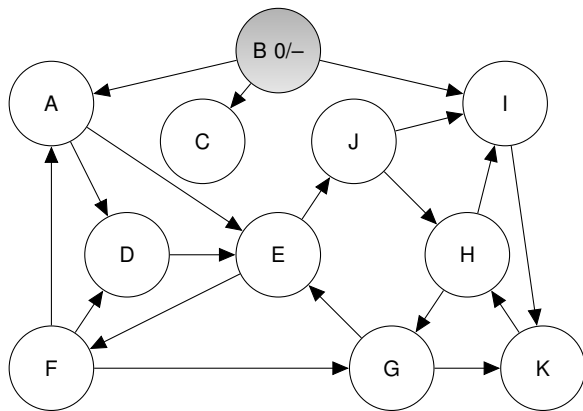
Etichettiamo i nodi con tempo di inizio e fine elaborazione

La visita genera un albero (o una foresta)

Etichettiamo gli archi che percorriamo come archi T (tree, in rosso nelle figure)

Per gli altri archi, distinguiamo tra archi F (forward, in verde), archi B (back, in blu) e archi C (cross, in giallo)

Visita in profondità



Archi:

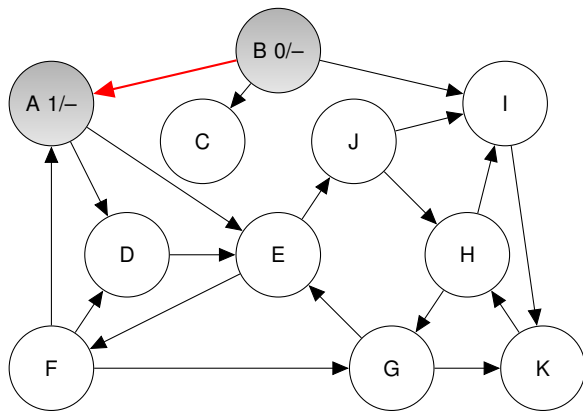
T

B

F

C

Visita in profondità



Archi:

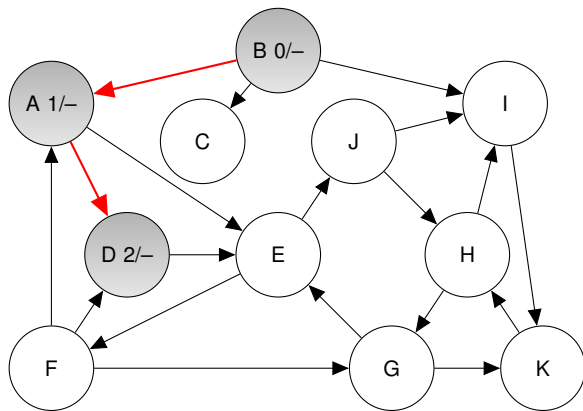
T

B

F

C

Visita in profondità



Archi:

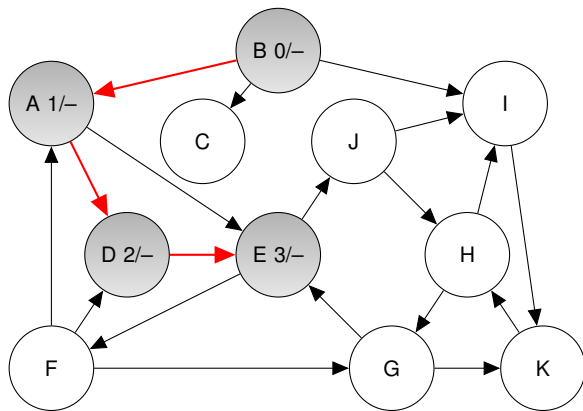
T

B

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Visita in profondità



Archi:

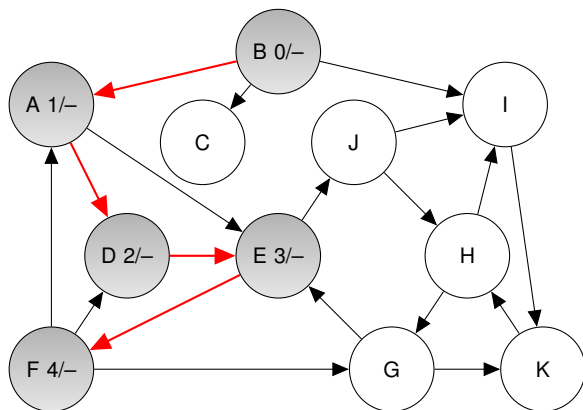
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F

C

Visita in profondità



Archi:

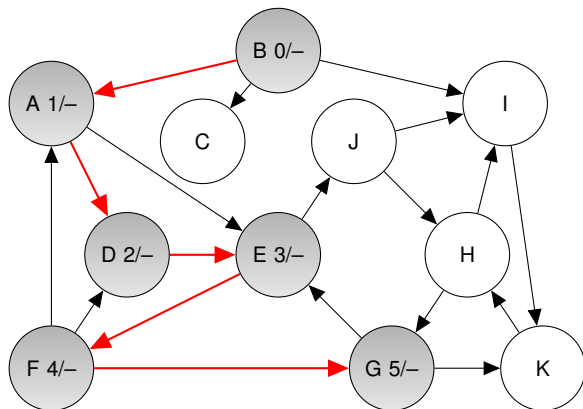
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Visita in profondità



Archi:

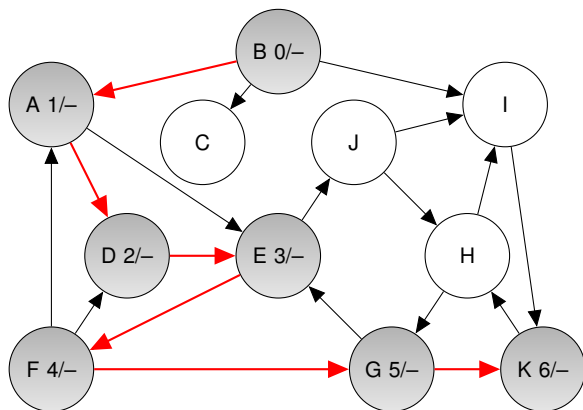
T

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F

C

Visita in profondità



Archi:

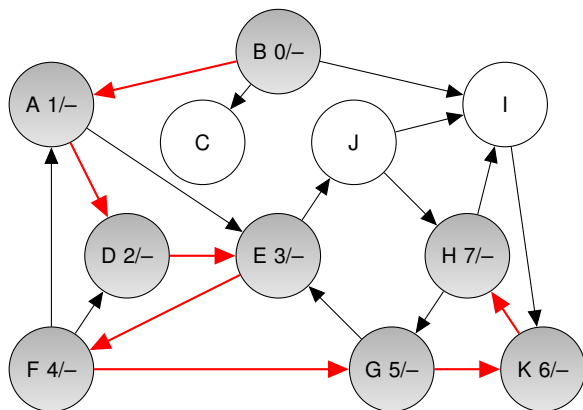
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Visita in profondità



Archi:

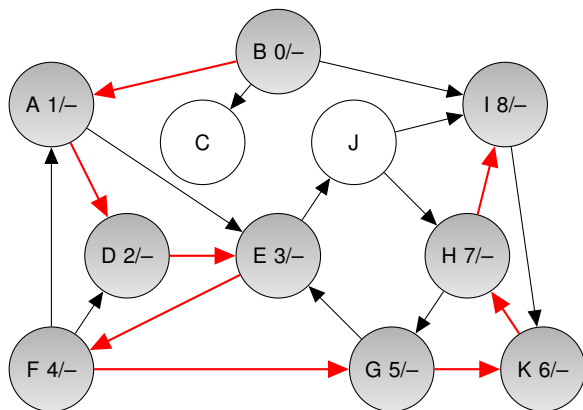
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Visita in profondità



Archi:

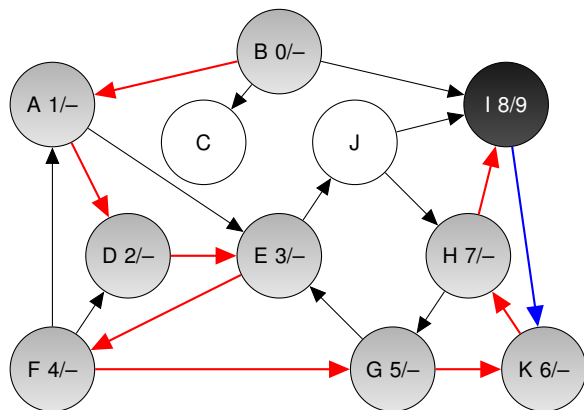
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Visita in profondità



Archi:

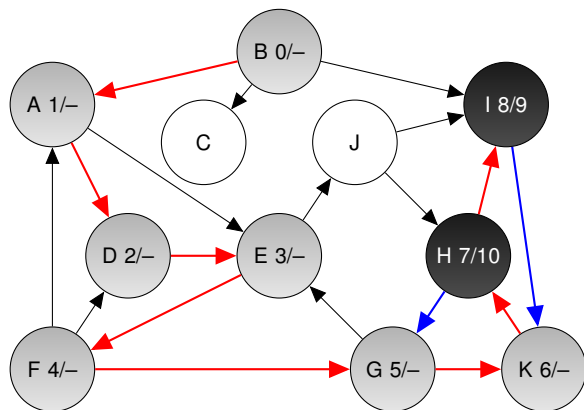
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Visita in profondità



Archi:

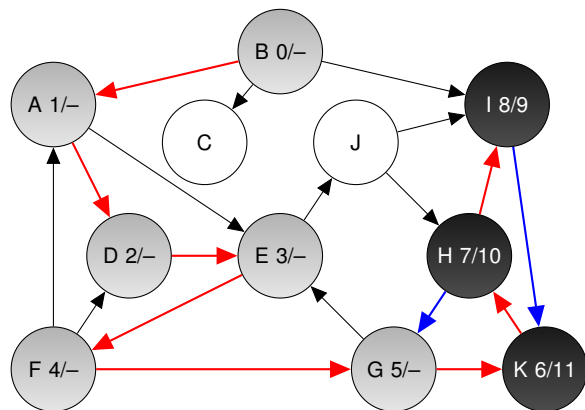
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Visita in profondità



Archi:

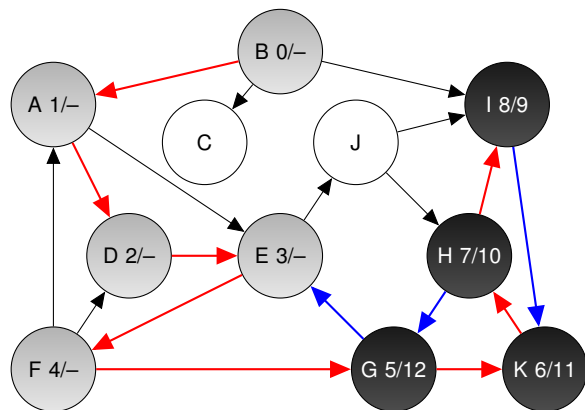
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Visita in profondità



Archi:

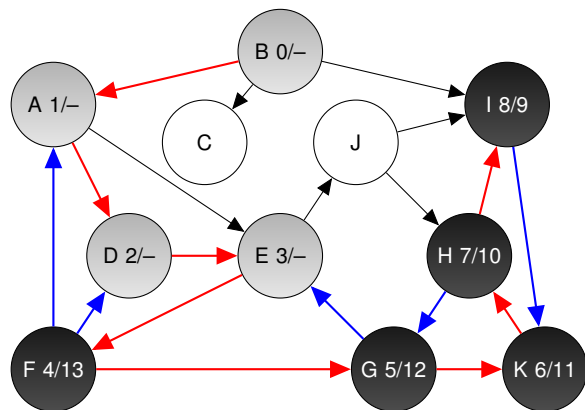
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Visita in profondità



Archi:

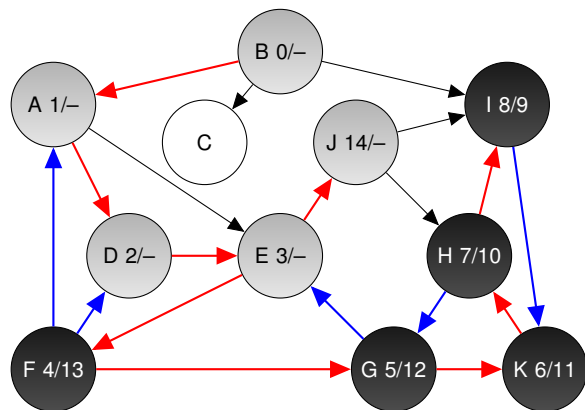
T

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Visita in profondità



Archi:

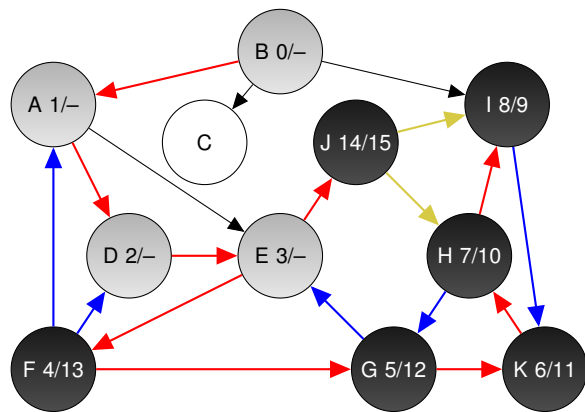
T

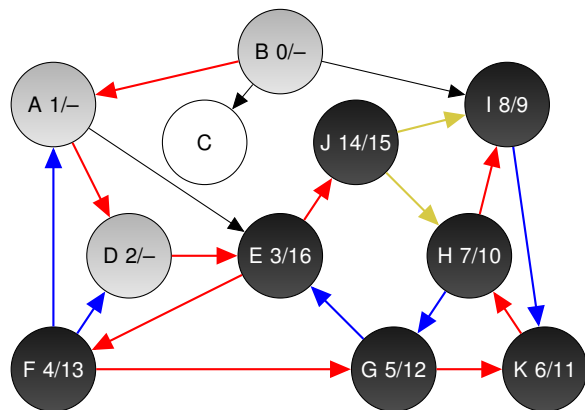
B

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Visita in profondità





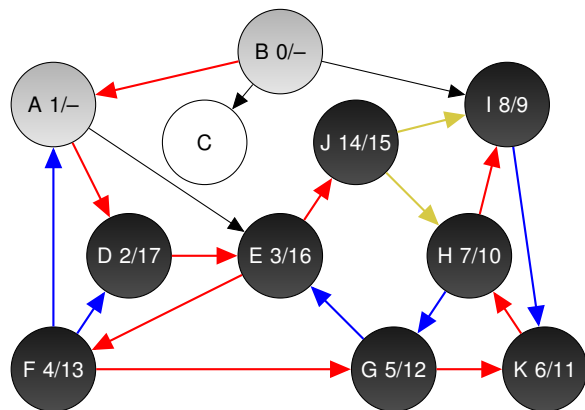
Archi:

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Archi:

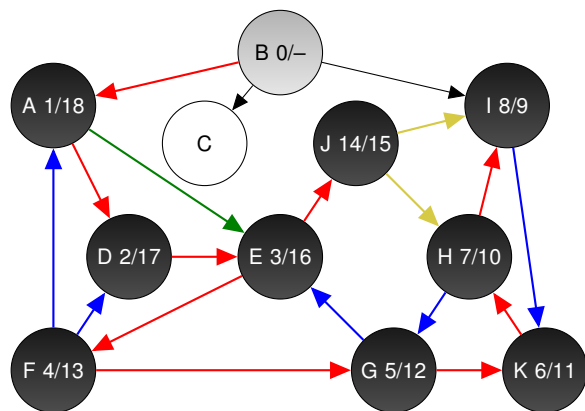
T

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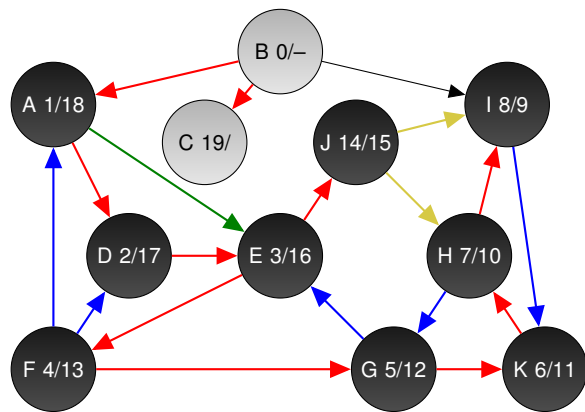
F

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Visita in profondità



Visita in profondità



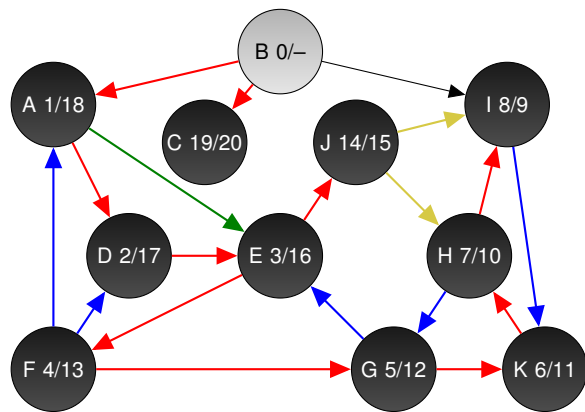
Archi:

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Archi:

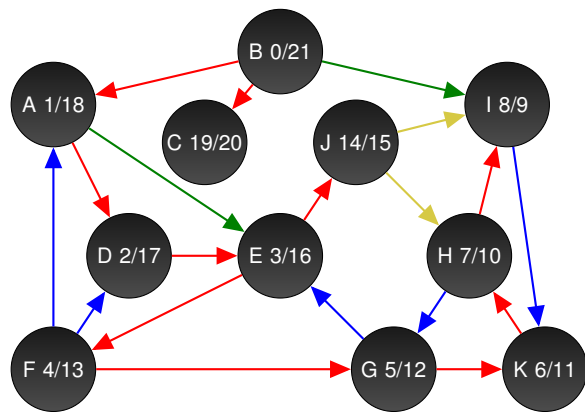
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Visita in profondità



Archi:

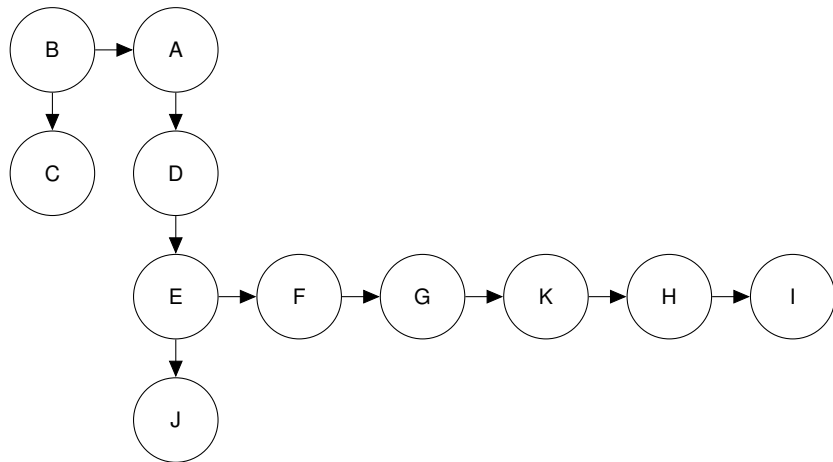
T

B

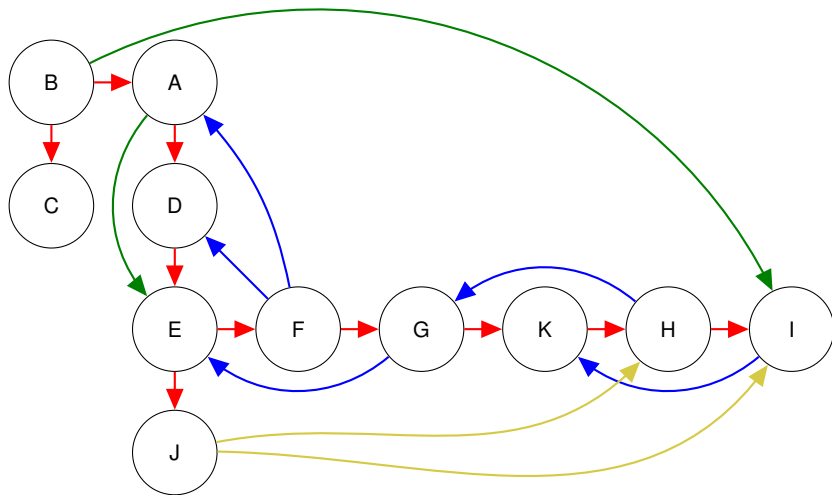
F

C

Albero della visita in profondità:

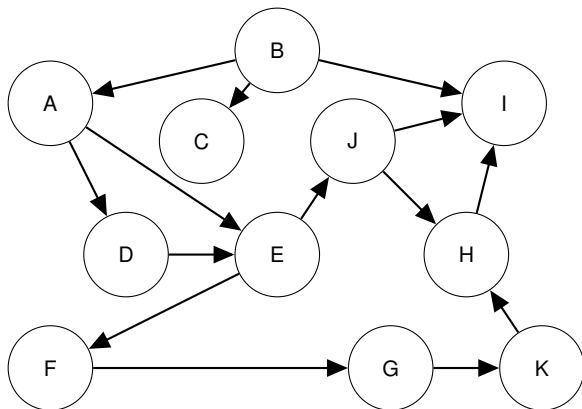


Completando il grafo:



DAG - Ordinamento topologico

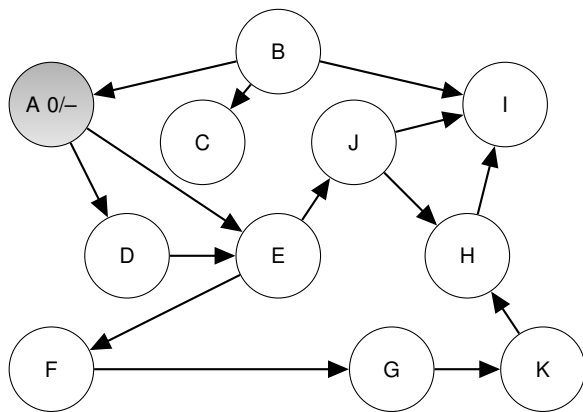
Consideriamo il grafo orientato aciclico



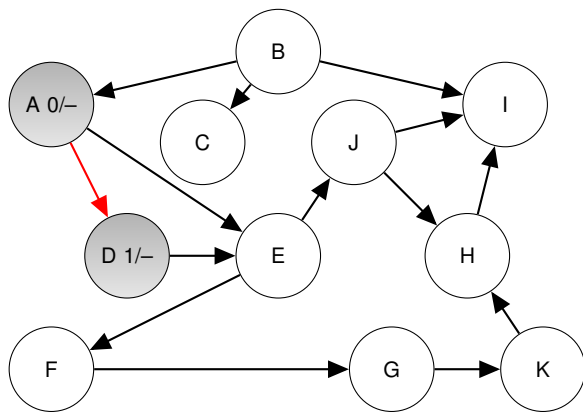
Per trovare un ordinamento topologico utilizziamo una visita DFS

Consideriamo i nodi ordinati lessicograficamente

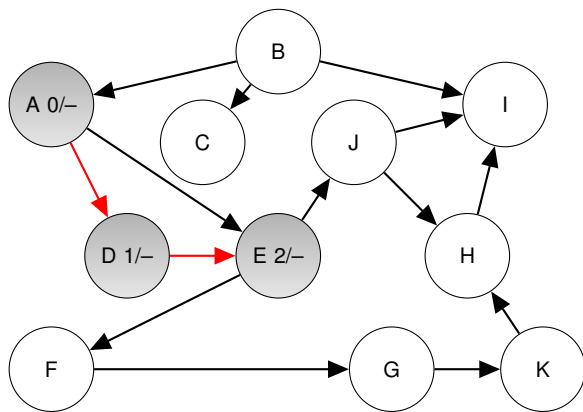
DAG - Ordinamento topologico



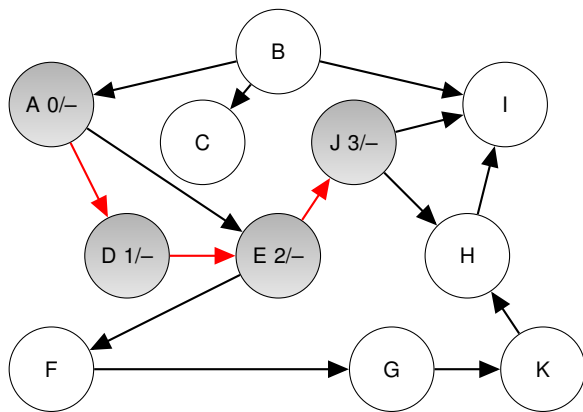
DAG - Ordinamento topologico



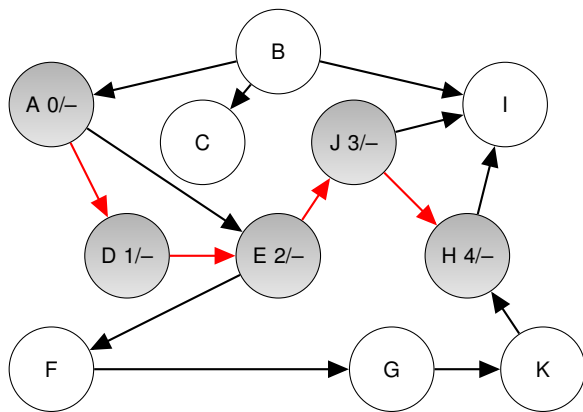
DAG - Ordinamento topologico



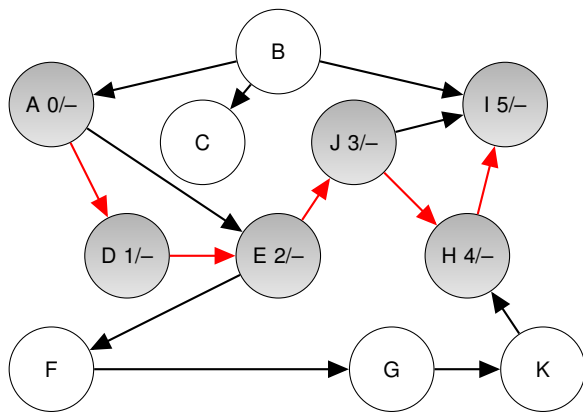
DAG - Ordinamento topologico



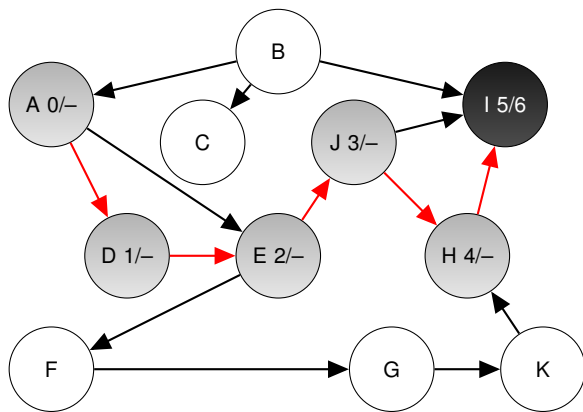
DAG - Ordinamento topologico



DAG - Ordinamento topologico

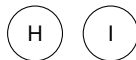
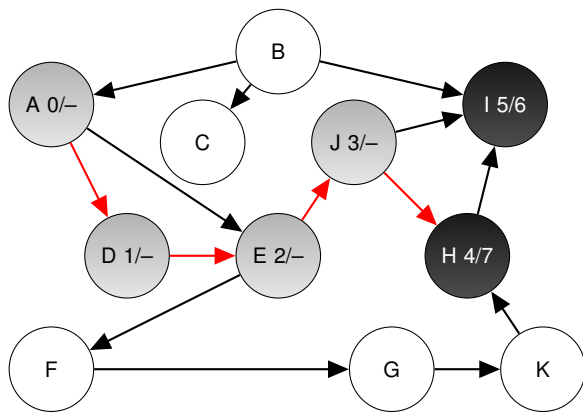


DAG - Ordinamento topologico

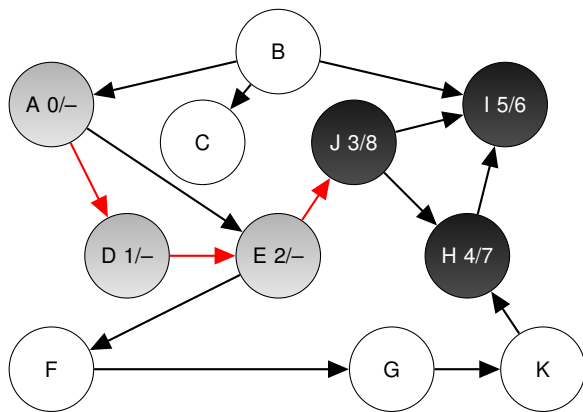


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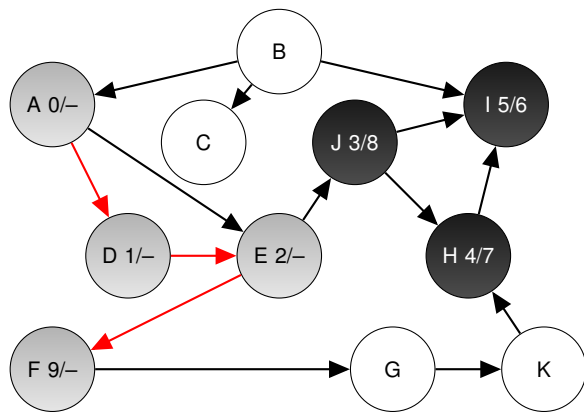
DAG - Ordinamento topologico



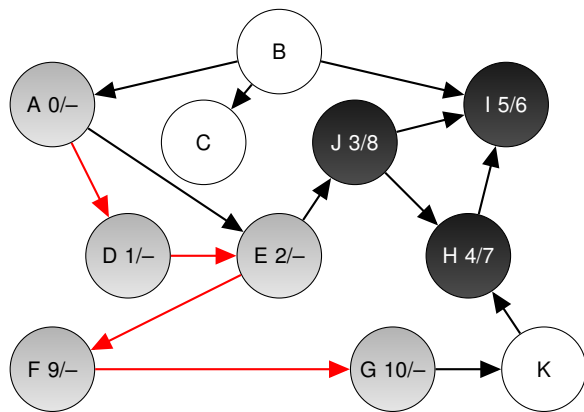
DAG - Ordinamento topologico



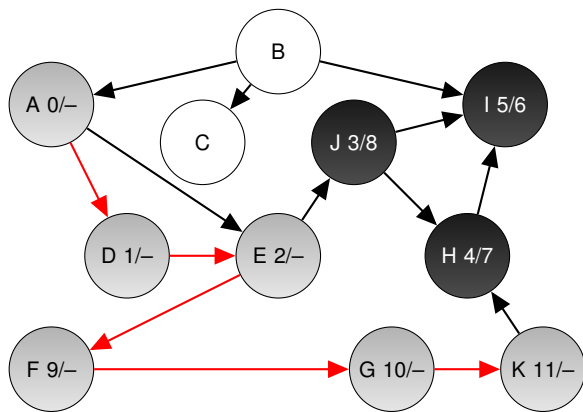
DAG - Ordinamento topologico



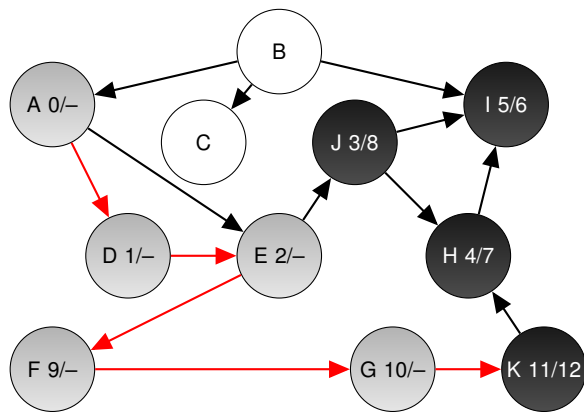
DAG - Ordinamento topologico



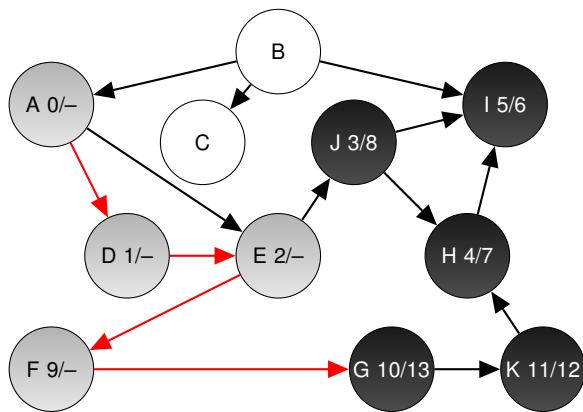
DAG - Ordinamento topologico



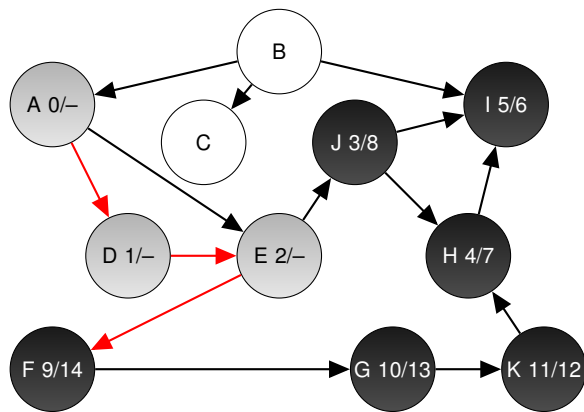
DAG - Ordinamento topologico



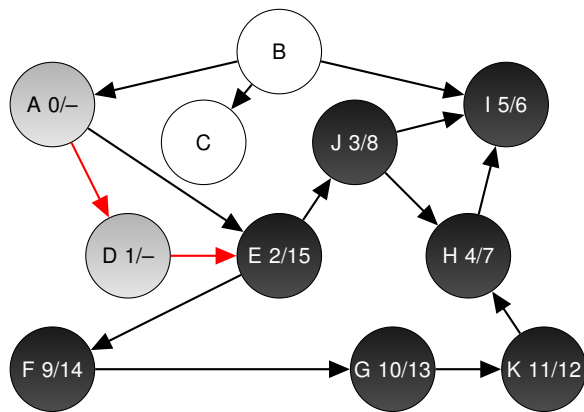
DAG - Ordinamento topologico



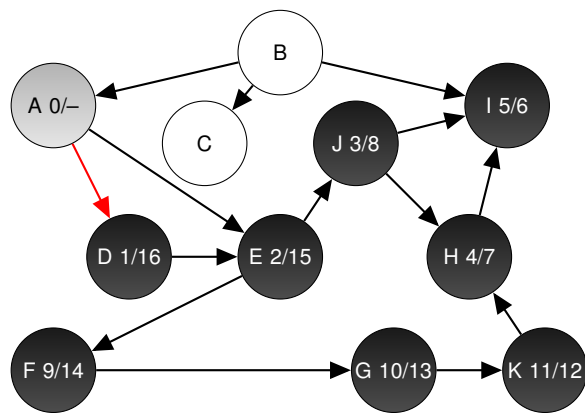
DAG - Ordinamento topologico



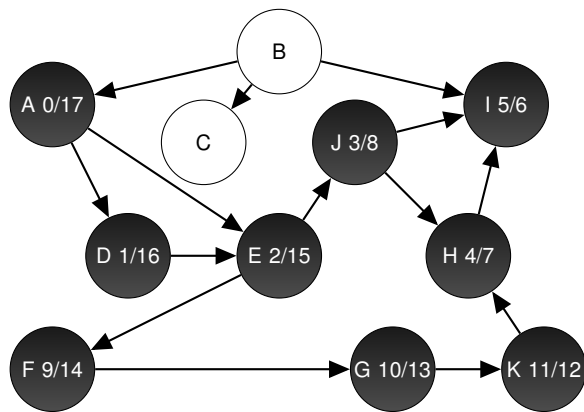
DAG - Ordinamento topologico



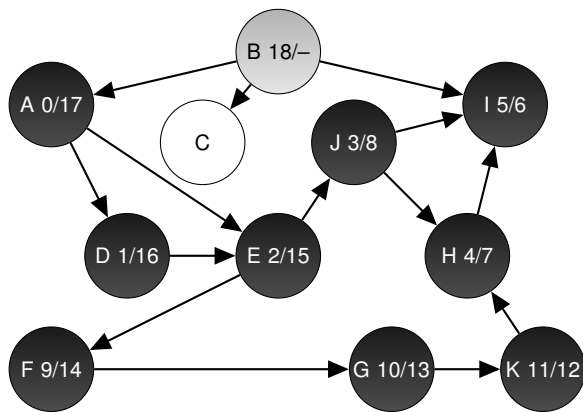
DAG - Ordinamento topologico



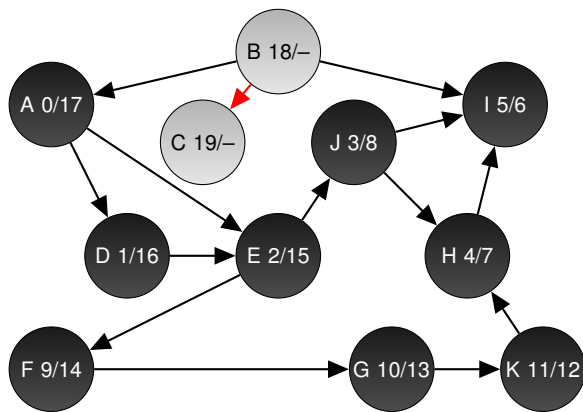
DAG - Ordinamento topologico



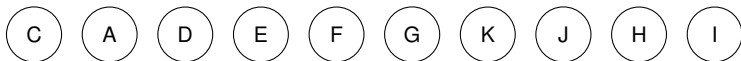
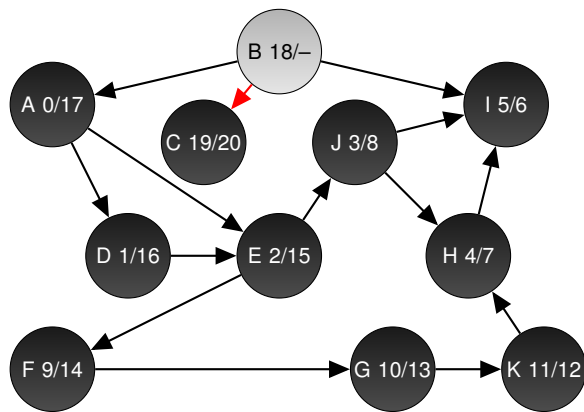
DAG - Ordinamento topologico



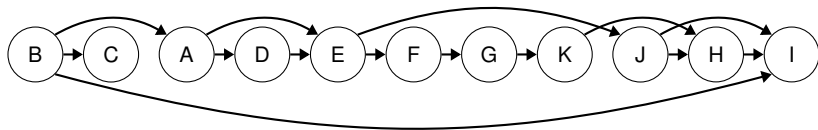
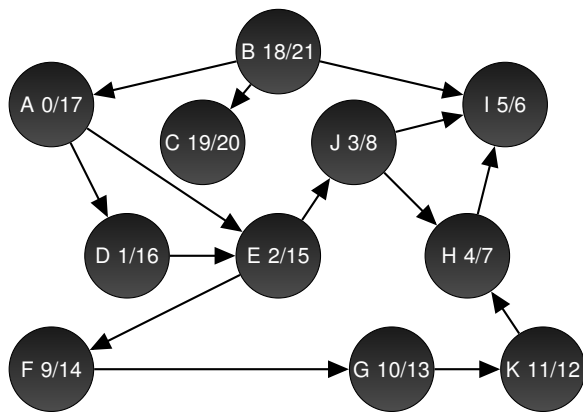
DAG - Ordinamento topologico



DAG - Ordinamento topologico

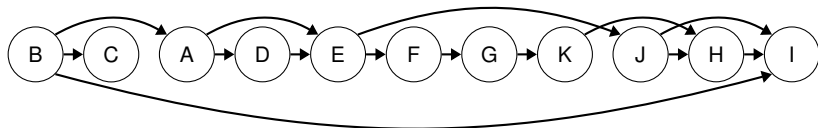


DAG - Ordinamento topologico



Vogliamo calcolare un cammino massimo

Partiamo dall'ordinamento topologico calcolato in precedenza:

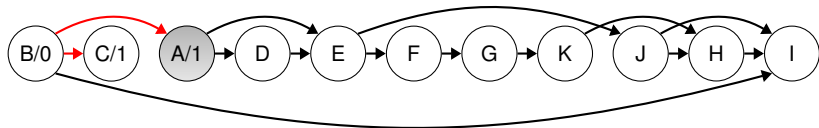
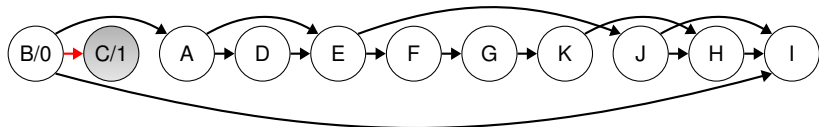
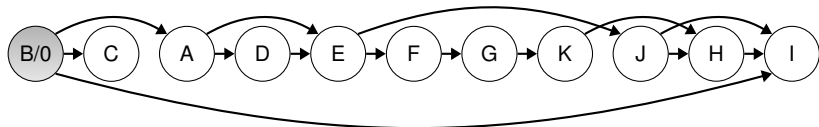


Percorriamo l'ordinamento topologico, calcolando, per ogni nodo:

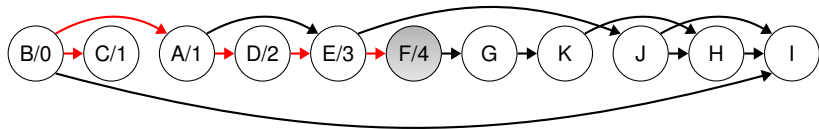
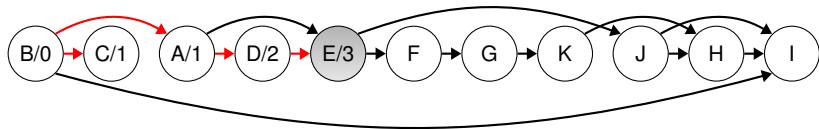
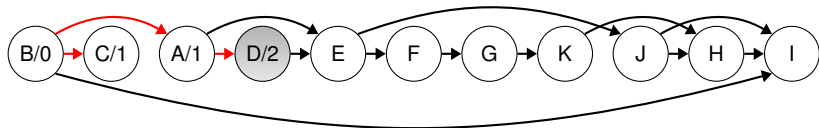
$$d(v) = \max_{u|(u,v) \in E} (d(u) + 1) \quad (1)$$

Se il nodo v non ha predecessori, mettiamo $d(v) = 0$

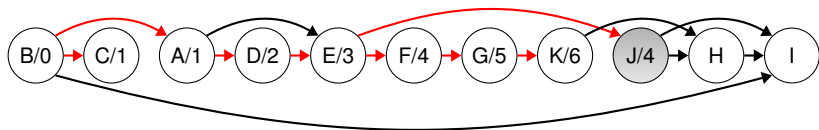
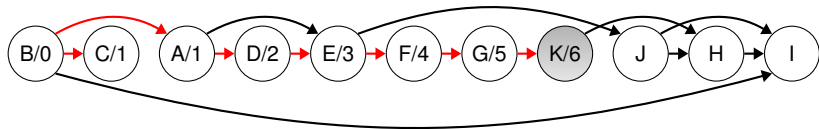
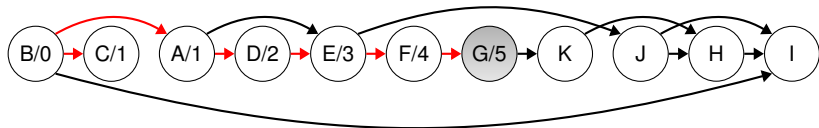
DAG - Cammini massimi



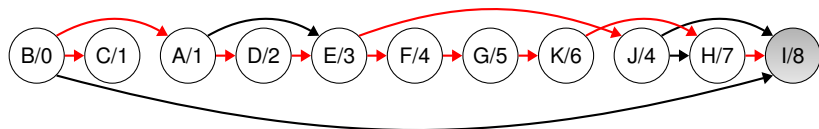
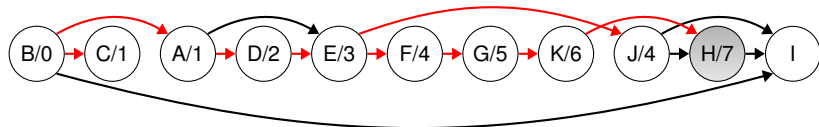
DAG - Cammini massimi



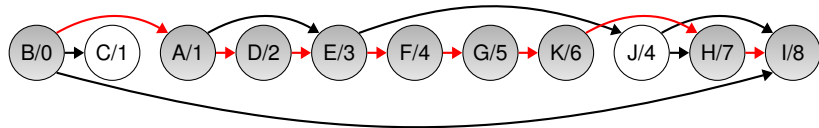
DAG - Cammini massimi



DAG - Cammini massimi



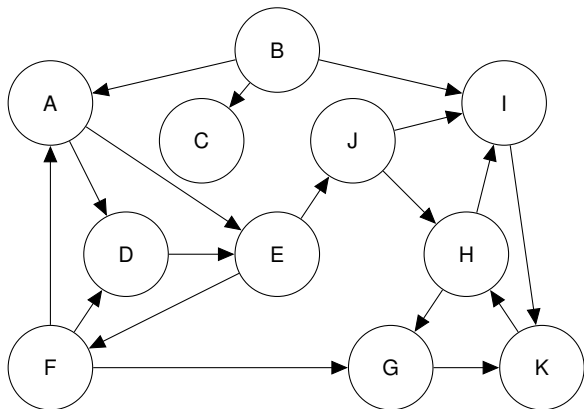
Cammino di lunghezza massima: nodo finale I ($\arg \max_v d(v)$)



Percorrendo all'indietro gli archi che portano a I:
B -> A -> D -> E -> F -> G -> K -> H -> I

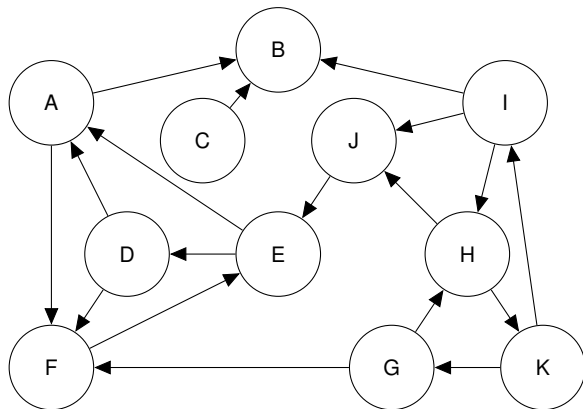
Componenti fortemente connesse

Consideriamo il seguente grafo (nota — il grafo è leggermente diverso da quello considerato nella prima parte):



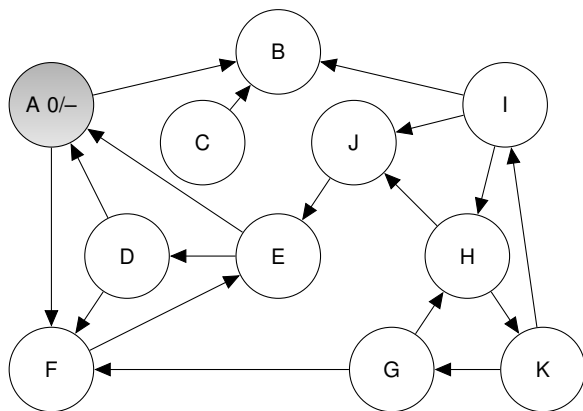
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



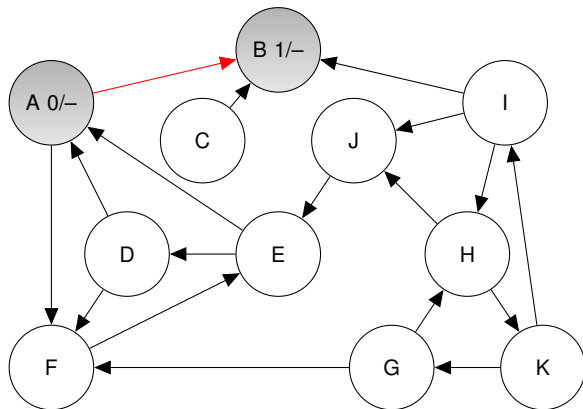
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



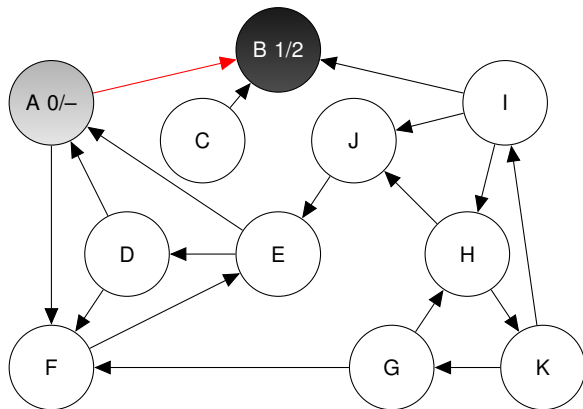
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



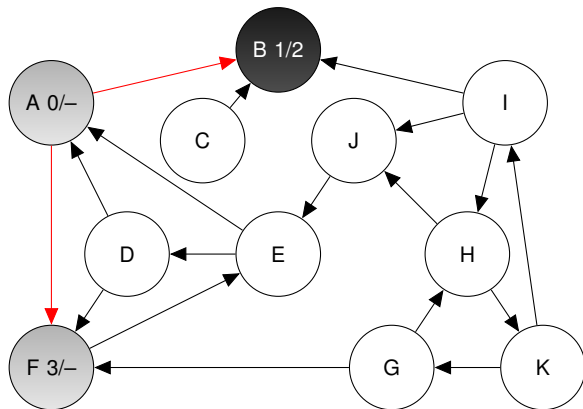
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



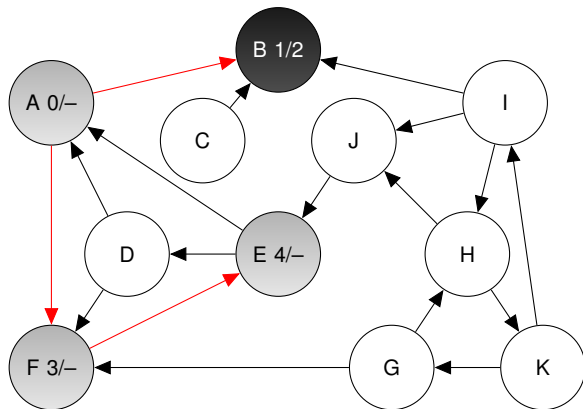
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



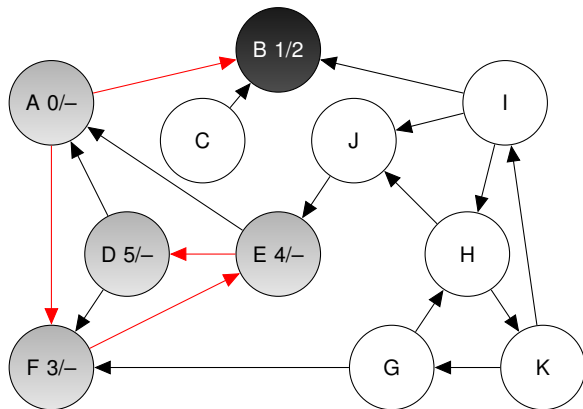
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



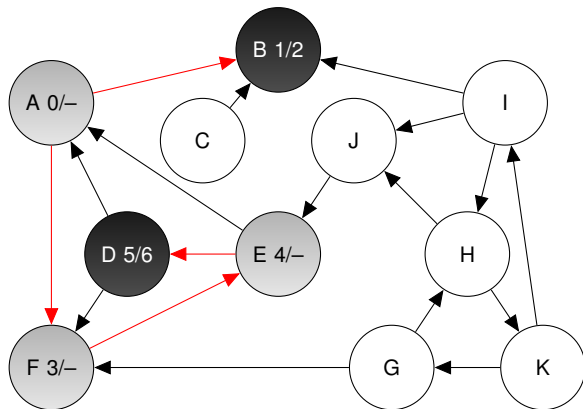
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



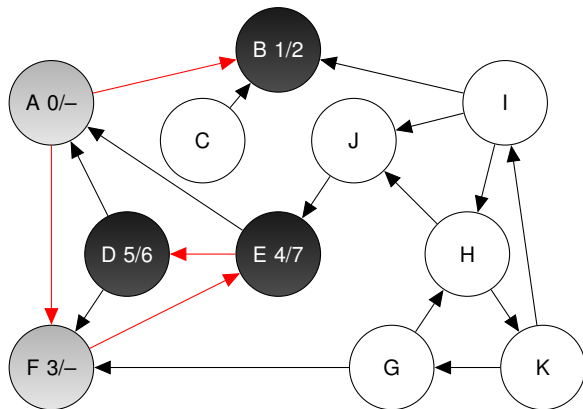
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



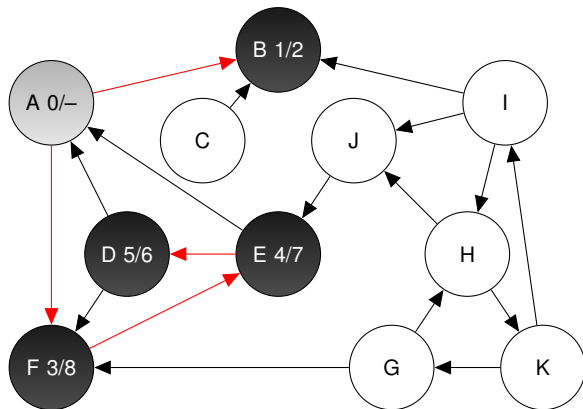
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



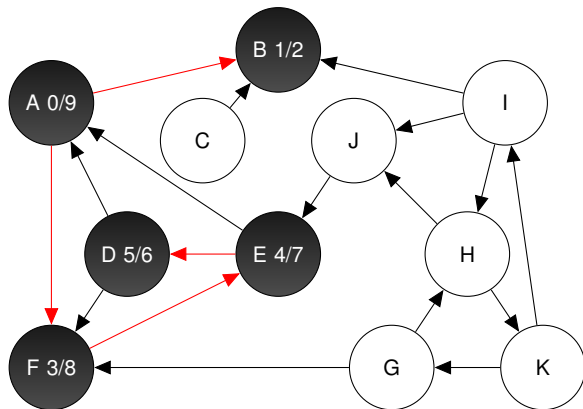
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



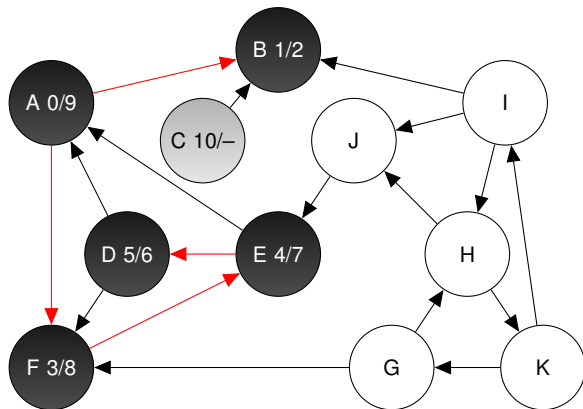
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



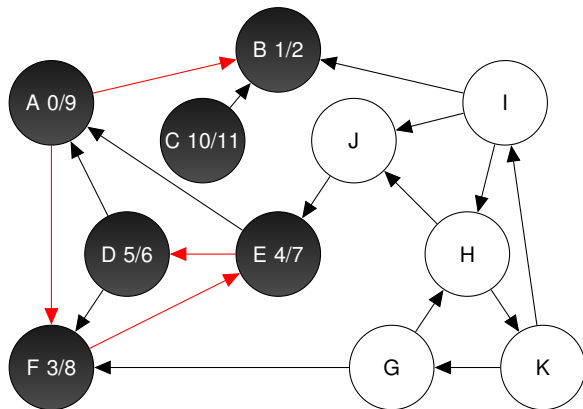
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



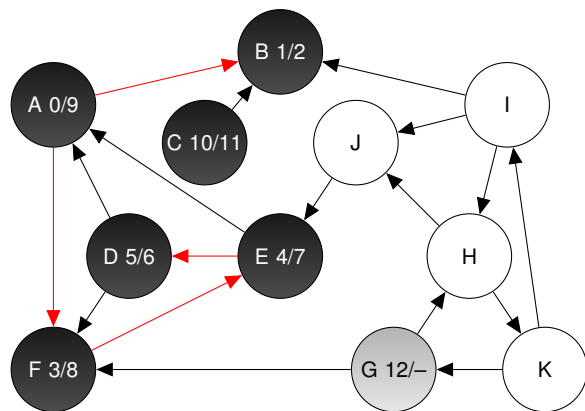
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



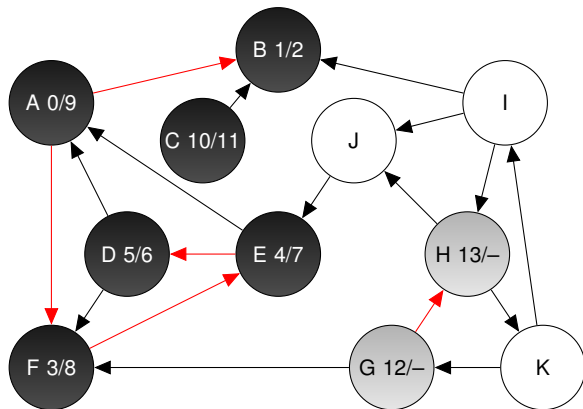
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



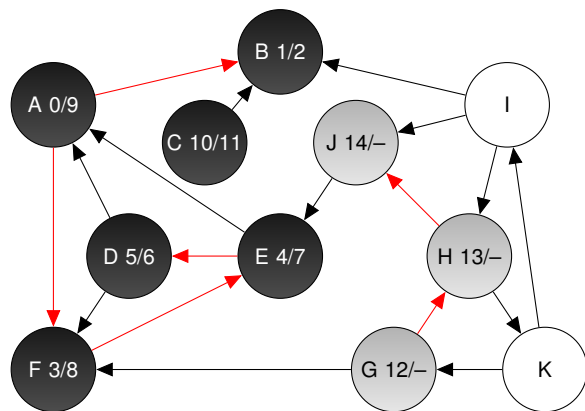
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



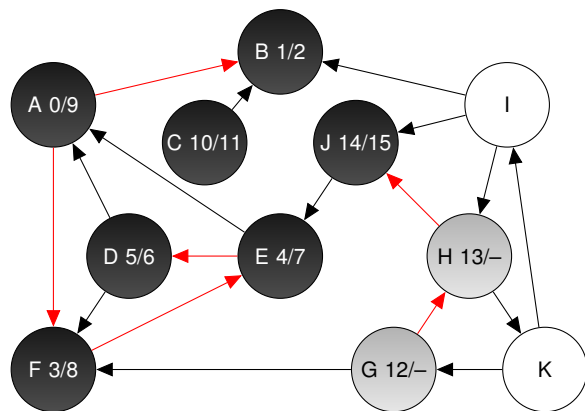
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



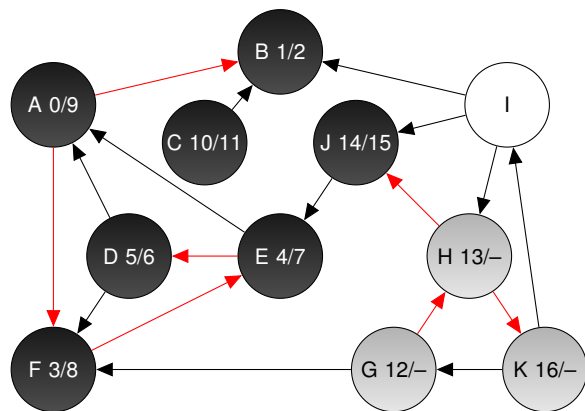
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



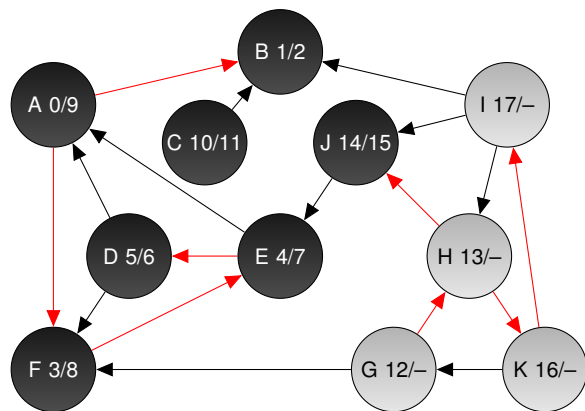
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



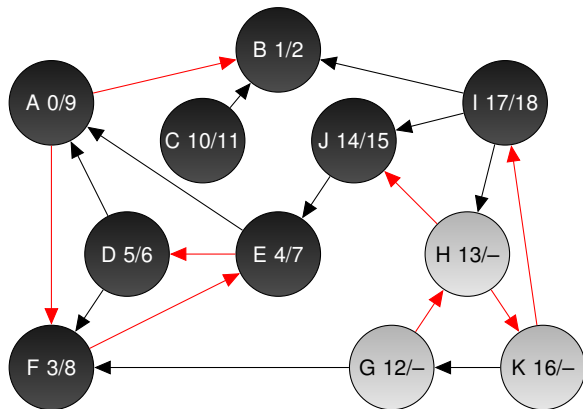
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



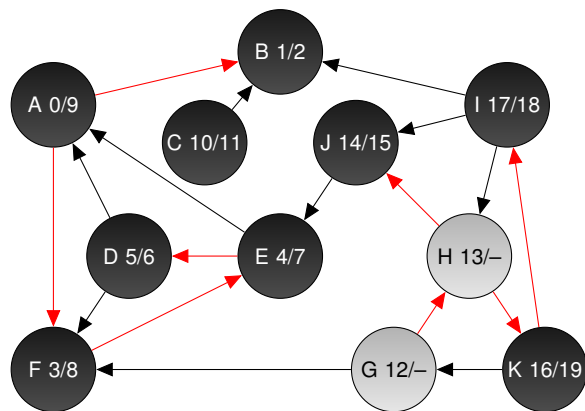
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



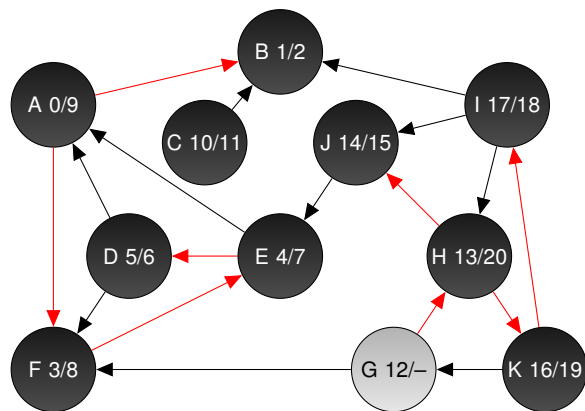
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



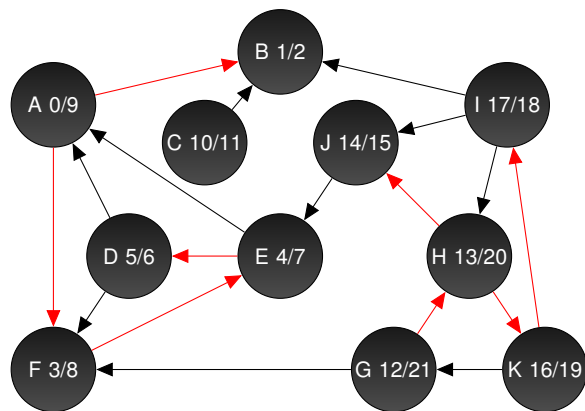
Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto



Componenti fortemente connesse

Primo passo: DFS sul grafo trasposto

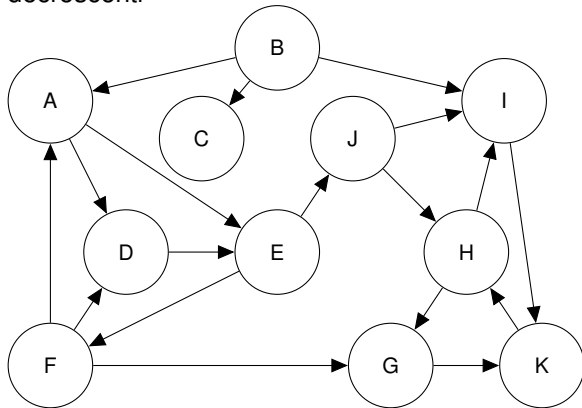


Componenti fortemente connesse

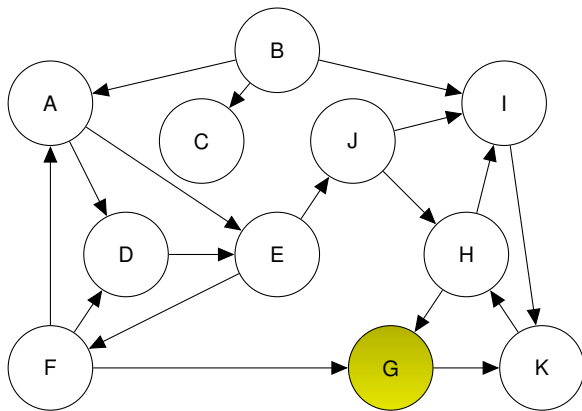
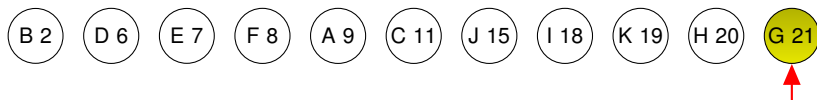
Ordinamento dei nodi per tempo di fine:



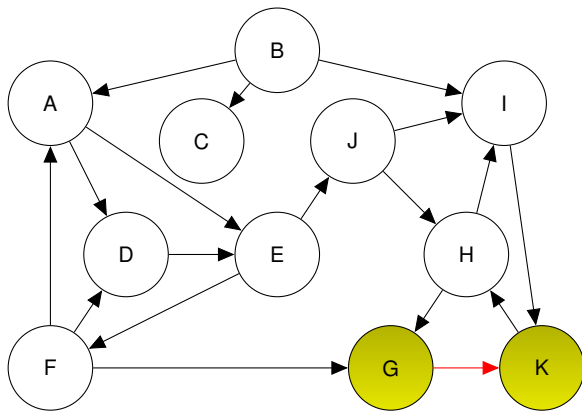
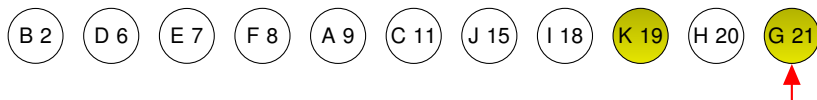
DFS su grafo originale secondo tempi di fine elaborazione
decrescenti



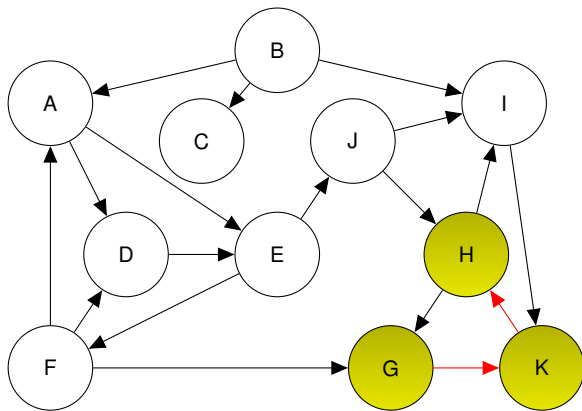
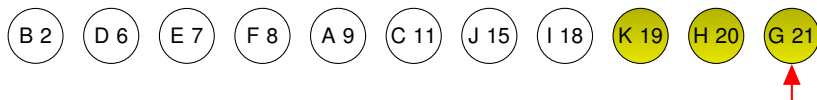
Componenti fortemente connesse



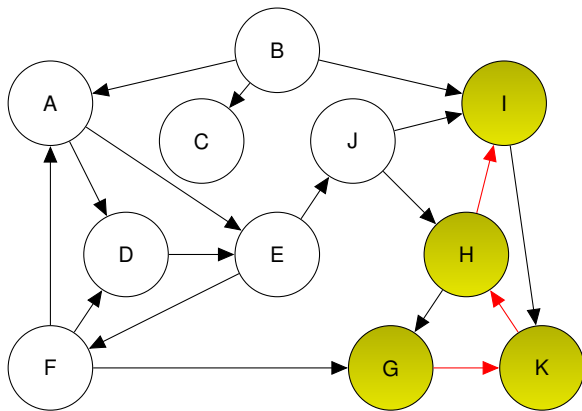
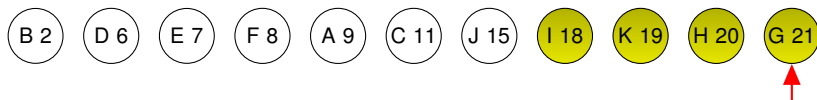
Componenti fortemente connesse



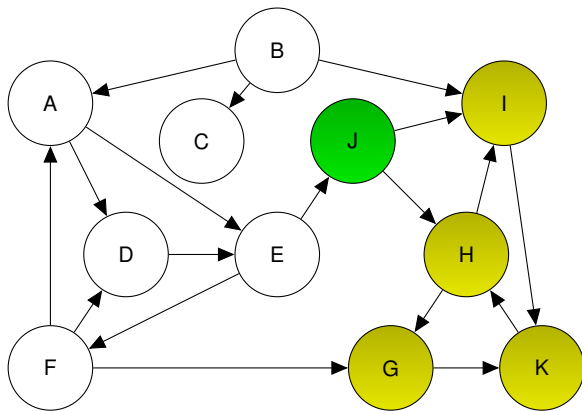
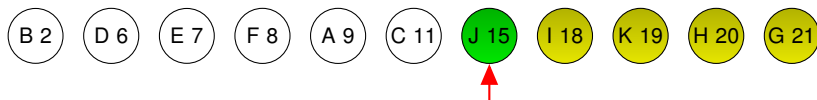
Componenti fortemente connesse



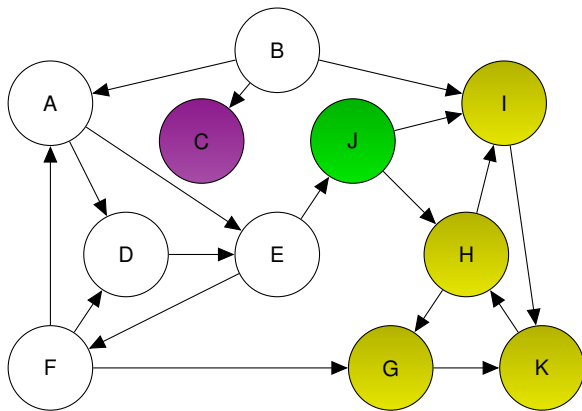
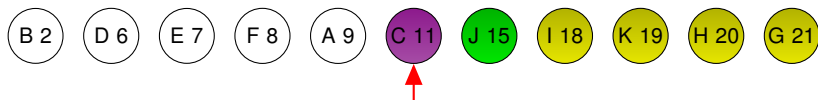
Componenti fortemente connesse



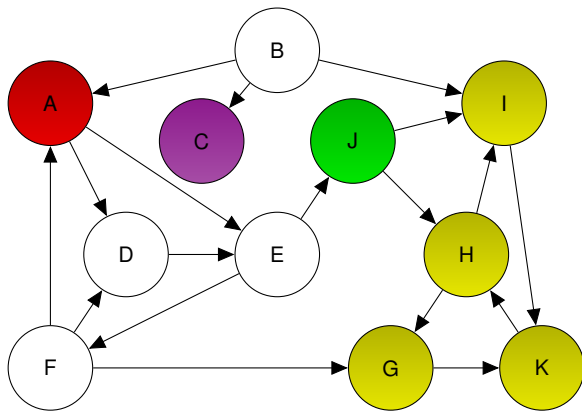
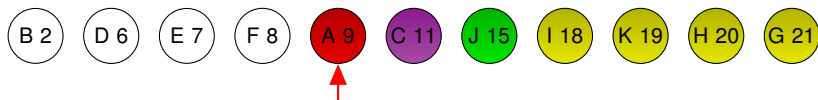
Componenti fortemente connesse



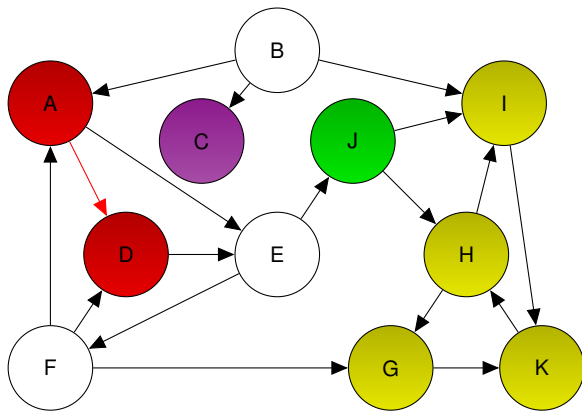
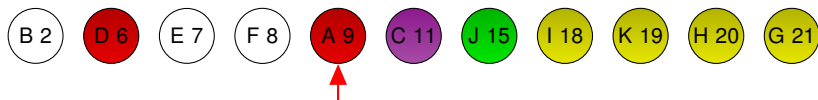
Componenti fortemente connesse



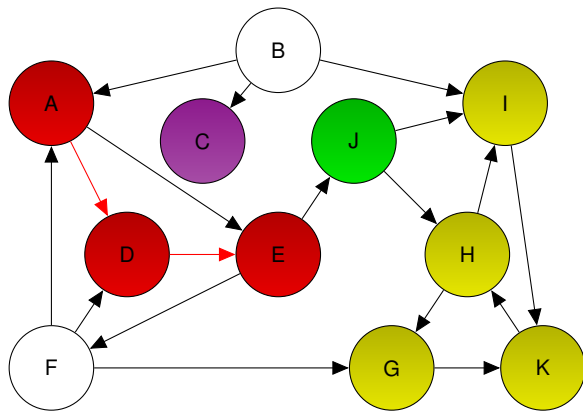
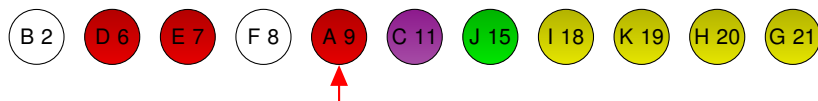
Componenti fortemente connesse



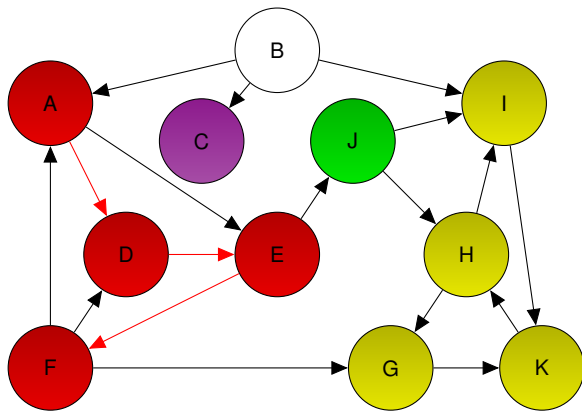
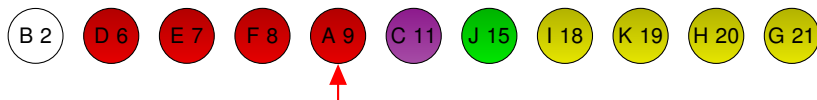
Componenti fortemente connesse



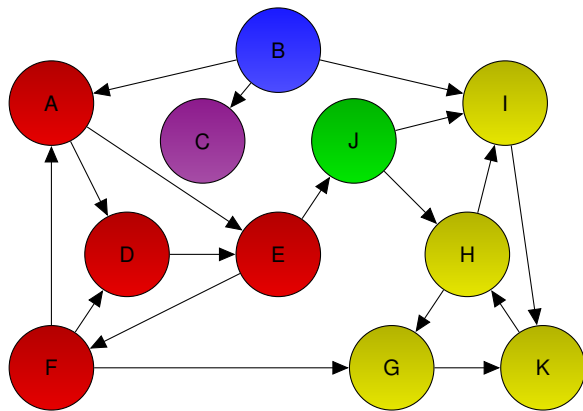
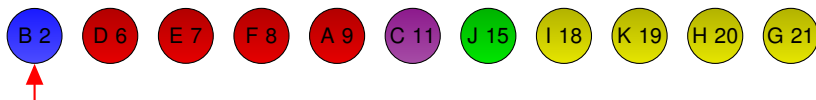
Componenti fortemente connesse



Componenti fortemente connesse



Componenti fortemente connesse



Longest increasing sequence

Data la sequenza

6	3	4	12	8	9	14	7	10	8
---	---	---	----	---	---	----	---	----	---

calcolare una sottosequenza crescente di lunghezza massima

Longest increasing sequence

0 1 2 3 4 5 6 7 8 9

6	3	4	12	8	9	14	7	10	8
---	---	---	----	---	---	----	---	----	---

LIS

1									
---	--	--	--	--	--	--	--	--	--

Parent

-1									
----	--	--	--	--	--	--	--	--	--

Longest increasing sequence

0 1 2 3 4 5 6 7 8 9

6	3	4	12	8	9	14	7	10	8
---	---	---	----	---	---	----	---	----	---

LIS

1	1								
---	---	--	--	--	--	--	--	--	--

Parent

-1	-1								
----	----	--	--	--	--	--	--	--	--

Longest increasing sequence

0 1 2 3 4 5 6 7 8 9

6	3	4	12	8	9	14	7	10	8
---	---	---	----	---	---	----	---	----	---

LIS

1	1	2							
---	---	---	--	--	--	--	--	--	--

Parent

-1	-1	1							
----	----	---	--	--	--	--	--	--	--

Longest increasing sequence

0 1 2 3 4 5 6 7 8 9

6	3	4	12	8	9	14	7	10	8
---	---	---	----	---	---	----	---	----	---

LIS

1	1	2	3						
---	---	---	---	--	--	--	--	--	--

Parent

-1	-1	1	2						
----	----	---	---	--	--	--	--	--	--

Longest increasing sequence

0 1 2 3 4 5 6 7 8 9

6	3	4	12	8	9	14	7	10	8
---	---	---	----	---	---	----	---	----	---

LIS

1	1	2	3	3					
---	---	---	---	---	--	--	--	--	--

Parent

-1	-1	1	2	2					
----	----	---	---	---	--	--	--	--	--

Longest increasing sequence

0 1 2 3 4 5 6 7 8 9

6	3	4	12	8	9	14	7	10	8
---	---	---	----	---	---	----	---	----	---

LIS

1	1	2	3	3	4				
---	---	---	---	---	---	--	--	--	--

Parent

-1	-1	1	2	2	4				
----	----	---	---	---	---	--	--	--	--

Longest increasing sequence

0 1 2 3 4 5 6 7 8 9

6	3	4	12	8	9	14	7	10	8
---	---	---	----	---	---	----	---	----	---

LIS

1	1	2	3	3	4	5			
---	---	---	---	---	---	---	--	--	--

Parent

-1	-1	1	2	2	4	5			
----	----	---	---	---	---	---	--	--	--

Longest increasing sequence

0 1 2 3 4 5 6 7 8 9

6	3	4	12	8	9	14	7	10	8
---	---	---	----	---	---	----	---	----	---

LIS

1	1	2	3	3	4	5	3		
---	---	---	---	---	---	---	---	--	--

Parent

-1	-1	1	2	2	4	5	2		
----	----	---	---	---	---	---	---	--	--

Longest increasing sequence

0 1 2 3 4 5 6 7 8 9

6	3	4	12	8	9	14	7	10	8
---	---	---	----	---	---	----	---	----	---

LIS

1	1	2	3	3	4	5	3	5	
---	---	---	---	---	---	---	---	---	--

Parent

-1	-1	1	2	2	4	5	2	5	
----	----	---	---	---	---	---	---	---	--

Longest increasing sequence

0 1 2 3 4 5 6 7 8 9

6	3	4	12	8	9	14	7	10	8
---	---	---	----	---	---	----	---	----	---

LIS

1	1	2	3	3	4	5	3	5	4
---	---	---	---	---	---	---	---	---	---

Parent

-1	-1	1	2	2	4	5	2	5	7
----	----	---	---	---	---	---	---	---	---

Longest increasing sequence

LIS: ultimo elemento in posizione 8 (massimo del vettore LIS)

Completiamo la sequenza seguendo all'indietro i riferimenti nel vettore Parent:

	0	1	2	3	4	5	6	7	8	9
	6	3	4	12	8	9	14	7	10	8
LIS	1	1	2	3	3	4	5	3	5	4
Parent	-1	-1	1	2	2	4	5	2	5	7

Longest increasing sequence

LIS: ultimo elemento in posizione 8 (massimo del vettore LIS)

Completiamo la sequenza seguendo all'indietro i riferimenti nel vettore Parent:

	0	1	2	3	4	5	6	7	8	9
	6	3	4	12	8	9	14	7	10	8
LIS	1	1	2	3	3	4	5	3	5	4
Parent	-1	-1	1	2	2	4	5	2	5	7

In alternativa, una seconda sequenza termina con l'elemento in posizione 6

Longest increasing sequence

Equivalente a ricerca cammino di lunghezza massima su DAG

Ogni elemento della sequenza è un nodo

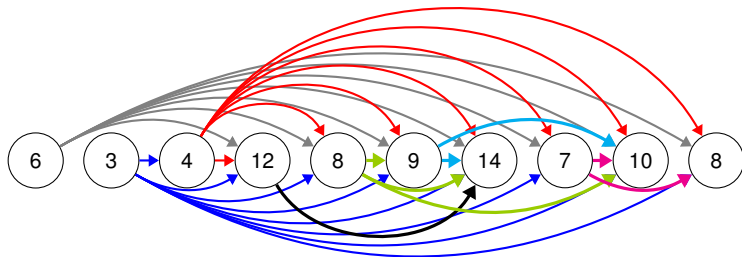
I nodi sono (topologicamente) ordinati secondo l'ordine della sequenza

Un arco (u, v) collega i nodi u e v se e solo se v segue u nell'ordinamento (topologico) e il valore di v è maggiore del valore di u

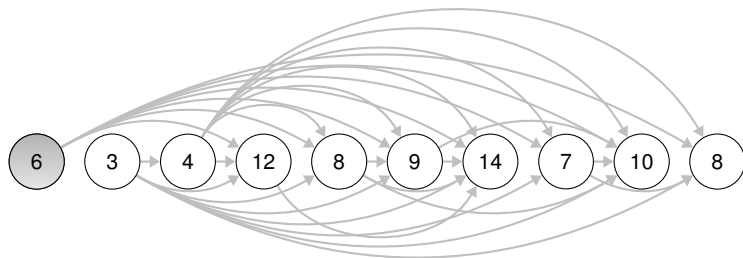
Longest increasing sequence

Applichiamo l'algoritmo per i cammini massimi, mettendo $d(v) = 1$ per i nodi senza predecessori

6	3	4	12	8	9	14	7	10	8
---	---	---	----	---	---	----	---	----	---

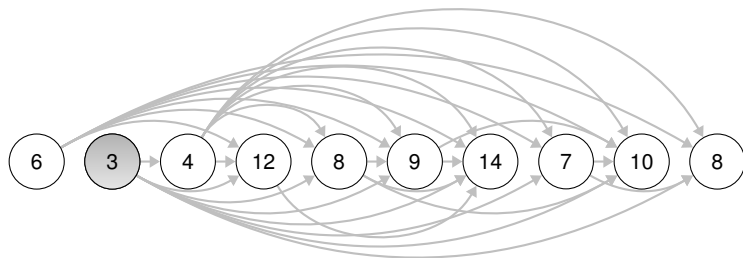


Longest increasing sequence



1

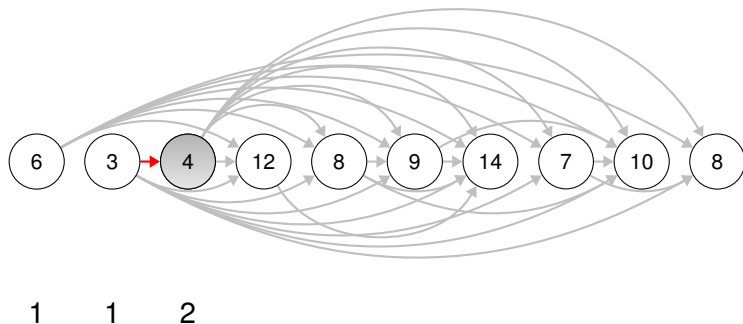
Longest increasing sequence



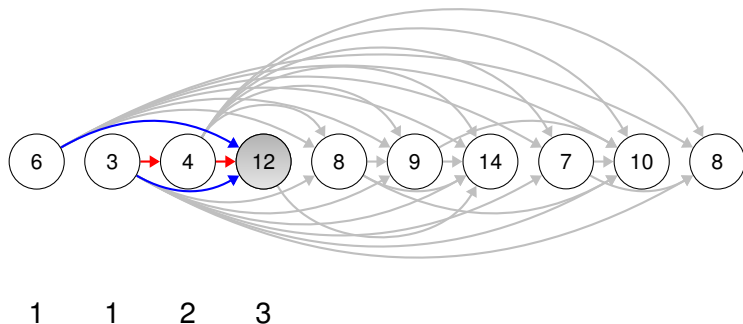
1

1

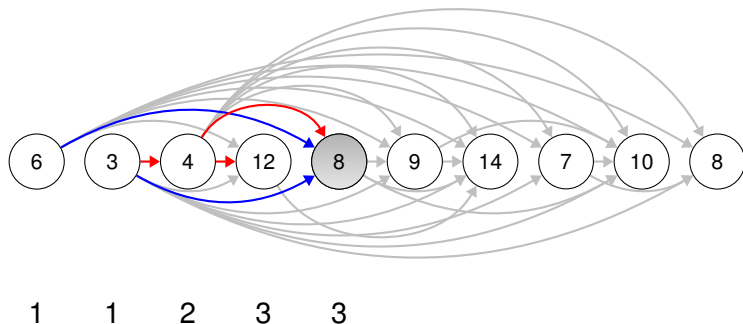
Longest increasing sequence



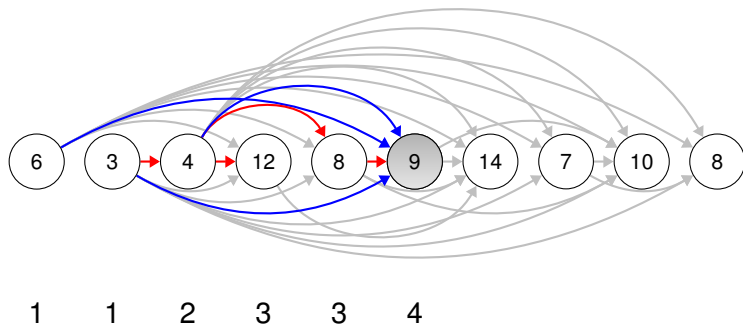
Longest increasing sequence



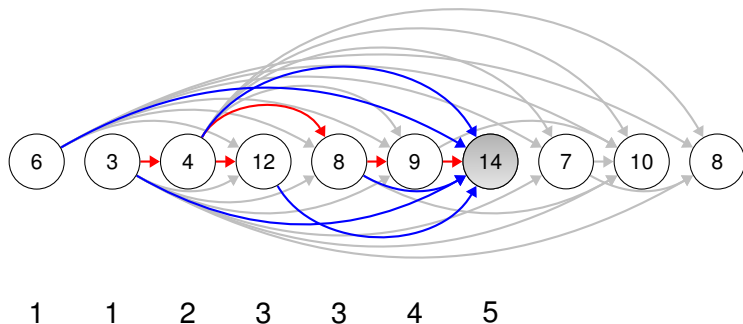
Longest increasing sequence



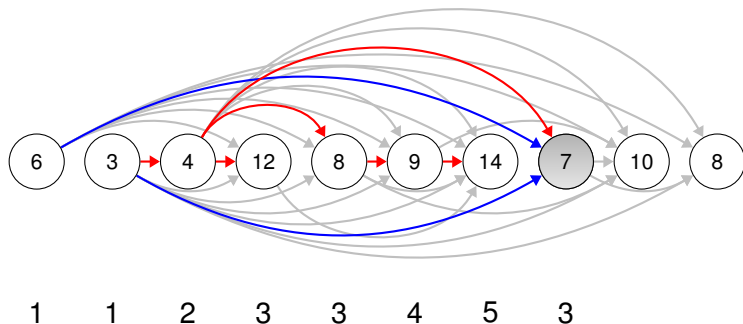
Longest increasing sequence



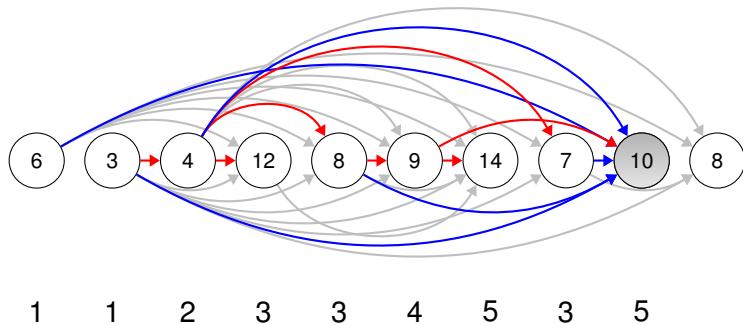
Longest increasing sequence



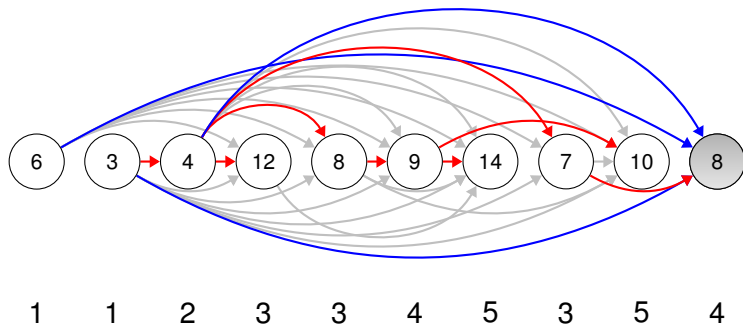
Longest increasing sequence



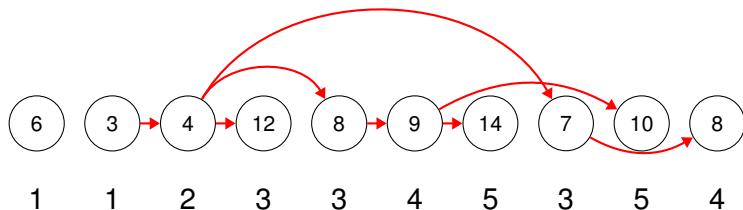
Longest increasing sequence



Longest increasing sequence

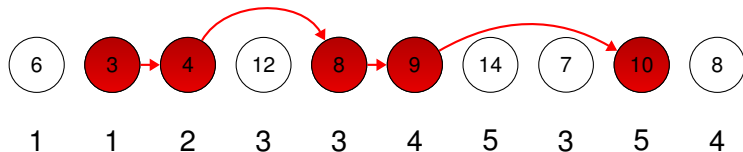


Longest increasing sequence



Una LIS termina in nodo 10 (alternativamente, in nodo 14)

Seguendo gli archi a ritroso possiamo ricostruire la sequenza



Longest increasing sequence

0 1 2 3 4 5 6 7 8 9

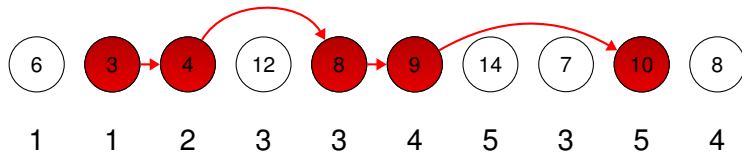
6	3	4	12	8	9	14	7	10	8
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LIS

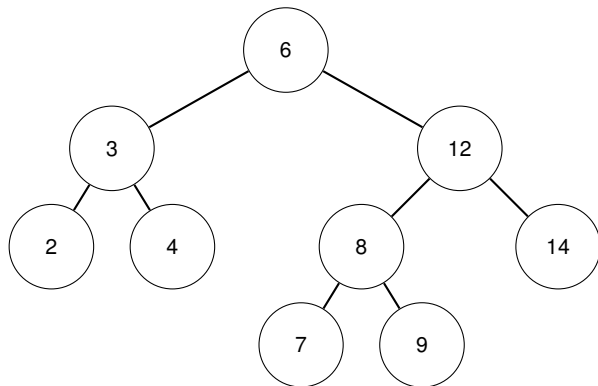
1	1	2	3	3	4	5	3	5	4
---	---	---	---	---	---	---	---	---	---

Parent

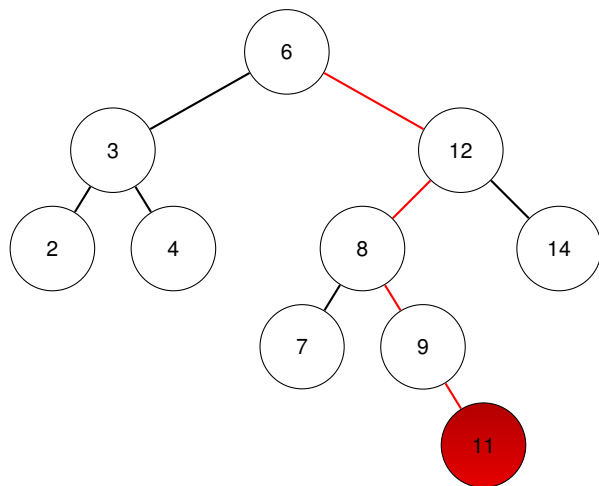
-1	-1	1	2	2	4	5	2	5	7
----	----	---	---	---	---	---	---	---	---



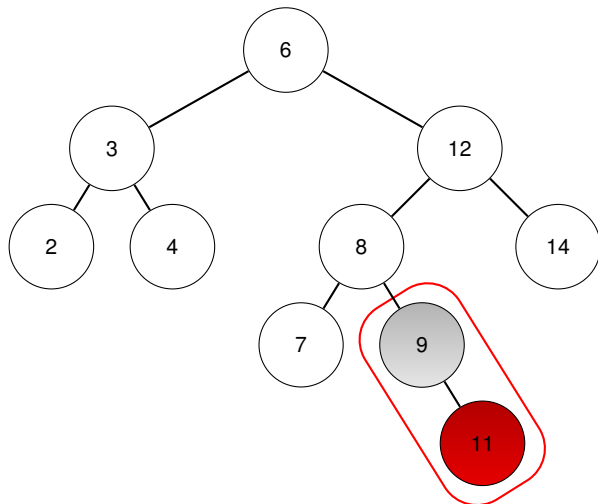
Inserimento in radice: dato il seguente BST, inserire il valore 11 in radice



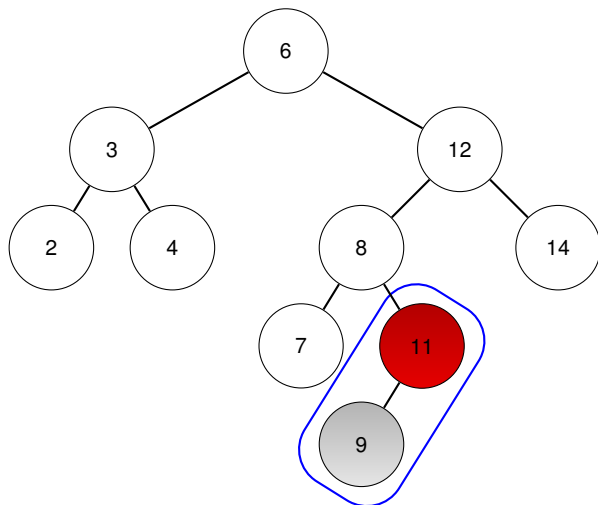
Inseriamo il nodo come foglia



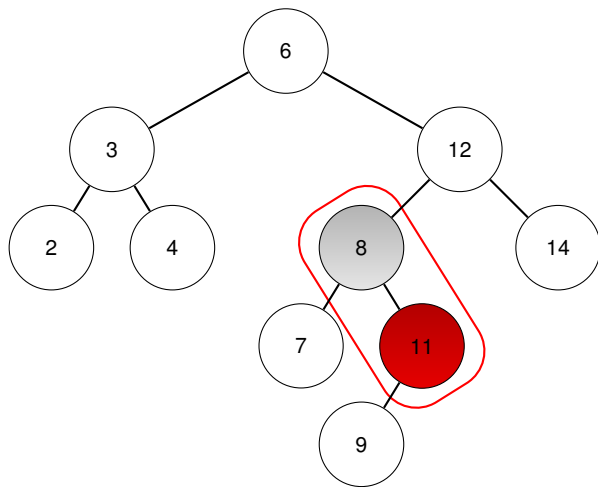
Effettuiamo le rotazioni per far diventare il nodo inserito la radice dell'albero



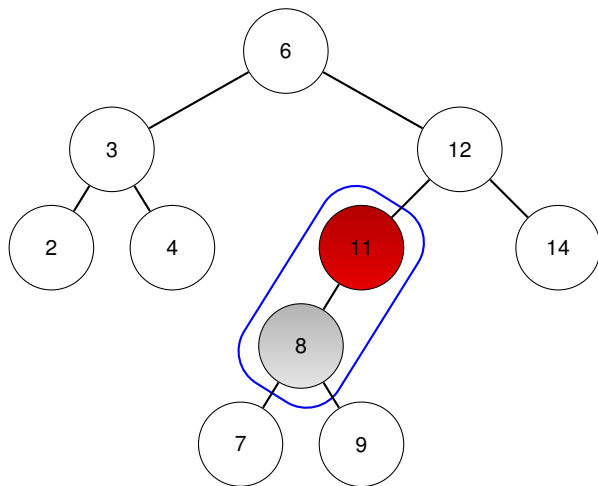
Effettuiamo le rotazioni per far diventare il nodo inserito la radice dell'albero



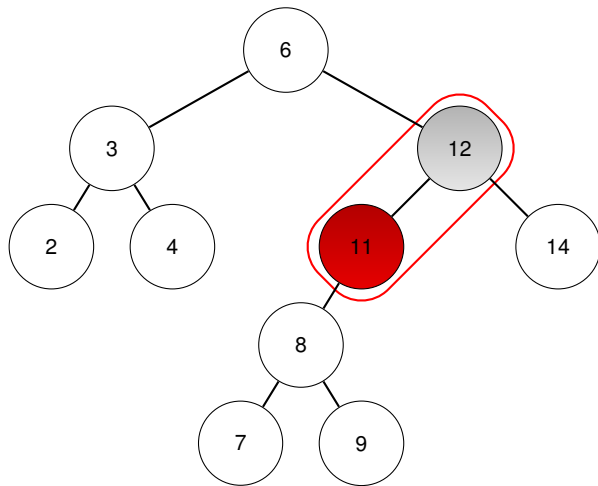
Effettuiamo le rotazioni per far diventare il nodo inserito la radice dell'albero



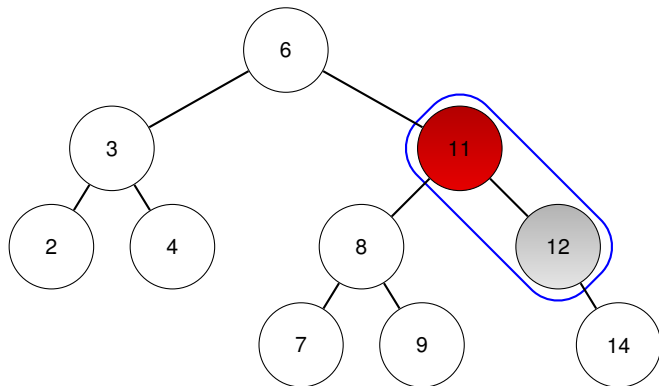
Effettuiamo le rotazioni per far diventare il nodo inserito la radice dell'albero



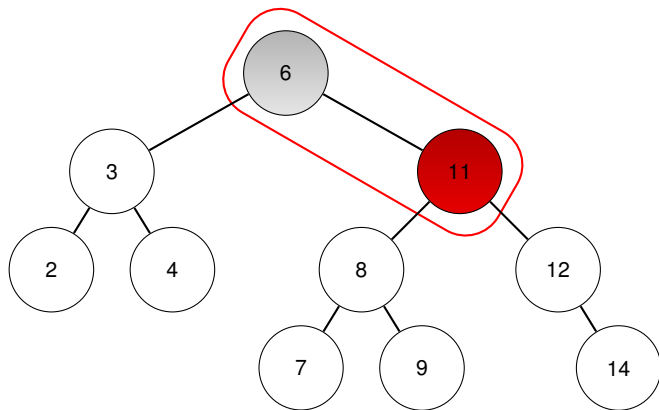
Effettuiamo le rotazioni per far diventare il nodo inserito la radice dell'albero



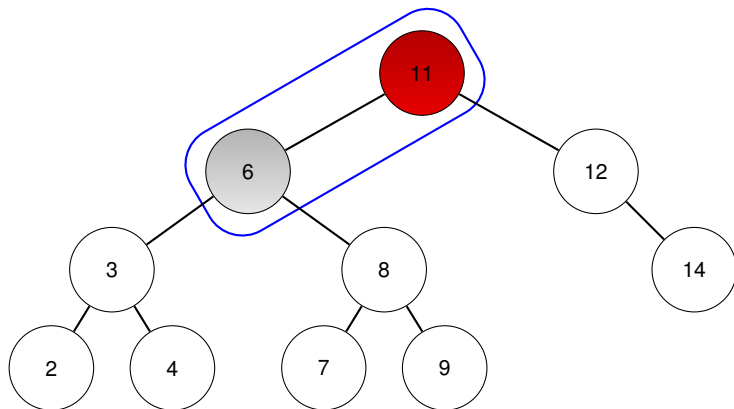
Effettuiamo le rotazioni per far diventare il nodo inserito la radice dell'albero



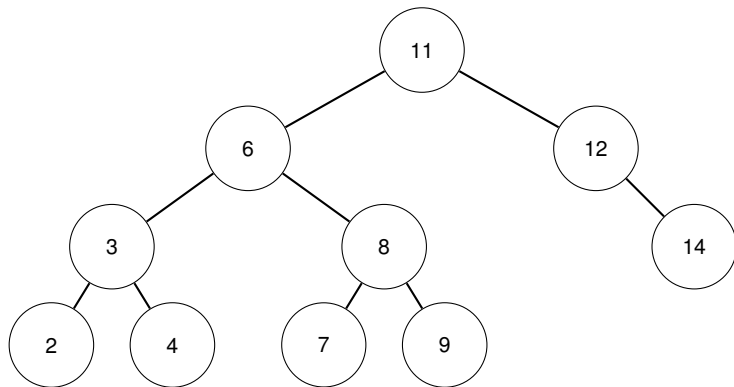
Effettuiamo le rotazioni per far diventare il nodo inserito la radice dell'albero



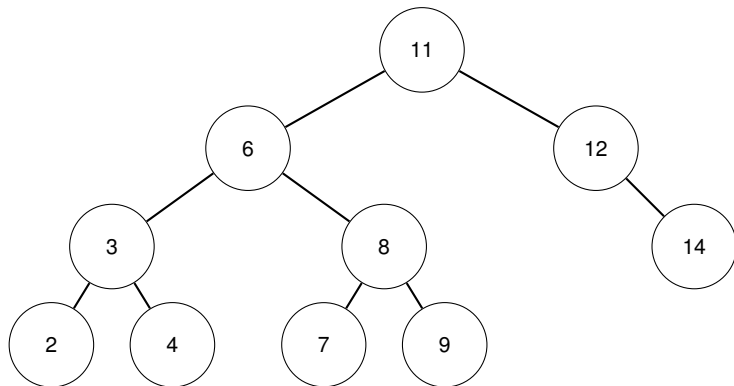
Effettuiamo le rotazioni per far diventare il nodo inserito la radice dell'albero



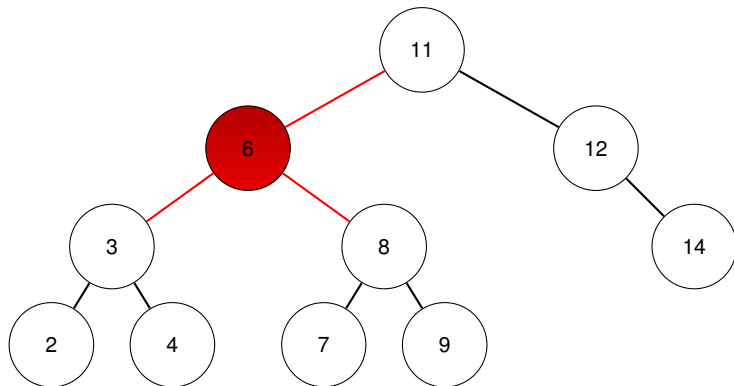
Effettuiamo le rotazioni per far diventare il nodo inserito la radice dell'albero



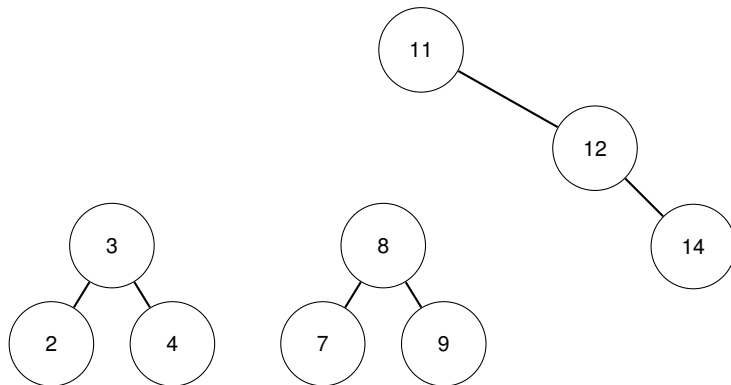
Cancellazione: vogliamo rimuovere il nodo 6



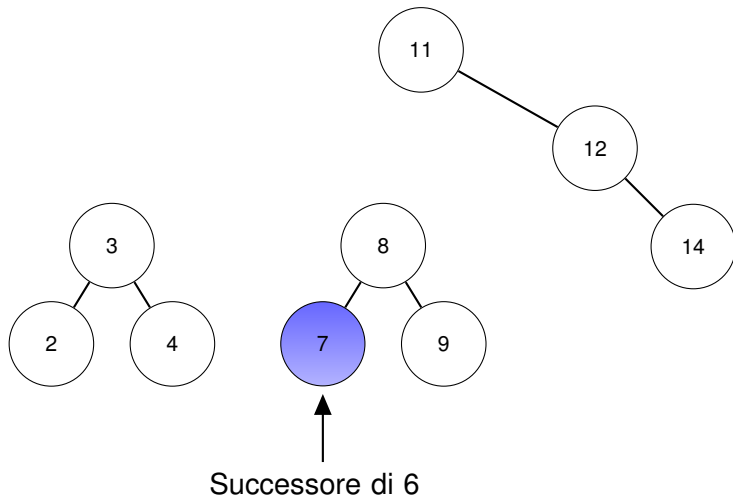
Cancellazione: vogliamo rimuovere il nodo 6



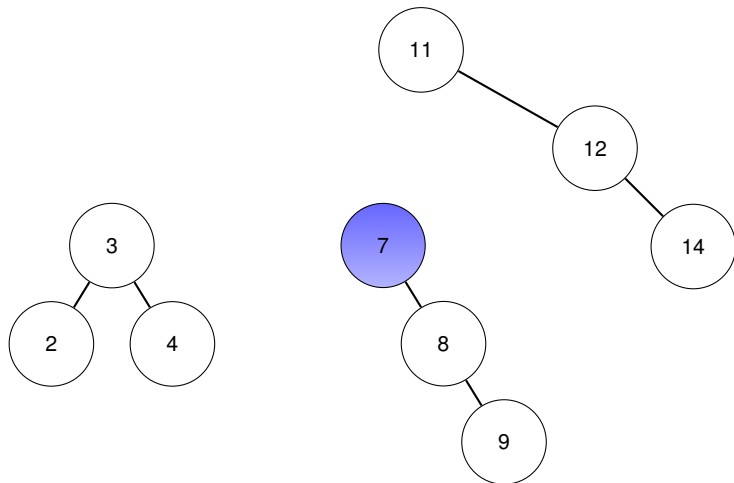
Cancellazione: vogliamo rimuovere il nodo 6



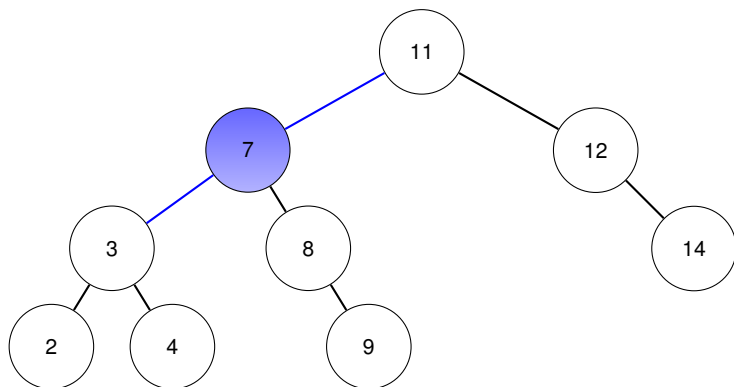
Cancellazione: vogliamo rimuovere il nodo 6



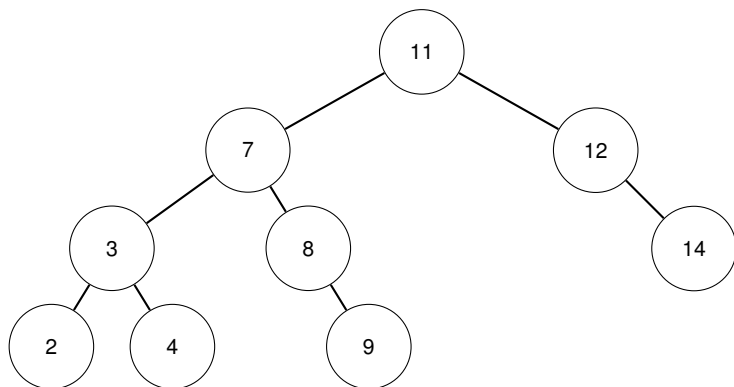
Partizionamento del sottoalbero DX rispetto al nodo 7



Ricostruzione dell'albero



Ricostruzione dell'albero

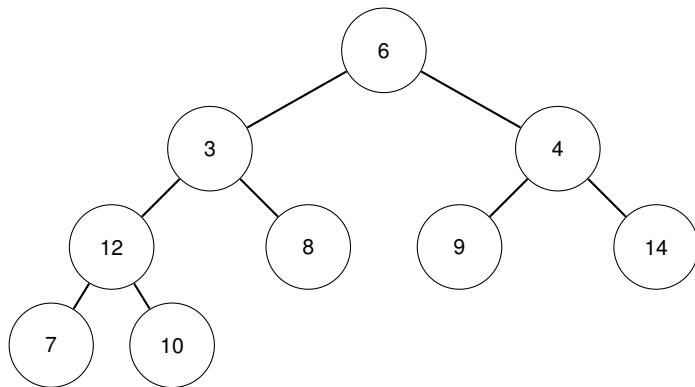


Ordinare tramite Heapsort il vettore

6	3	4	12	8	9	14	7	10
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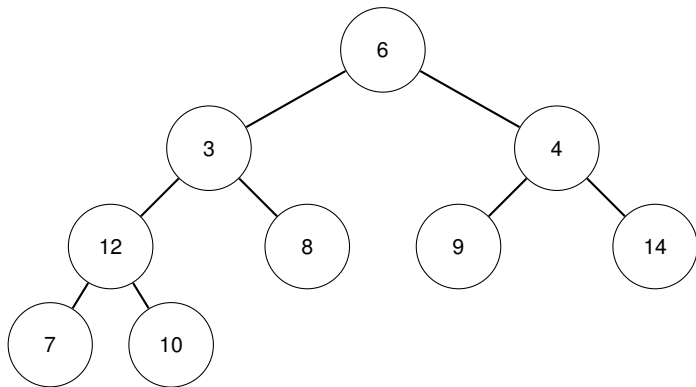
Ordinare tramite Heapsort il vettore

6	3	4	12	8	9	14	7	10
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Heapsort

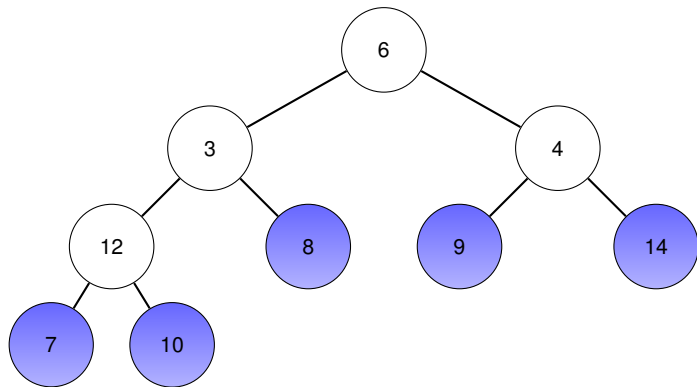
Heapbuild: trasformazione dell'albero in un heap



6	3	4	12	8	9	14	7	10
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Heapsort

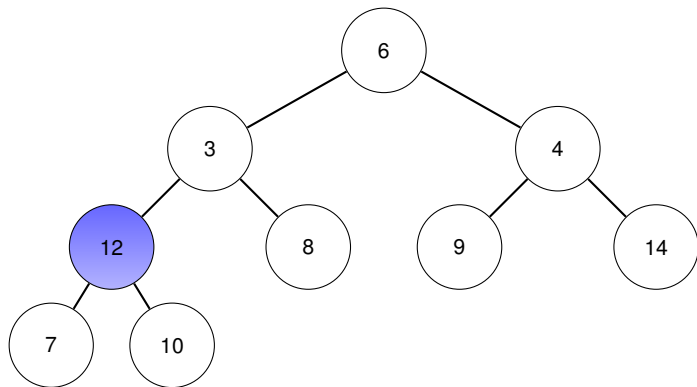
Le foglie sono già heap



6	3	4	12	8	9	14	7	10
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Heapsort

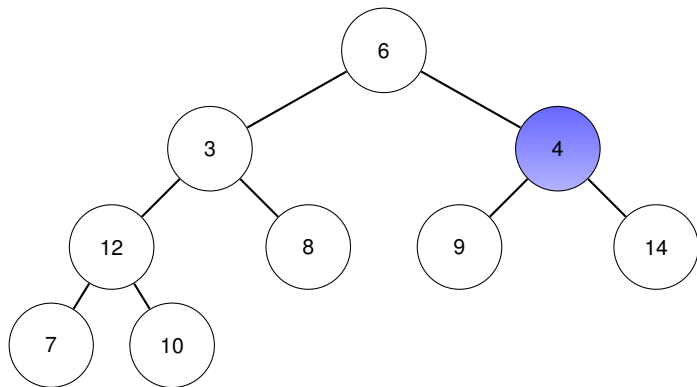
Applichiamo di HEAPify dal primo nodo non foglia fino alla radice



6	3	4	12	8	9	14	7	10
---	---	---	----	---	---	----	---	----

Heapsort

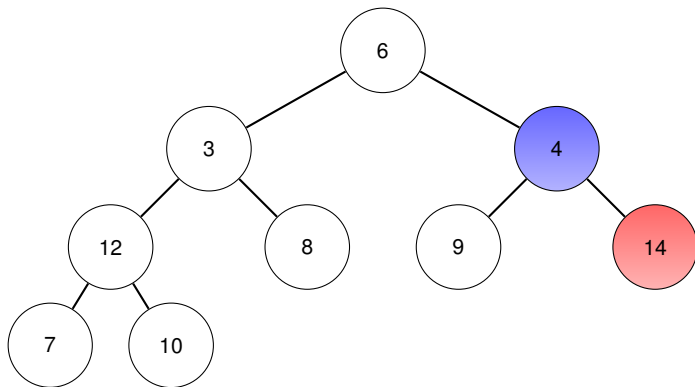
Applichiamo di HEAPify dal primo nodo non foglia fino alla radice



6	3	4	12	8	9	14	7	10
---	---	---	----	---	---	----	---	----

Heapsort

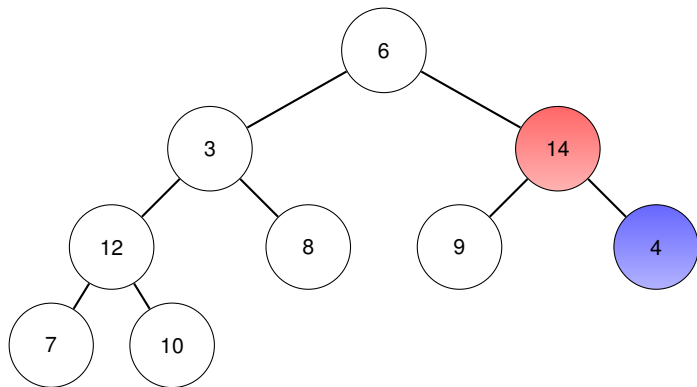
Applichiamo di HEAPify dal primo nodo non foglia fino alla radice



6	3	4	12	8	9	14	7	10
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Heapsort

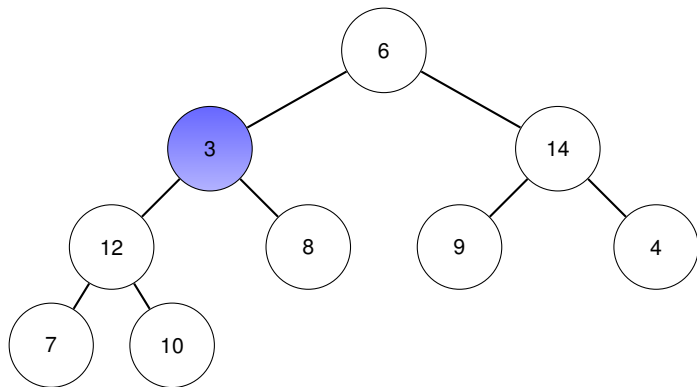
Applichiamo di HEAPify dal primo nodo non foglia fino alla radice



6	3	14	12	8	9	4	7	10
---	---	----	----	---	---	---	---	----

Heapsort

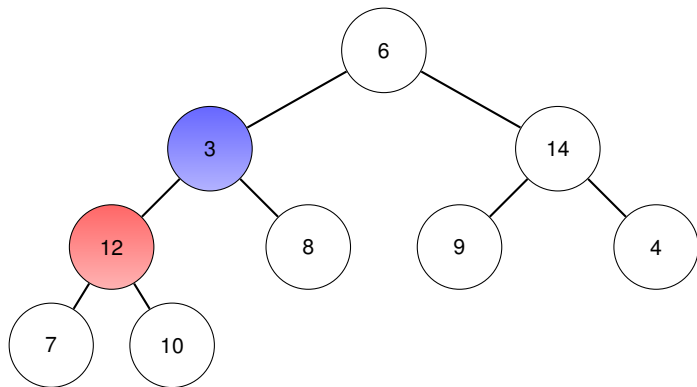
Applichiamo di HEAPify dal primo nodo non foglia fino alla radice



6	3	14	12	8	9	4	7	10
---	---	----	----	---	---	---	---	----

Heapsort

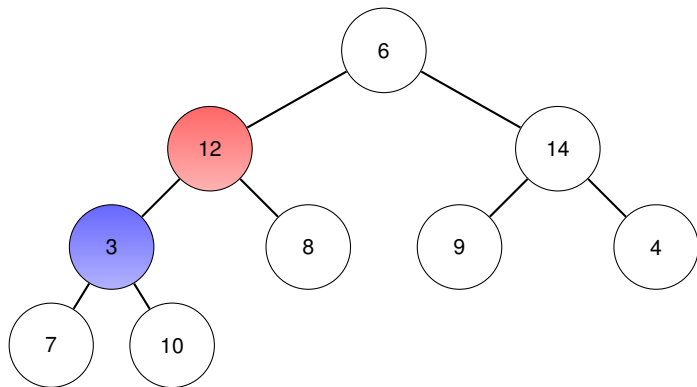
Applichiamo di HEAPify dal primo nodo non foglia fino alla radice



6	3	14	12	8	9	4	7	10
---	---	----	----	---	---	---	---	----

Heapsort

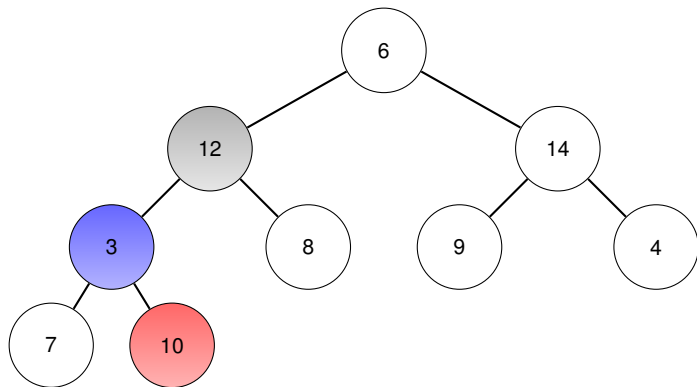
Applichiamo di HEAPify dal primo nodo non foglia fino alla radice



6	12	14	3	8	9	4	7	10
---	----	----	---	---	---	---	---	----

Heapsort

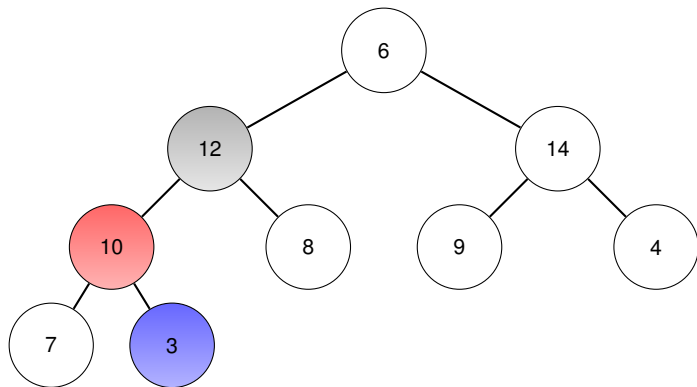
Applichiamo di HEAPify dal primo nodo non foglia fino alla radice



6	12	14	3	8	9	4	7	10
---	----	----	---	---	---	---	---	----

Heapsort

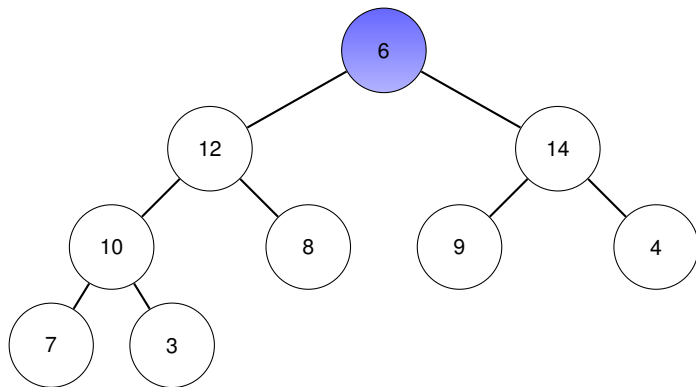
Applichiamo di HEAPIfy dal primo nodo non foglia fino alla radice



6	12	14	10	8	9	4	7	3
---	----	----	----	---	---	---	---	---

Heapsort

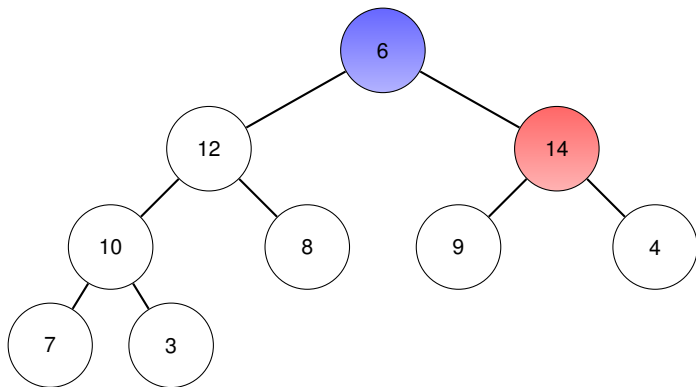
Applichiamo di HEAPIfy dal primo nodo non foglia fino alla radice



6	12	14	10	8	9	4	7	3
---	----	----	----	---	---	---	---	---

Heapsort

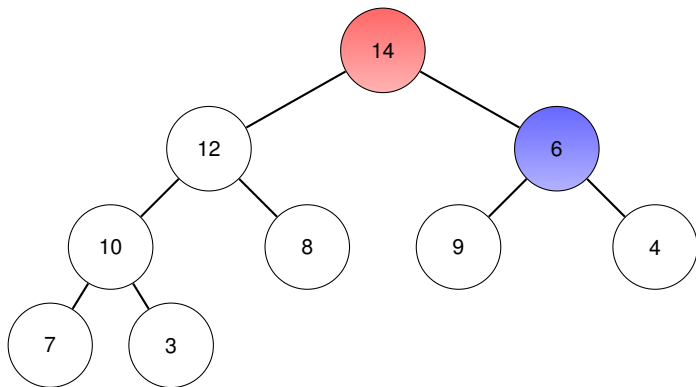
Applichiamo di HEAPIfy dal primo nodo non foglia fino alla radice



6	12	14	10	8	9	4	7	3
---	----	----	----	---	---	---	---	---

Heapsort

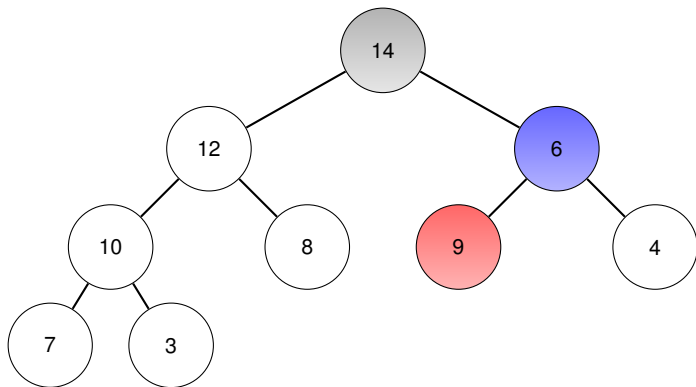
Applichiamo di HEAPIfy dal primo nodo non foglia fino alla radice



14	12	6	10	8	9	4	7	3
----	----	---	----	---	---	---	---	---

Heapsort

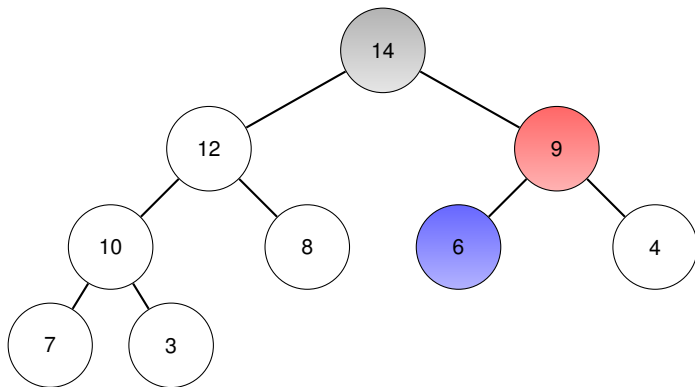
Applichiamo di HEAPIfy dal primo nodo non foglia fino alla radice



14	12	6	10	8	9	4	7	3
----	----	---	----	---	---	---	---	---

Heapsort

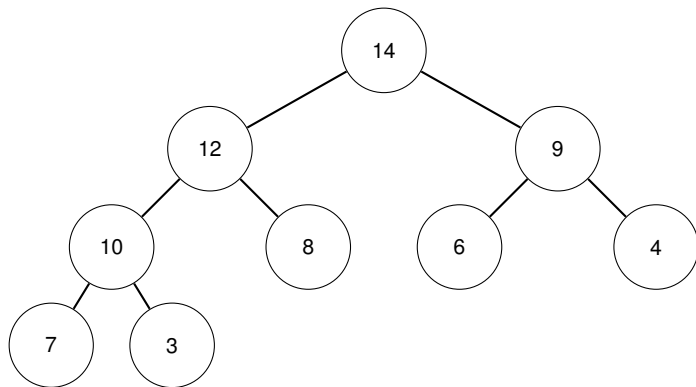
Applichiamo di HEAPIfy dal primo nodo non foglia fino alla radice



14	12	9	10	8	6	4	7	3
----	----	---	----	---	---	---	---	---

Heapsort

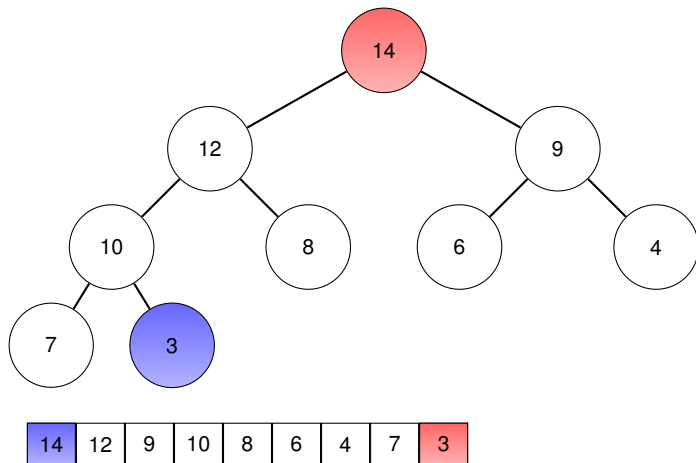
Heap:



14	12	9	10	8	6	4	7	3
----	----	---	----	---	---	---	---	---

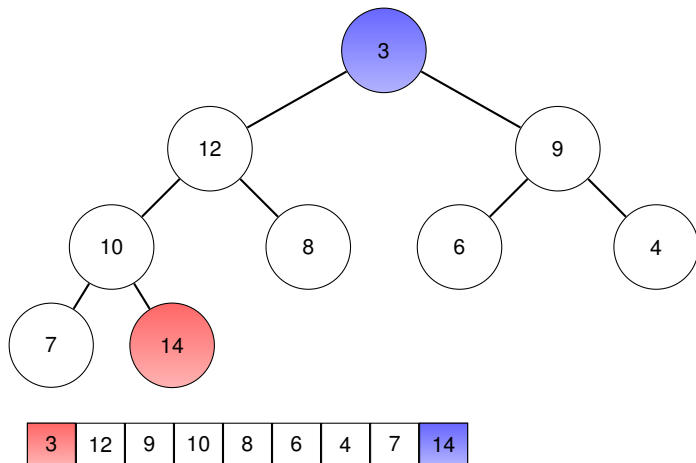
Heapsort

Scambio primo e ultimo elemento:



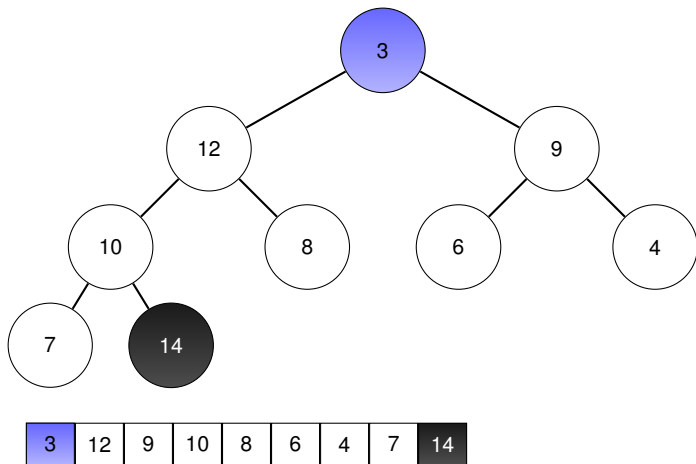
Heapsort

Scambio primo e ultimo elemento:



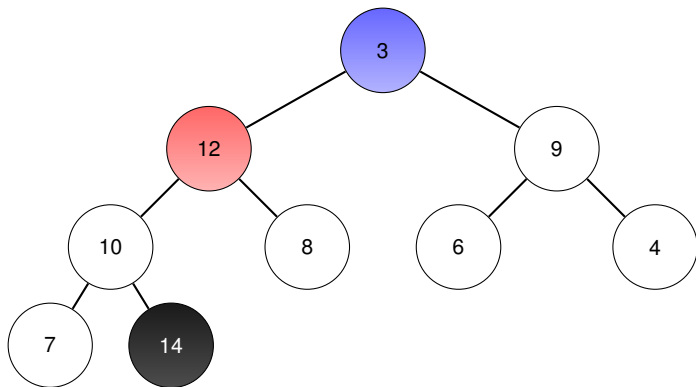
Heapsort

Ricostruzione Heap:



Heapsort

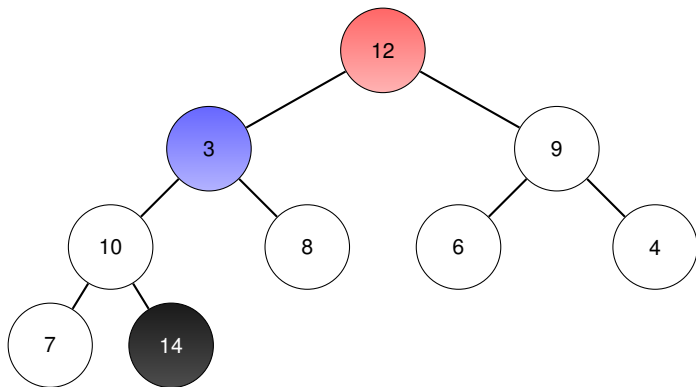
Ricostruzione Heap:



3	12	9	10	8	6	4	7	14
---	----	---	----	---	---	---	---	----

Heapsort

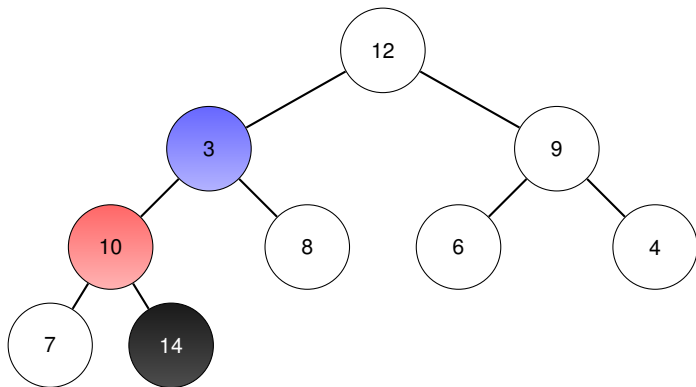
Ricostruzione Heap:



12	3	9	10	8	6	4	7	14
----	---	---	----	---	---	---	---	----

Heapsort

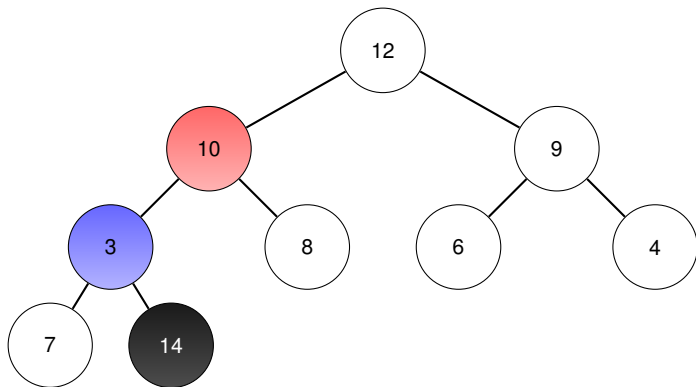
Ricostruzione Heap:



12	3	9	10	8	6	4	7	14
----	---	---	----	---	---	---	---	----

Heapsort

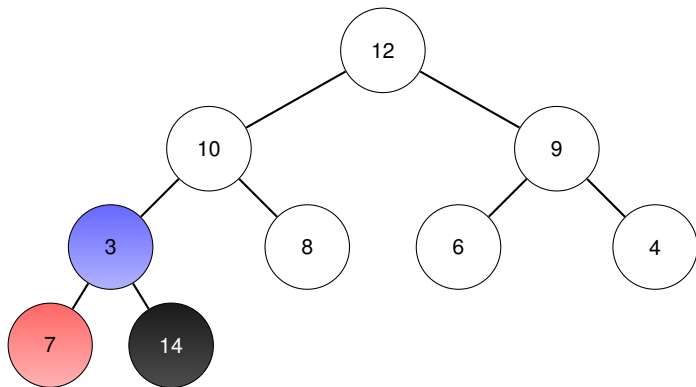
Ricostruzione Heap:



12	10	9	3	8	6	4	7	14
----	----	---	---	---	---	---	---	----

Heapsort

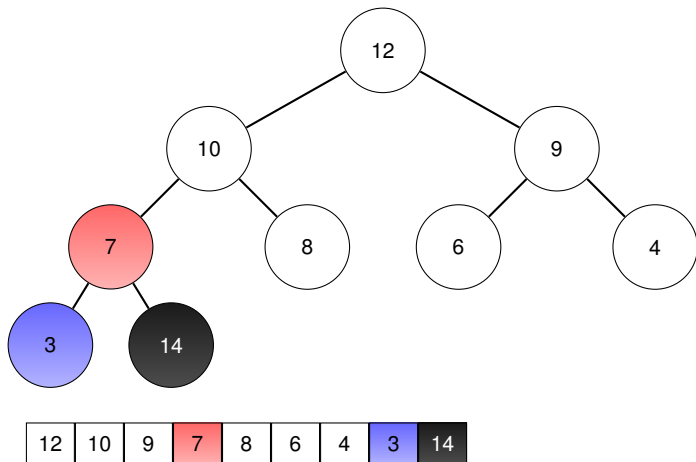
Ricostruzione Heap:



12	10	9	3	8	6	4	7	14
----	----	---	---	---	---	---	---	----

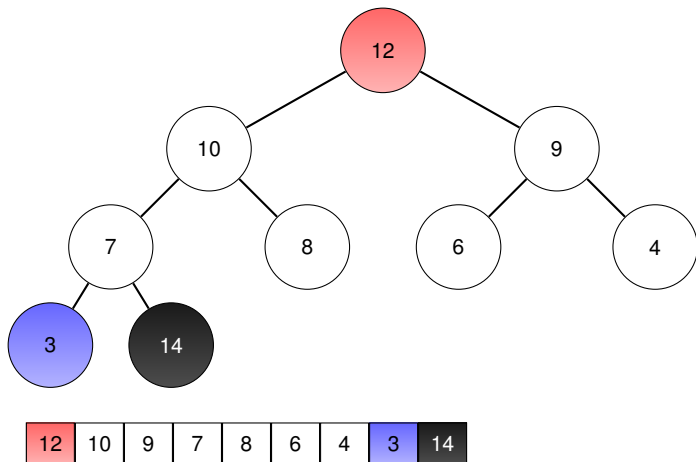
Heapsort

Ricostruzione Heap:



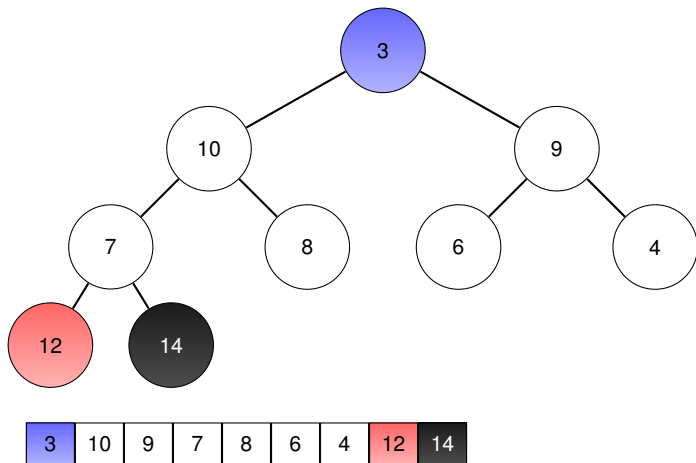
Heapsort

Scambio primo e ultimo elemento:



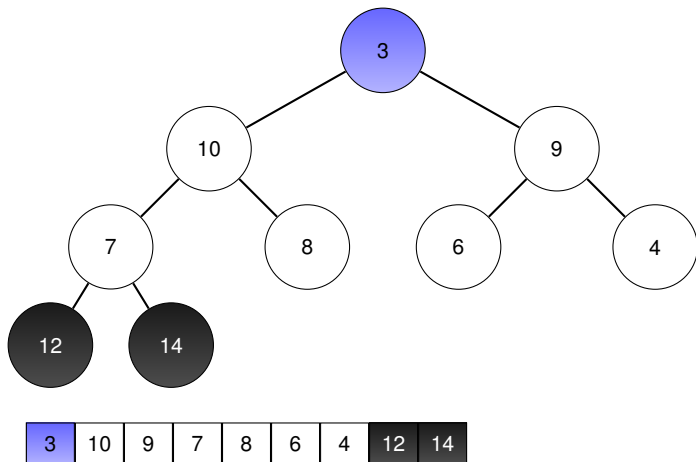
Heapsort

Scambio primo e ultimo elemento:



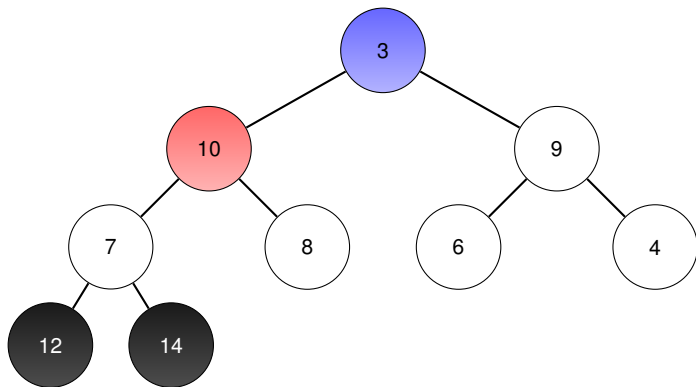
Heapsort

Ricostruzione Heap:



Heapsort

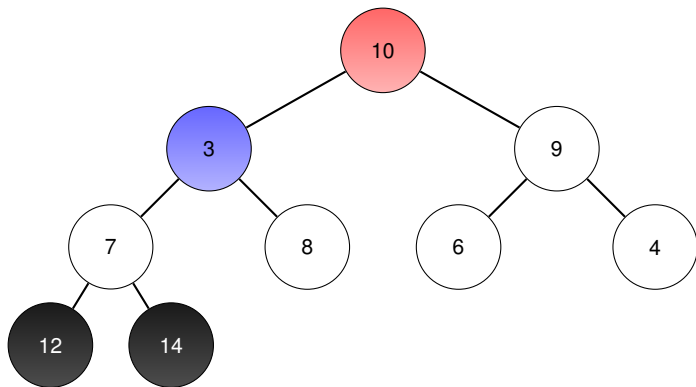
Ricostruzione Heap:



3	10	9	7	8	6	4	12	14
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Heapsort

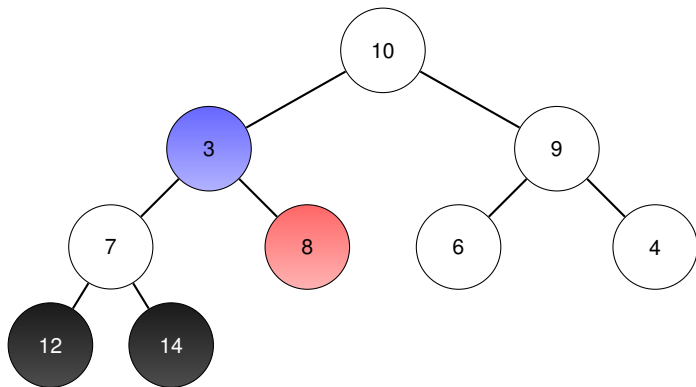
Ricostruzione Heap:



10	3	9	7	8	6	4	12	14
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Heapsort

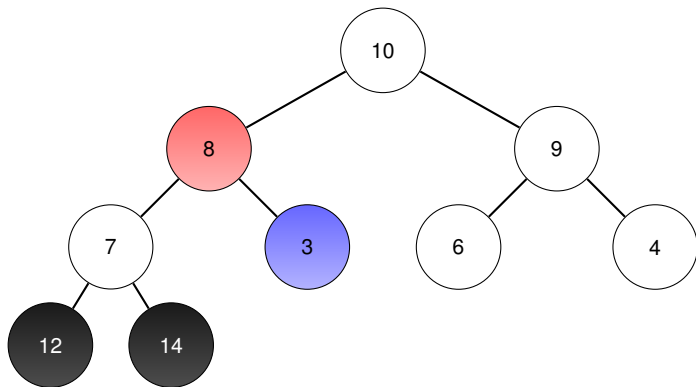
Ricostruzione Heap:



10	3	9	7	8	6	4	12	14
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Heapsort

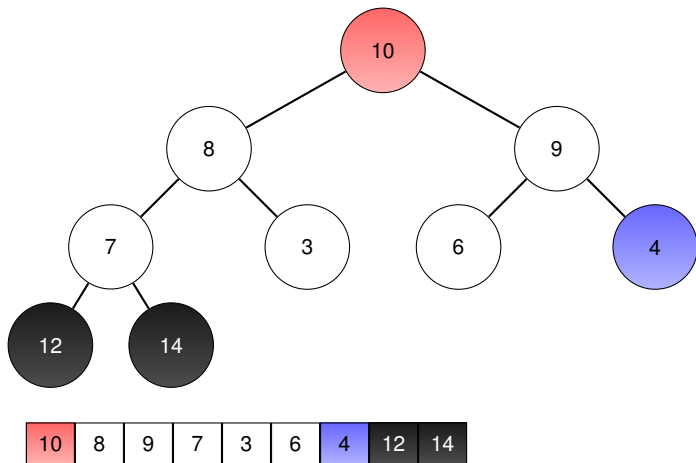
Ricostruzione Heap:



10	8	9	7	3	6	4	12	14
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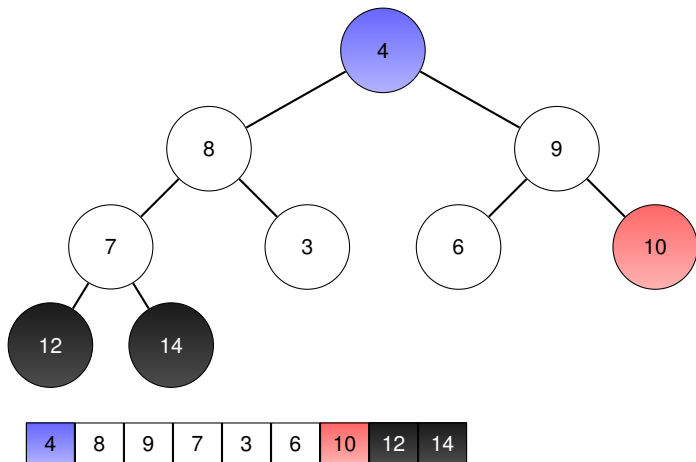
Heapsort

Scambio primo e ultimo elemento:



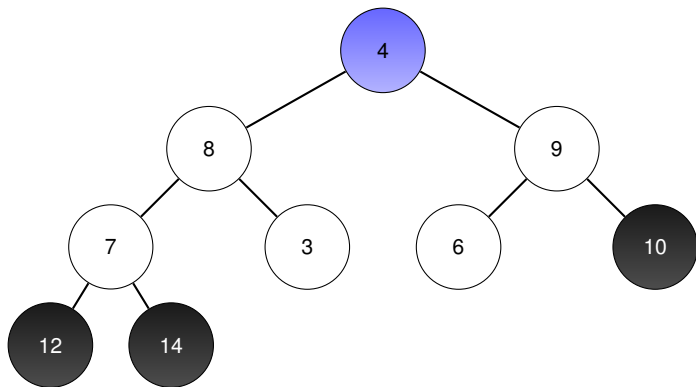
Heapsort

Scambio primo e ultimo elemento:



Heapsort

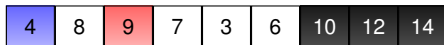
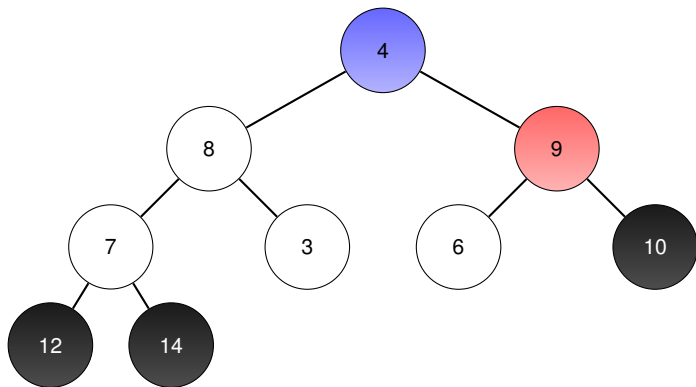
Ricostruzione Heap:



4	8	9	7	3	6	10	12	14
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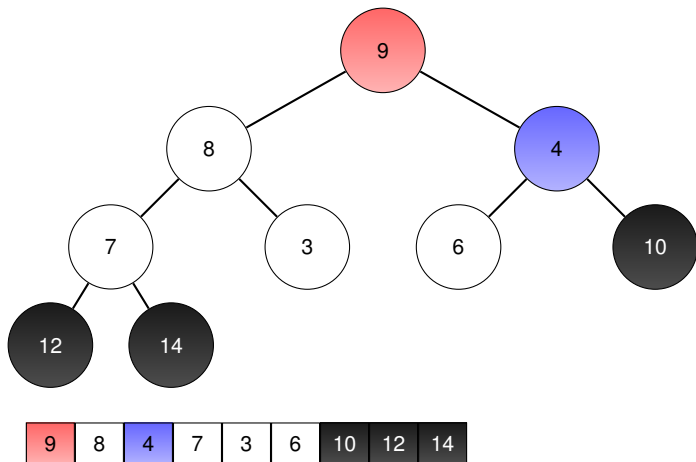
Heapsort

Ricostruzione Heap:



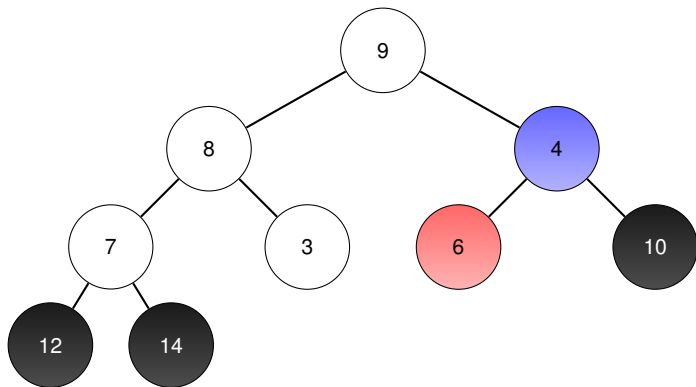
Heapsort

Ricostruzione Heap:



Heapsort

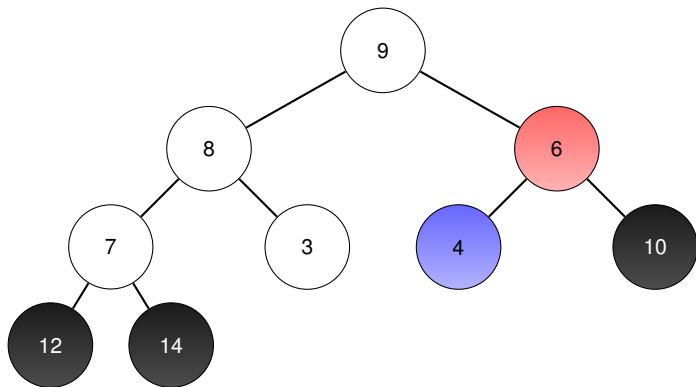
Ricostruzione Heap:



9	8	4	7	3	6	10	12	14
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Heapsort

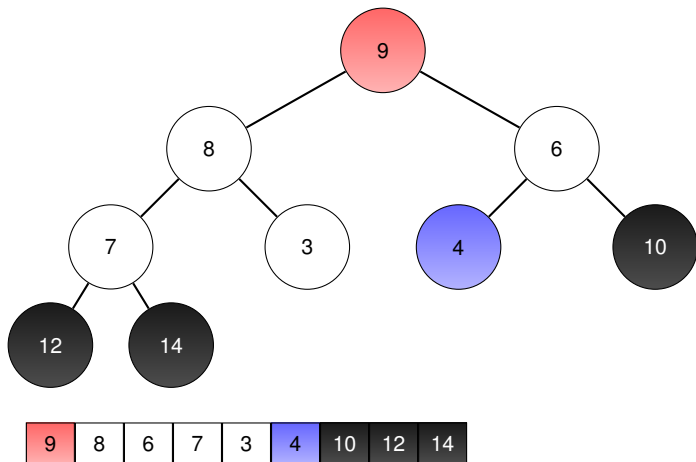
Ricostruzione Heap:



9	8	6	7	3	4	10	12	14
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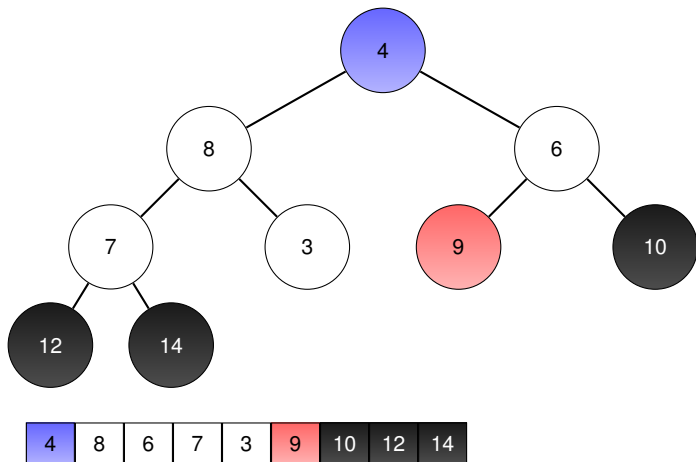
Heapsort

Scambio primo e ultimo elemento:



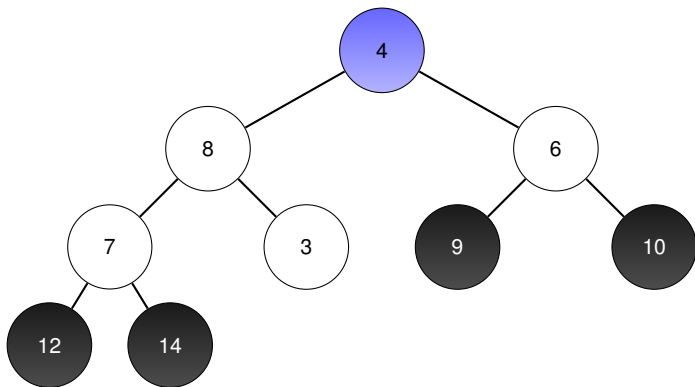
Heapsort

Scambio primo e ultimo elemento:



Heapsort

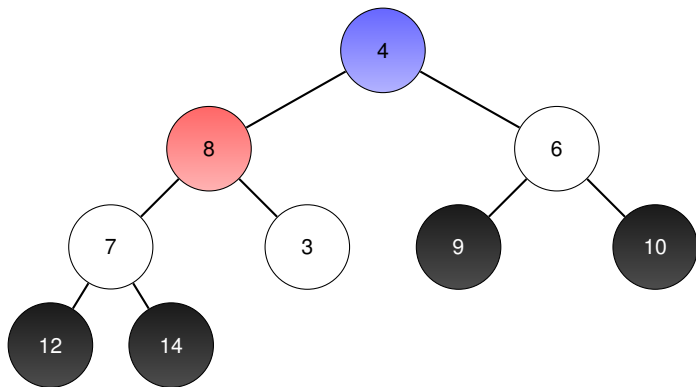
Ricostruzione Heap:



4	8	6	7	3	9	10	12	14
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Heapsort

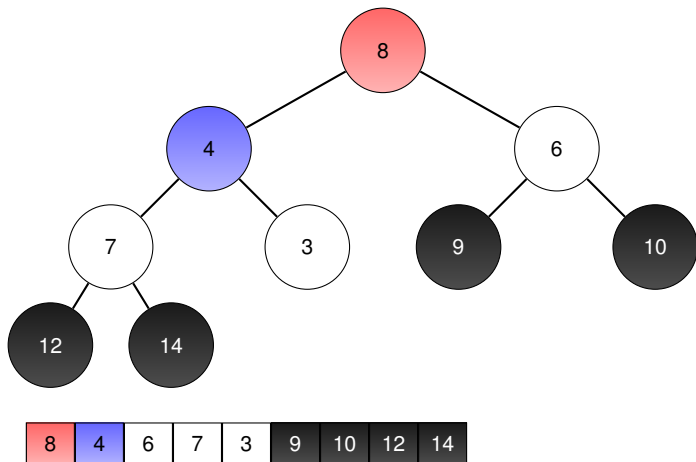
Ricostruzione Heap:



4	8	6	7	3	9	10	12	14
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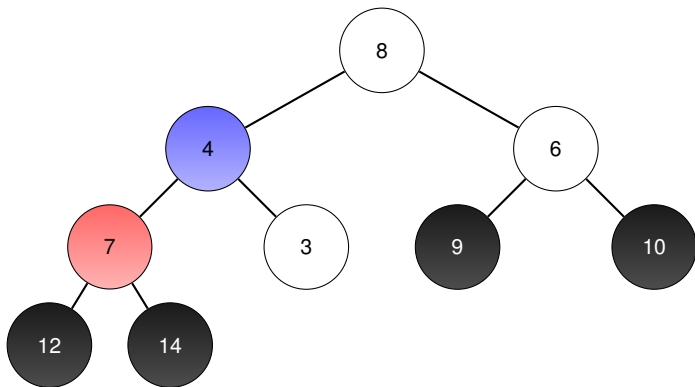
Heapsort

Ricostruzione Heap:



Heapsort

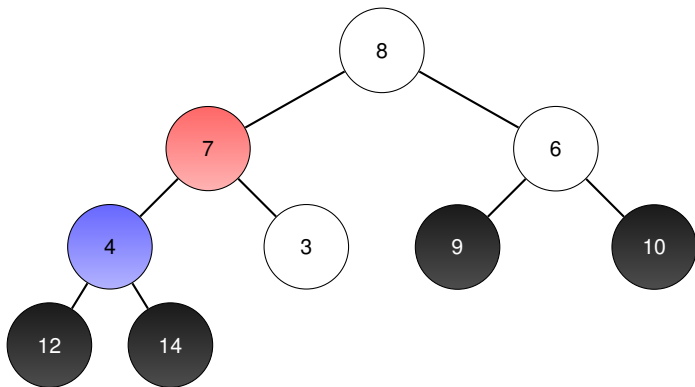
Ricostruzione Heap:



8	4	6	7	3	9	10	12	14
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Heapsort

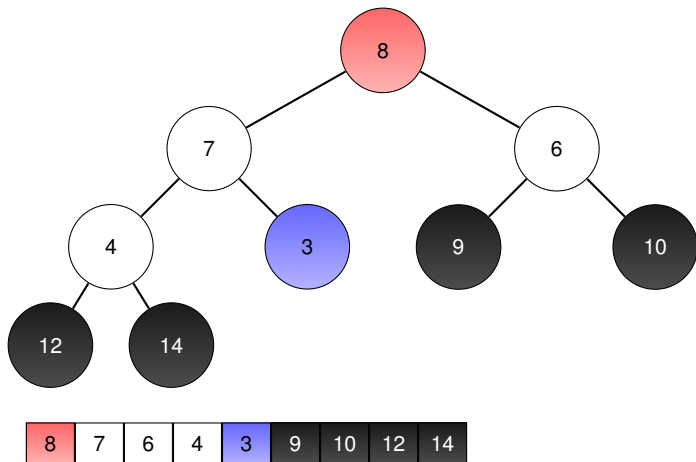
Ricostruzione Heap:



8	7	6	4	3	9	10	12	14
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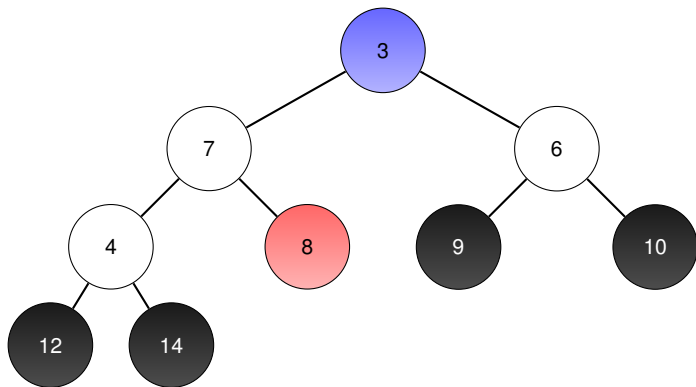
Heapsort

Scambio primo e ultimo elemento:



Heapsort

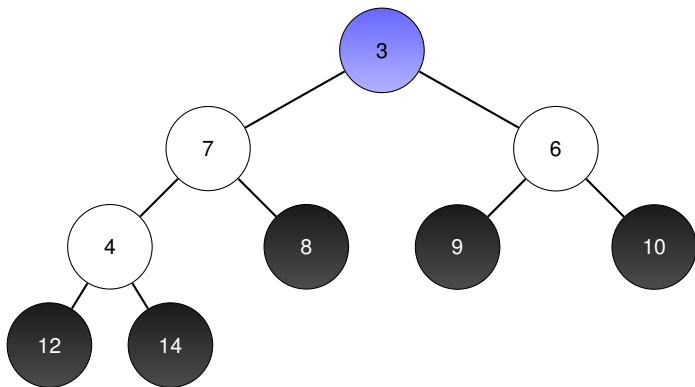
Scambio primo e ultimo elemento:



3	7	6	4	8	9	10	12	14
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Heapsort

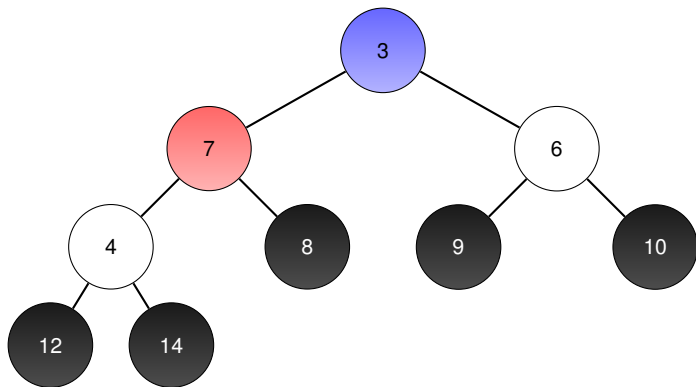
Ricostruzione Heap:



3	7	6	4	8	9	10	12	14
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Heapsort

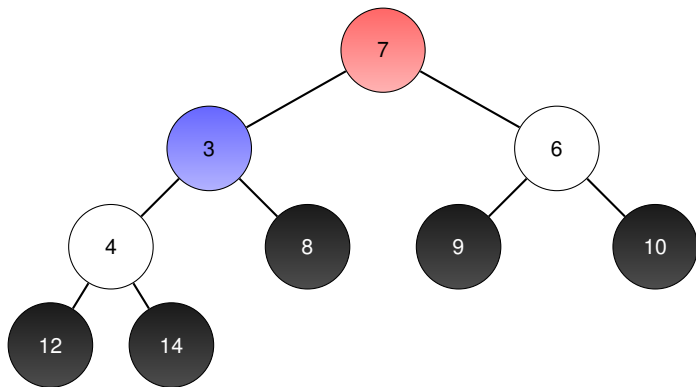
Ricostruzione Heap:



3	7	6	4	8	9	10	12	14
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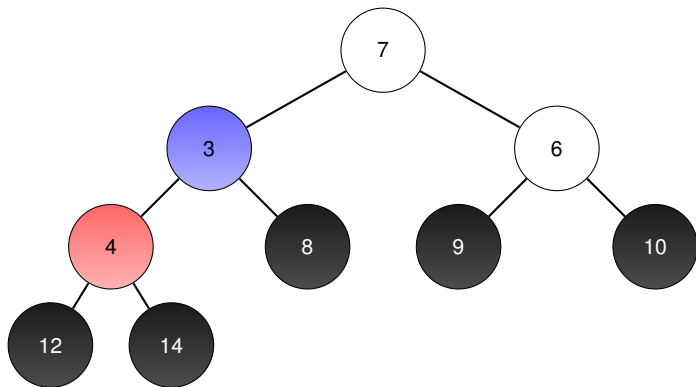
Heapsort

Ricostruzione Heap:



Heapsort

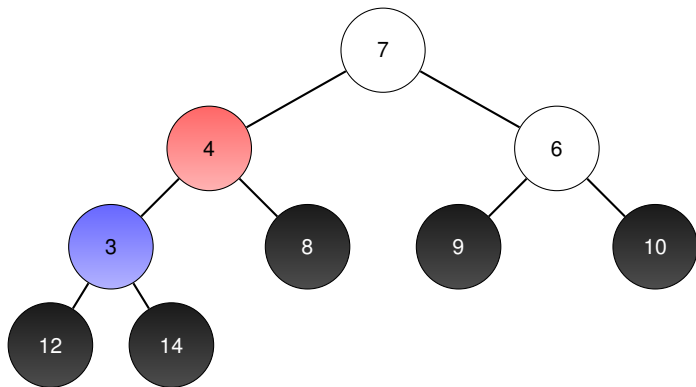
Ricostruzione Heap:



7	3	6	4	8	9	10	12	14
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Heapsort

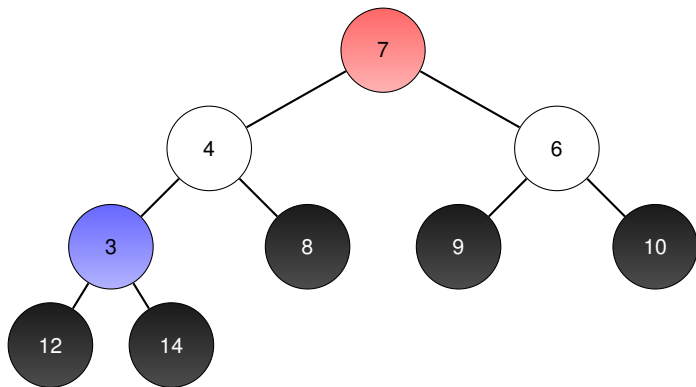
Ricostruzione Heap:



7	4	6	3	8	9	10	12	14
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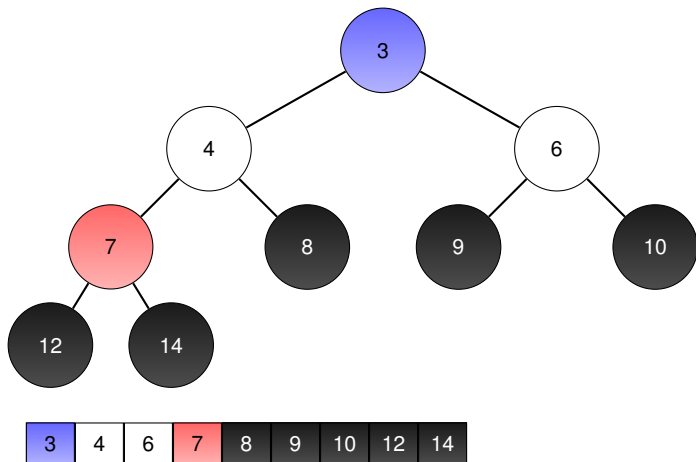
Heapsort

Scambio primo e ultimo elemento:



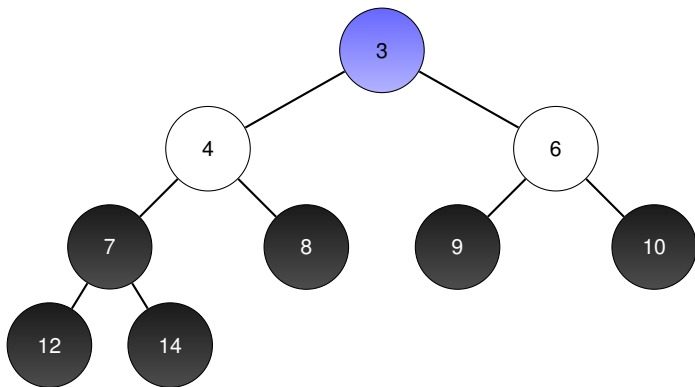
Heapsort

Scambio primo e ultimo elemento:



Heapsort

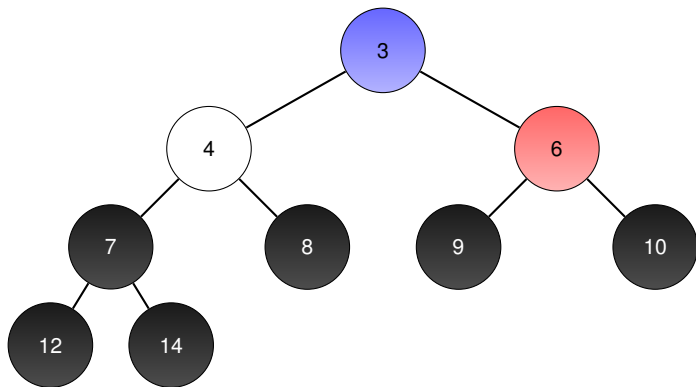
Ricostruzione Heap:



3	4	6	7	8	9	10	12	14
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Heapsort

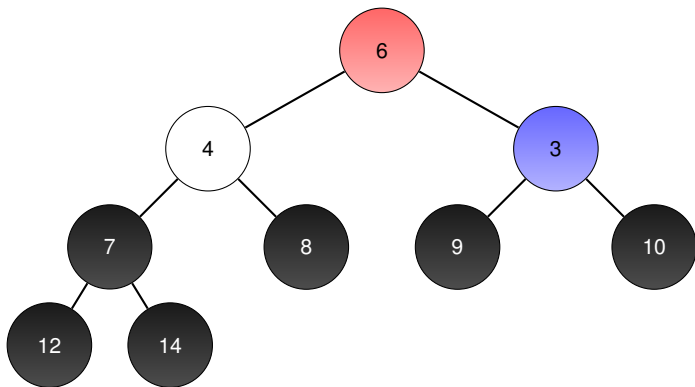
Ricostruzione Heap:



3	4	6	7	8	9	10	12	14
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Heapsort

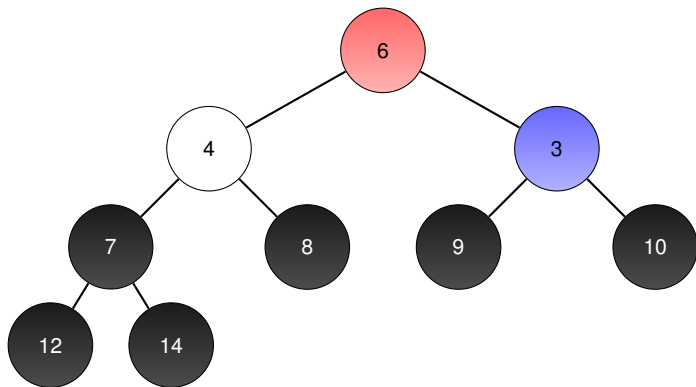
Ricostruzione Heap:



6	4	3	7	8	9	10	12	14
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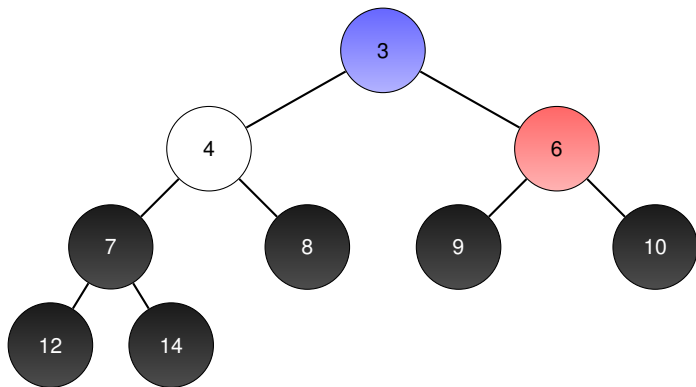
Heapsort

Scambio primo e ultimo elemento:



Heapsort

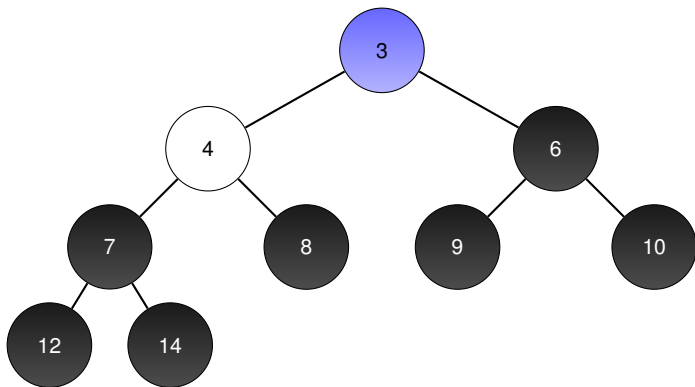
Scambio primo e ultimo elemento:



3	4	6	7	8	9	10	12	14
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Heapsort

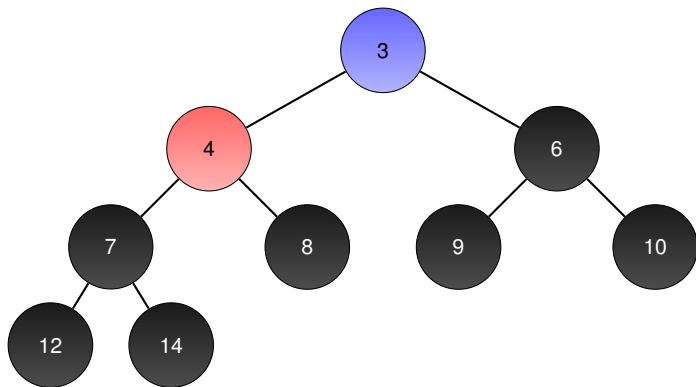
Ricostruzione Heap:



3	4	6	7	8	9	10	12	14
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Heapsort

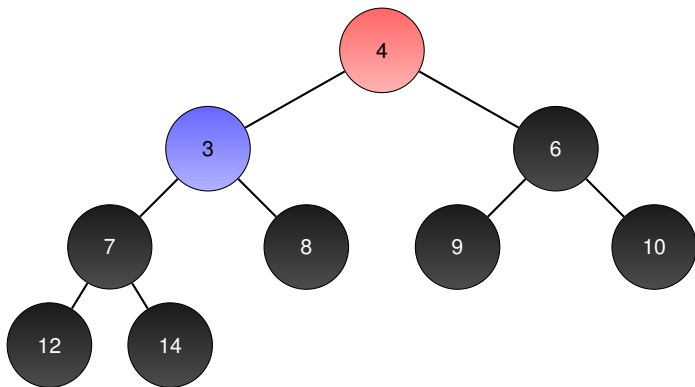
Ricostruzione Heap:



3	4	6	7	8	9	10	12	14
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Heapsort

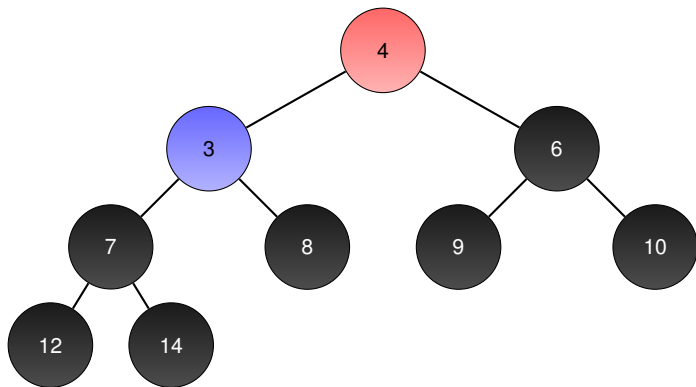
Ricostruzione Heap:



4	3	6	7	8	9	10	12	14
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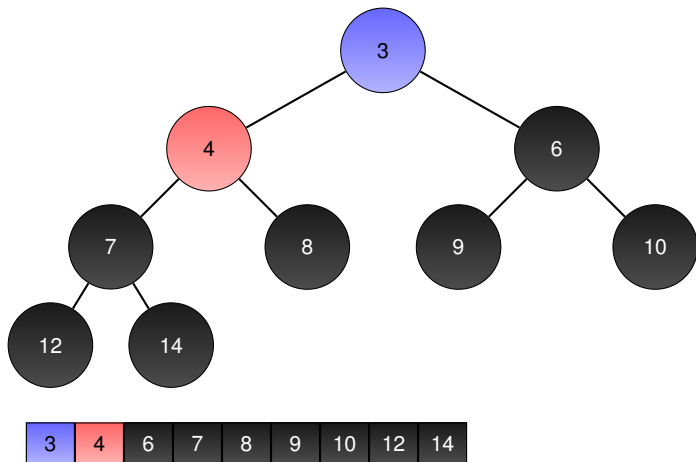
Heapsort

Scambio primo e ultimo elemento:



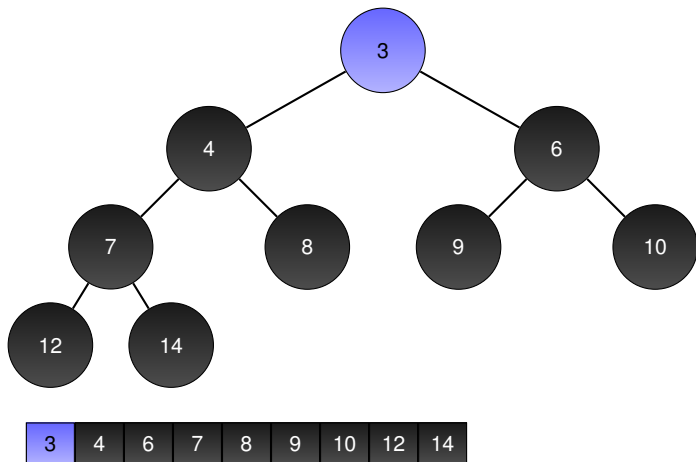
Heapsort

Scambio primo e ultimo elemento:



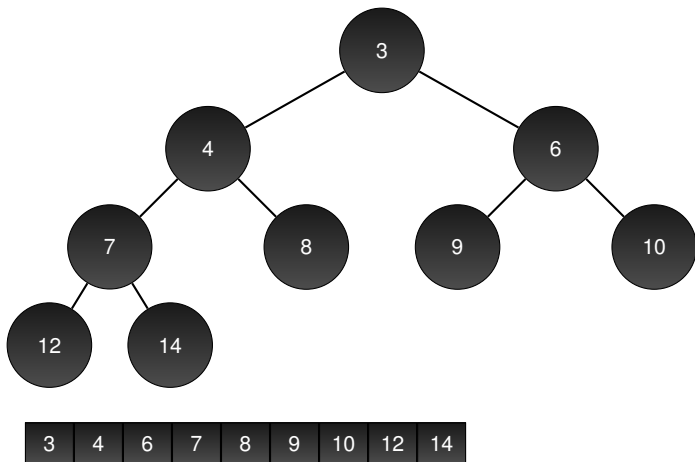
Heapsort

Ricostruzione Heap: 1 solo elemento $\implies$ già heap

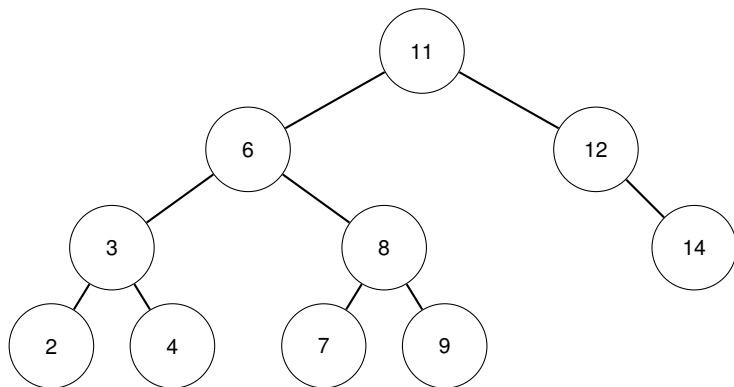


Heapsort

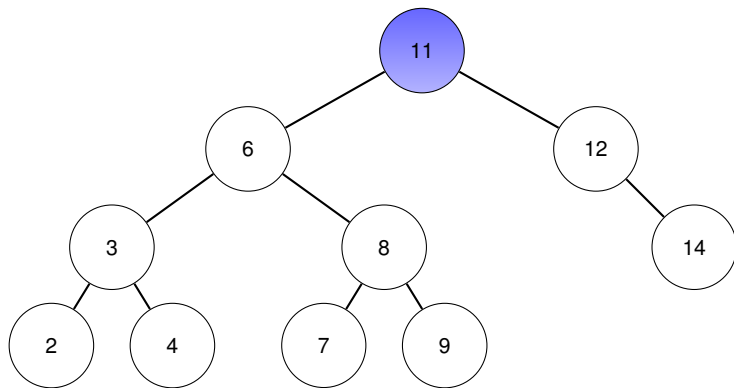
Elementi esauriti $\Rightarrow$ array ordinato:



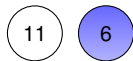
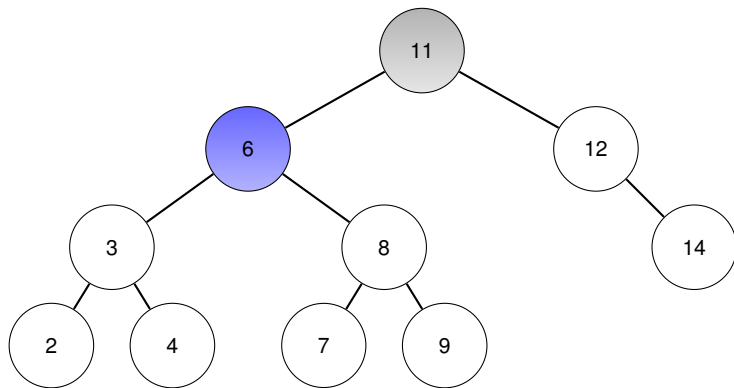
Effettuare le visite pre-order, in-order e post-order del seguente albero:



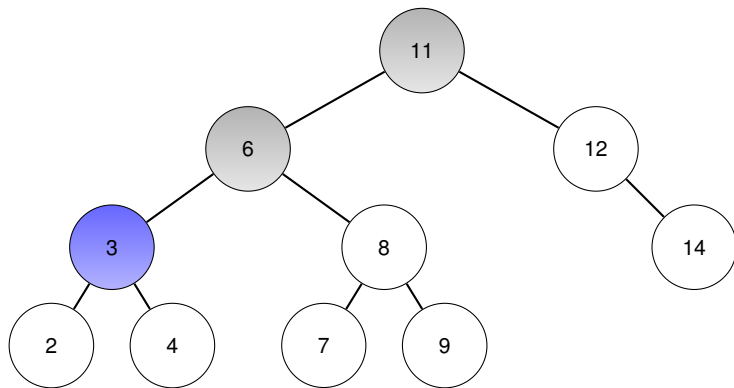
Visite di alberi — pre-order



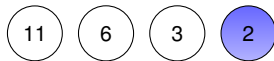
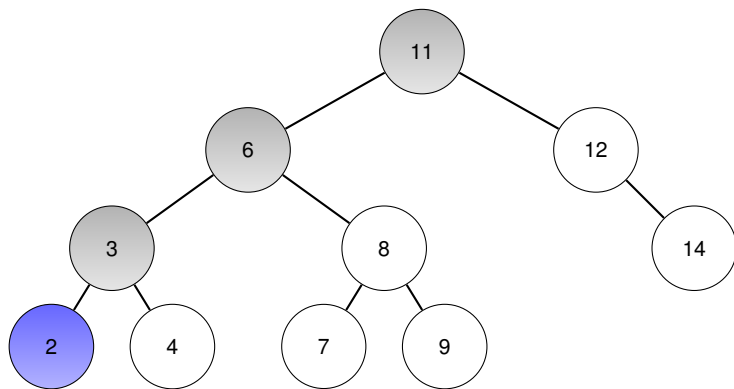
Visite di alberi — pre-order



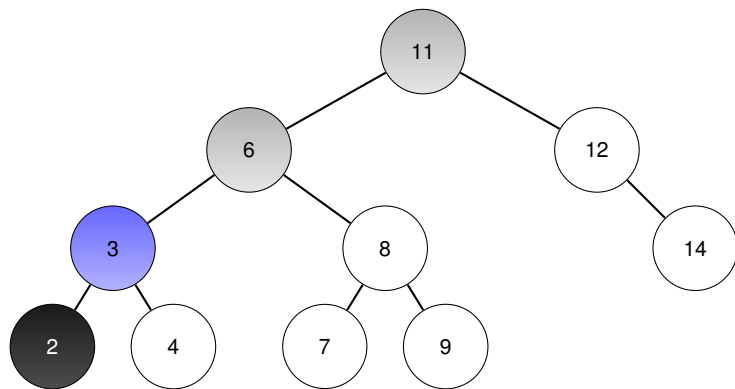
Visite di alberi — pre-order



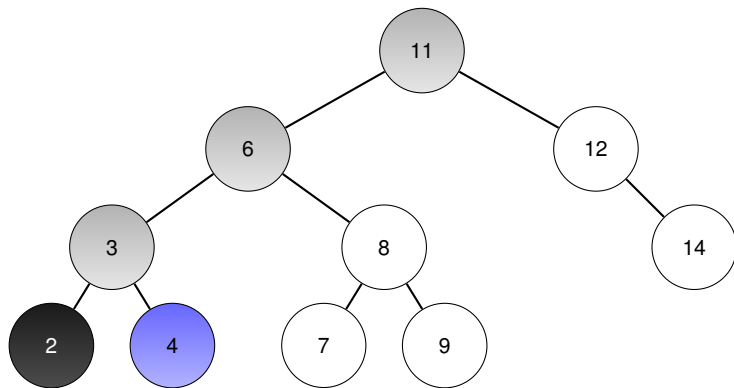
Visite di alberi — pre-order



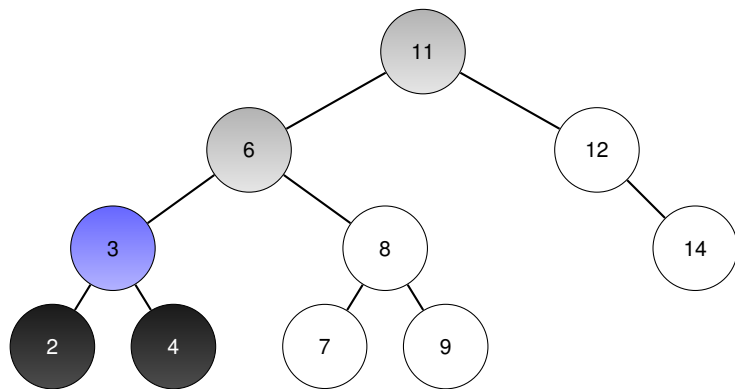
Visite di alberi — pre-order



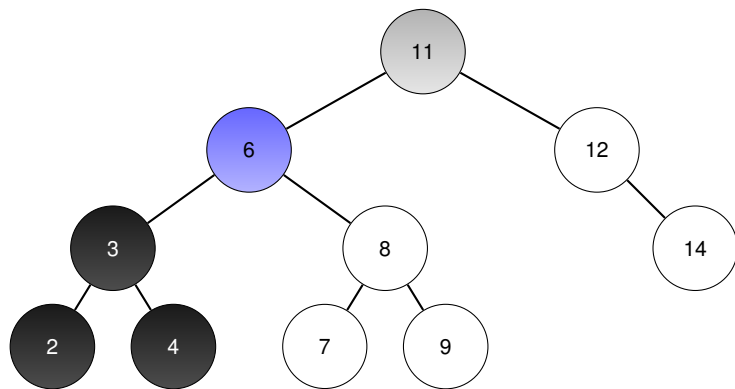
Visite di alberi — pre-order



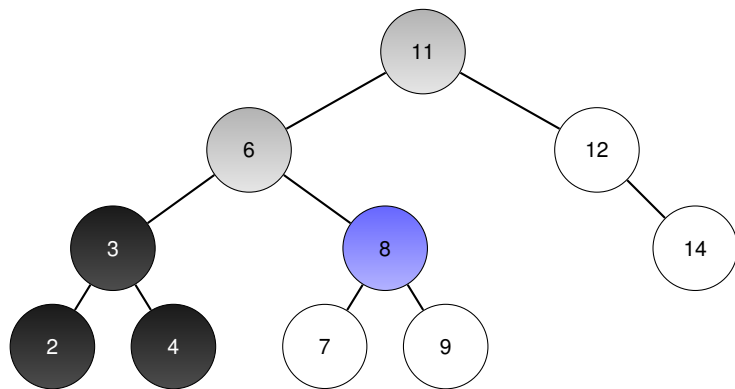
Visite di alberi — pre-order



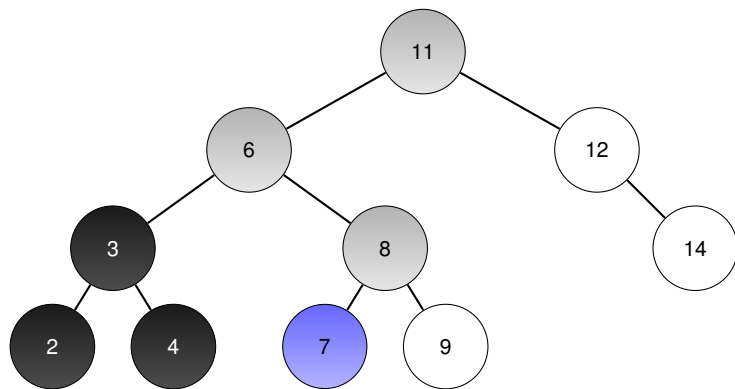
Visite di alberi — pre-order



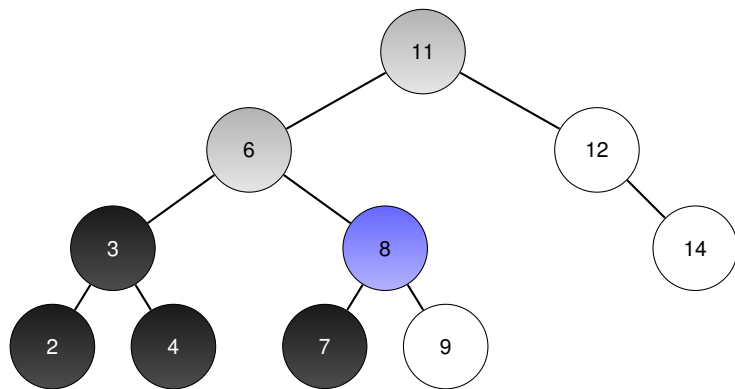
Visite di alberi — pre-order



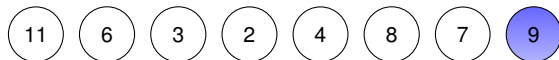
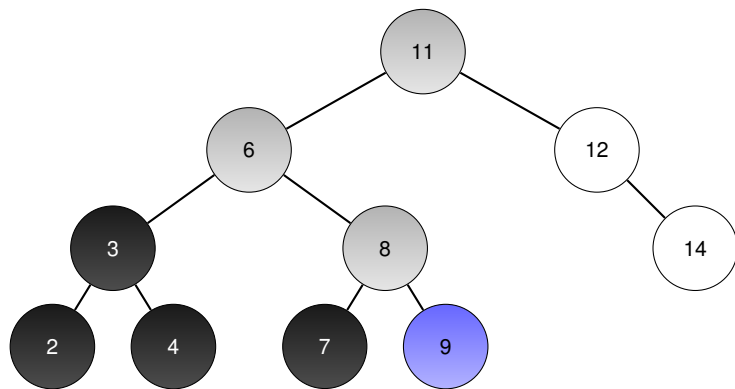
Visite di alberi — pre-order



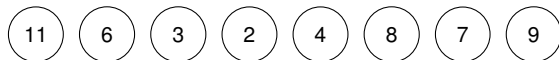
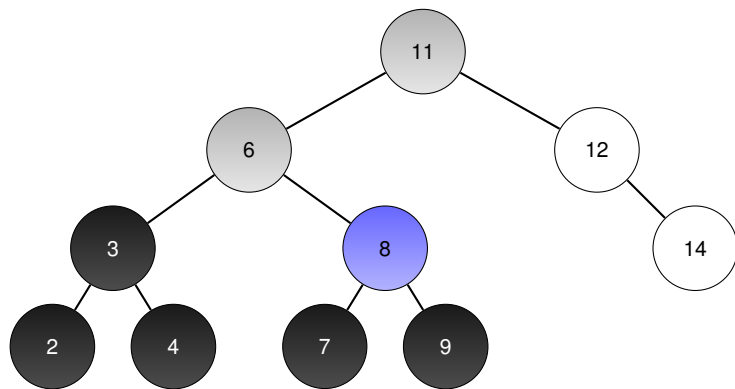
Visite di alberi — pre-order



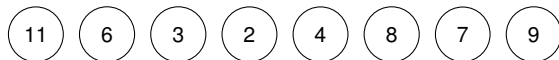
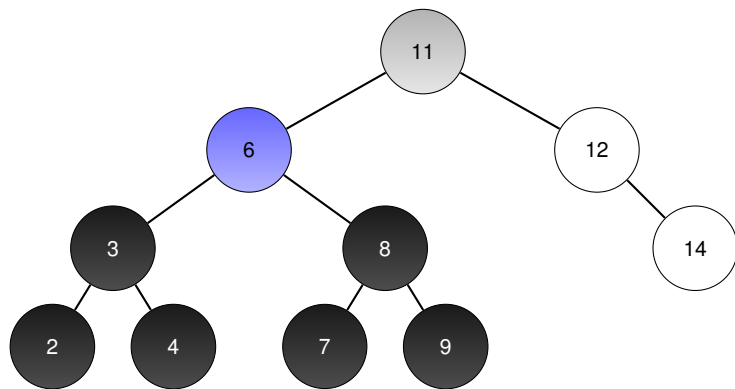
Visite di alberi — pre-order



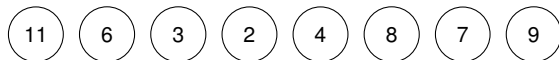
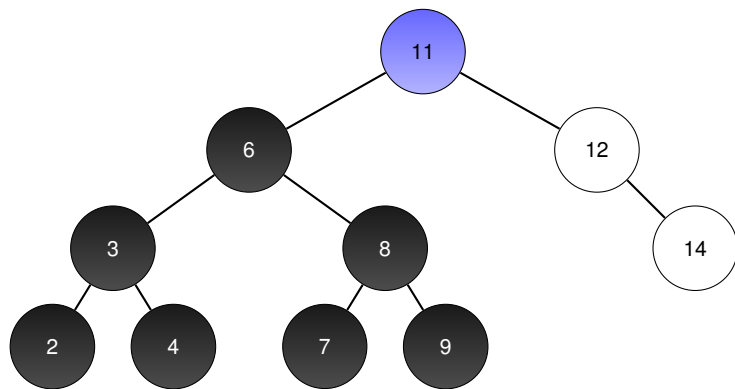
Visite di alberi — pre-order



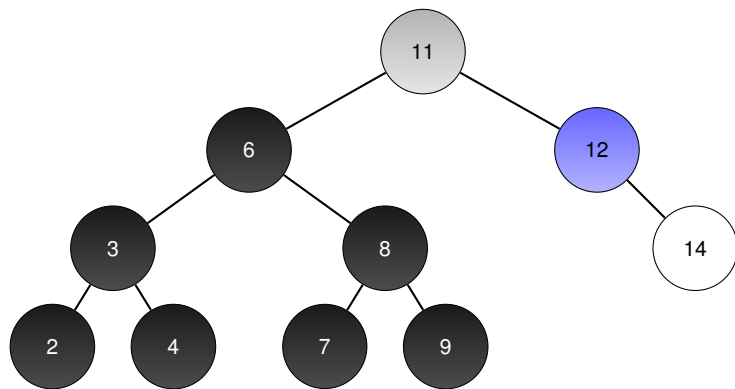
Visite di alberi — pre-order



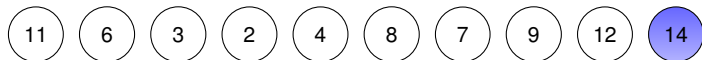
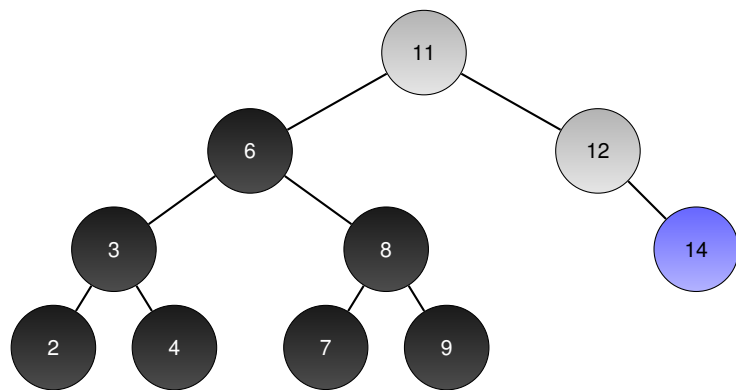
Visite di alberi — pre-order



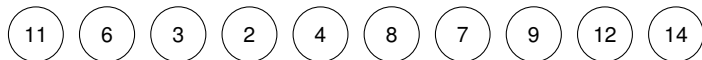
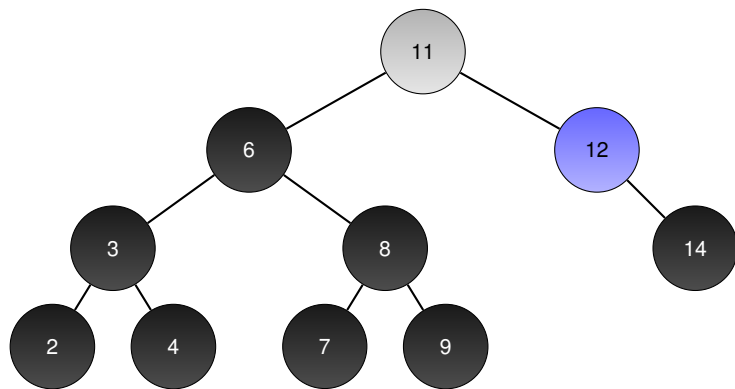
Visite di alberi — pre-order



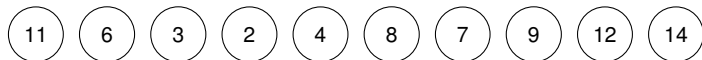
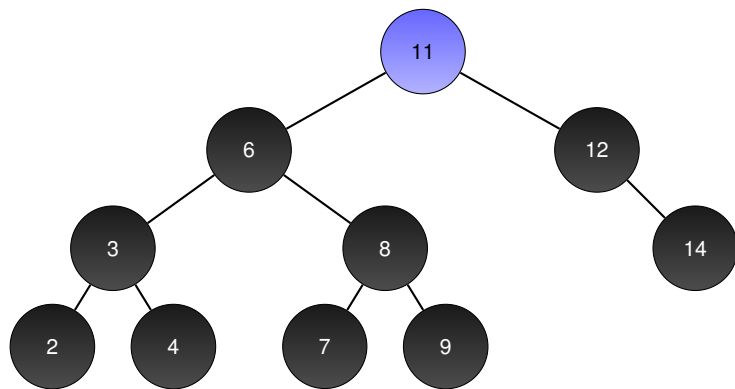
Visite di alberi — pre-order



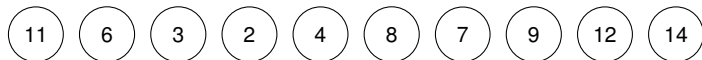
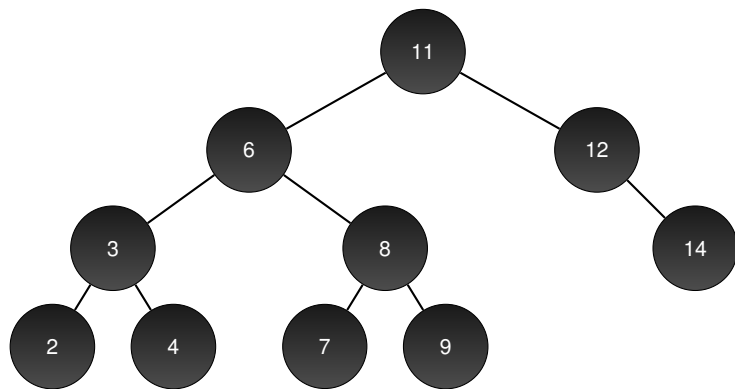
Visite di alberi — pre-order



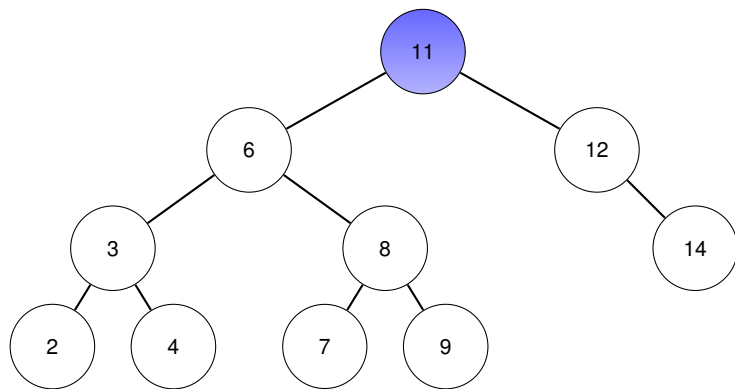
Visite di alberi — pre-order



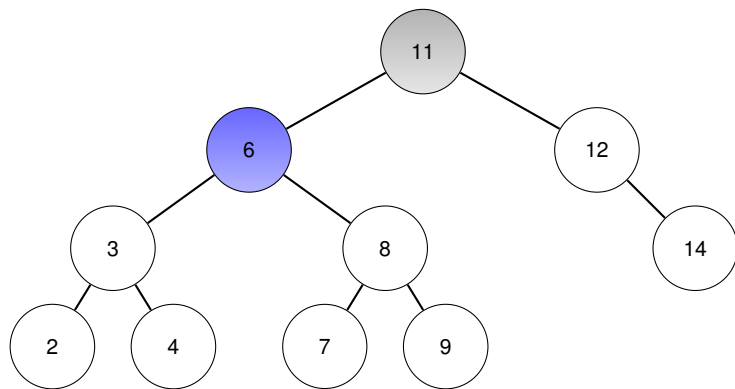
Visite di alberi — pre-order



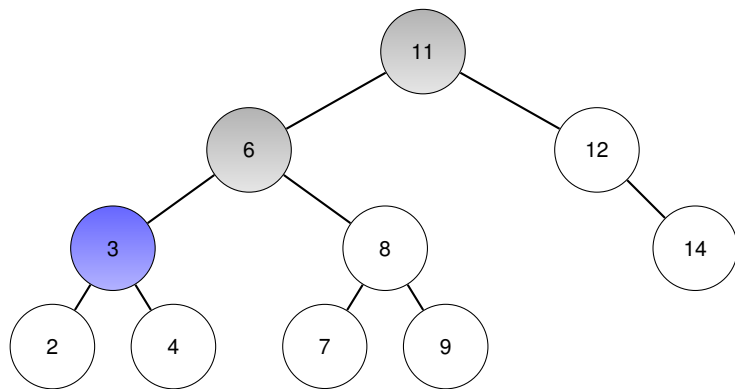
Visite di alberi — in-order



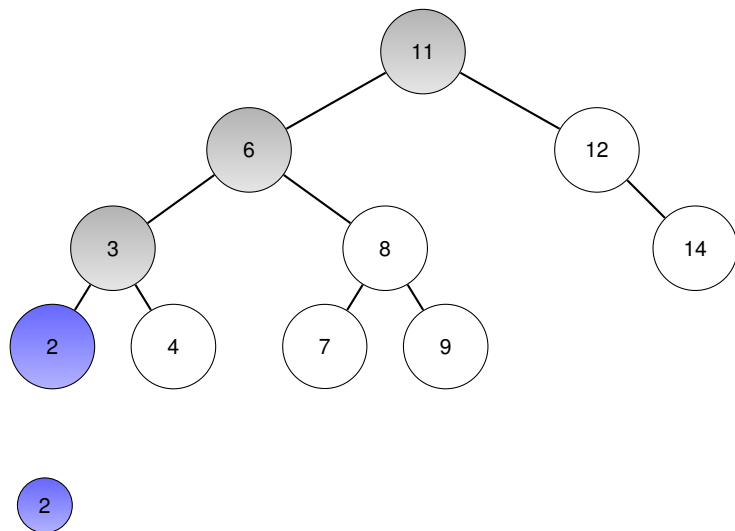
Visite di alberi — in-order



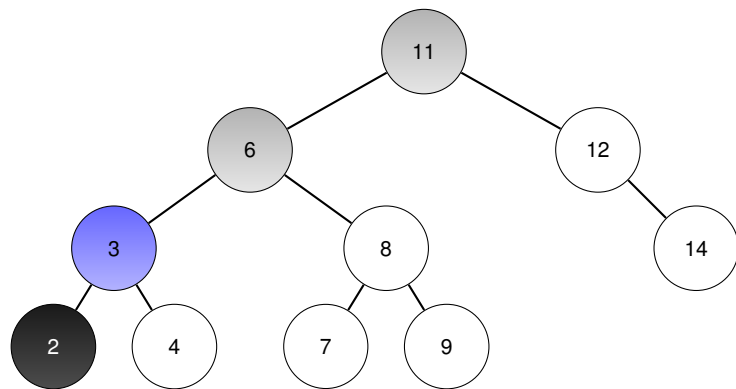
Visite di alberi — in-order



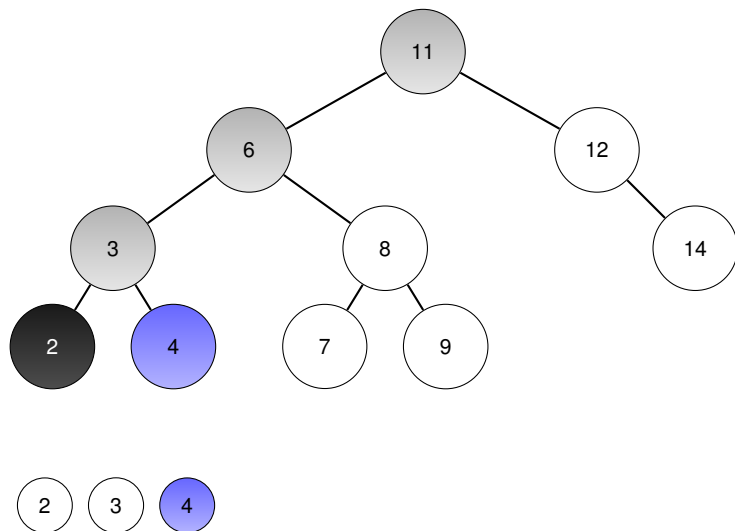
Visite di alberi — in-order



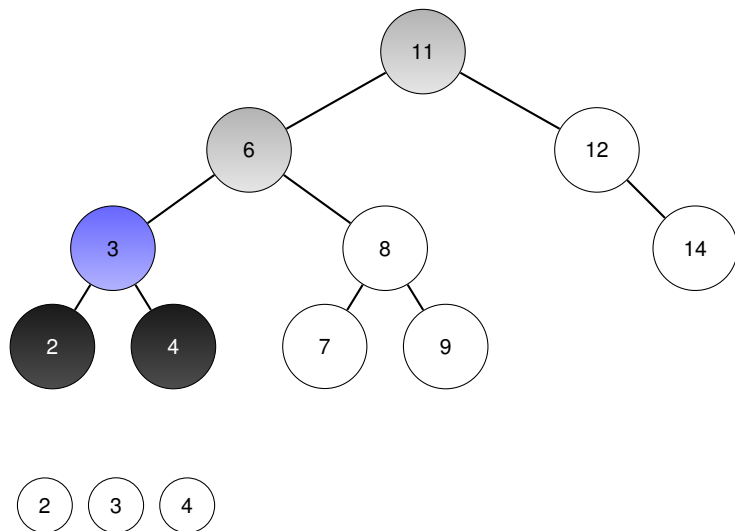
Visite di alberi — in-order



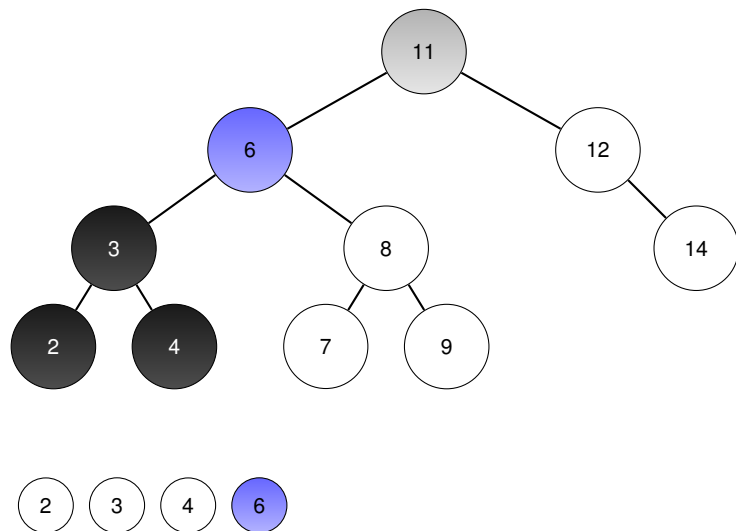
Visite di alberi — in-order



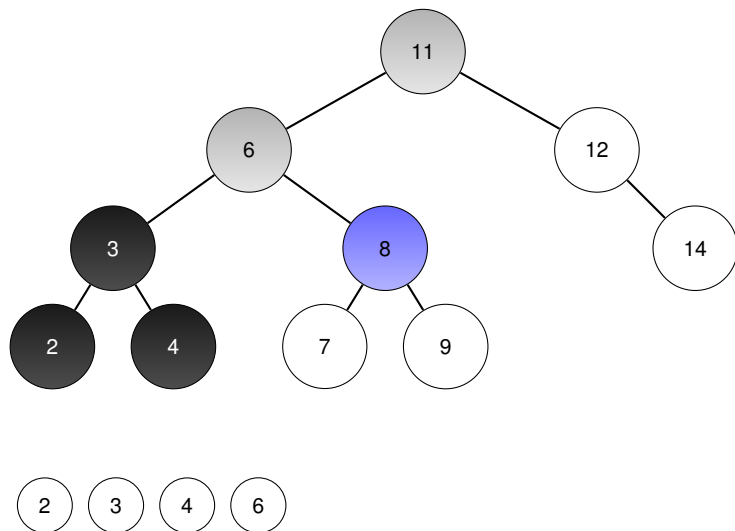
Visite di alberi — in-order



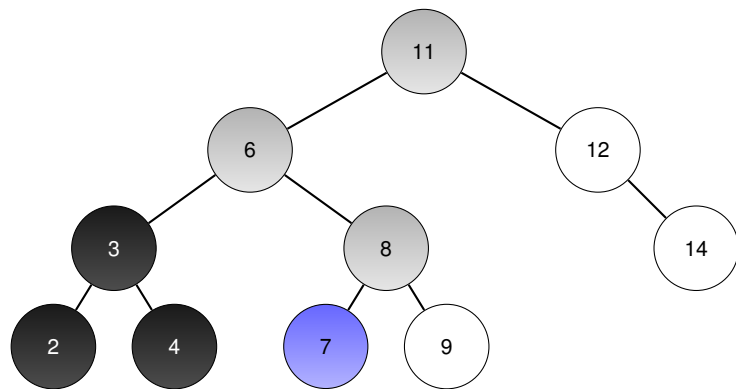
Visite di alberi — in-order



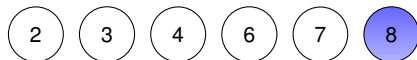
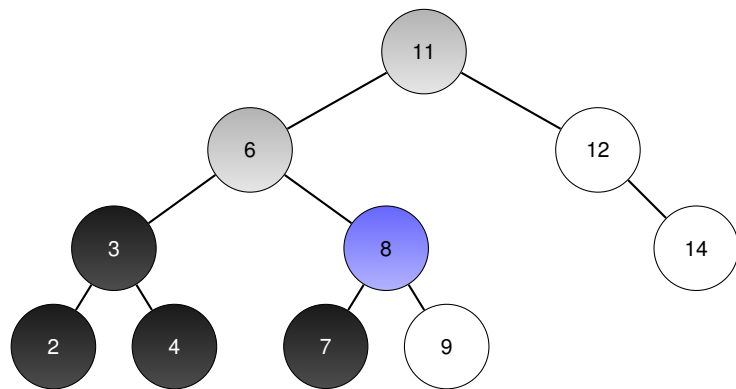
Visite di alberi — in-order



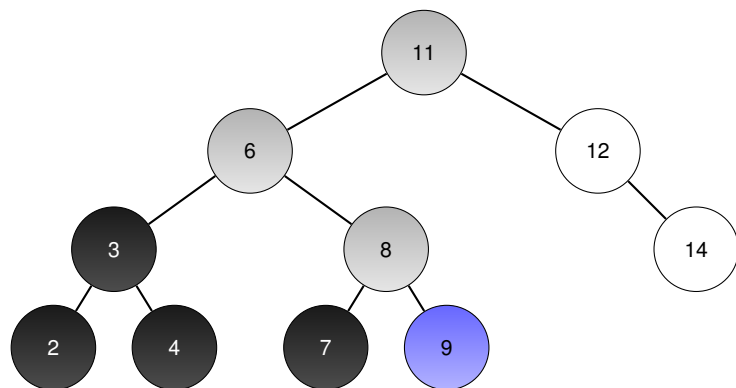
Visite di alberi — in-order



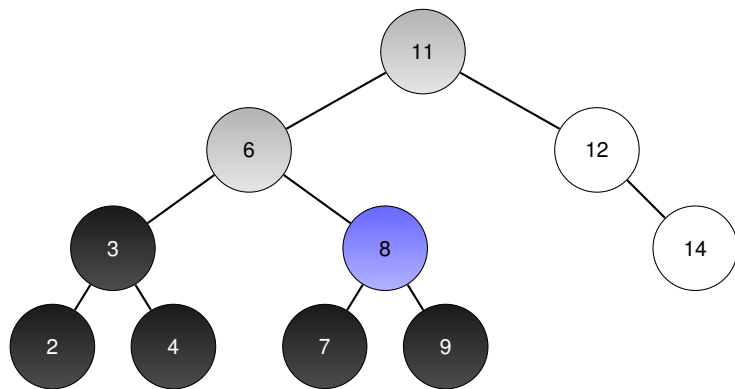
Visite di alberi — in-order



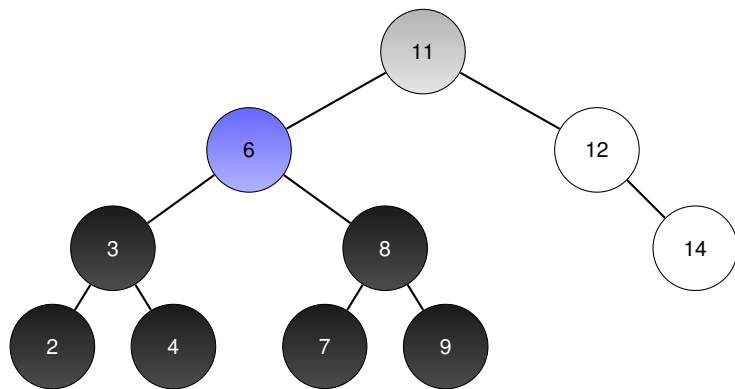
Visite di alberi — in-order



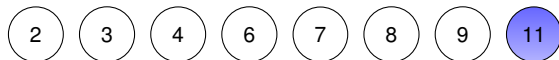
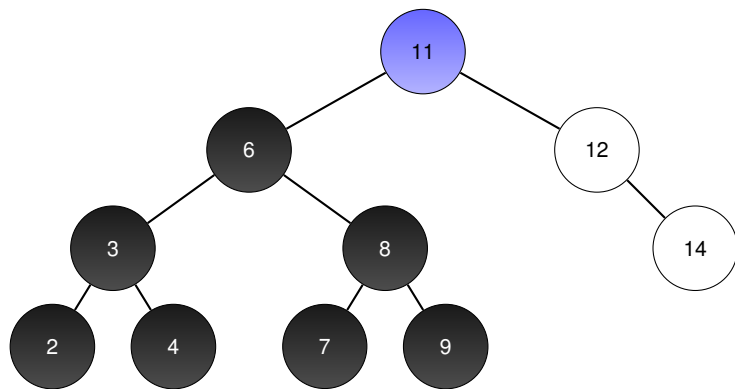
Visite di alberi — in-order



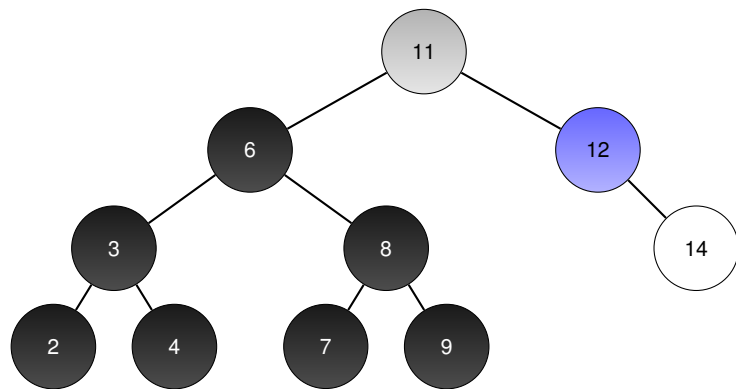
Visite di alberi — in-order



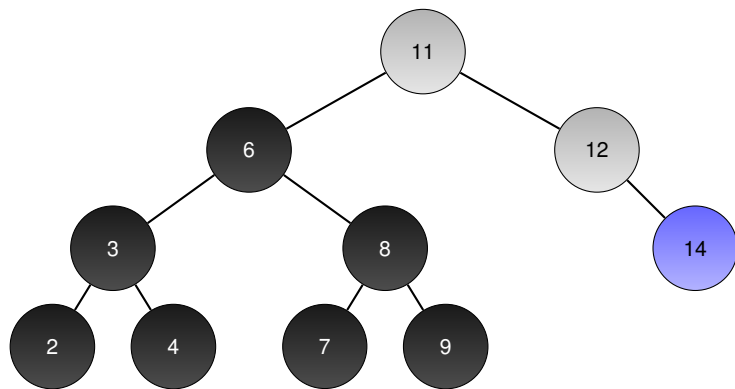
Visite di alberi — in-order



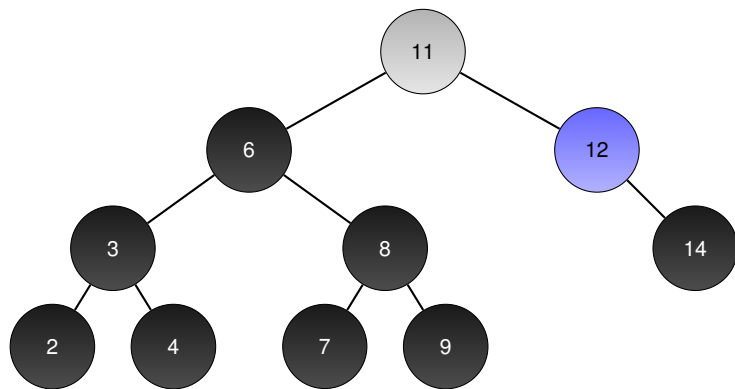
Visite di alberi — in-order



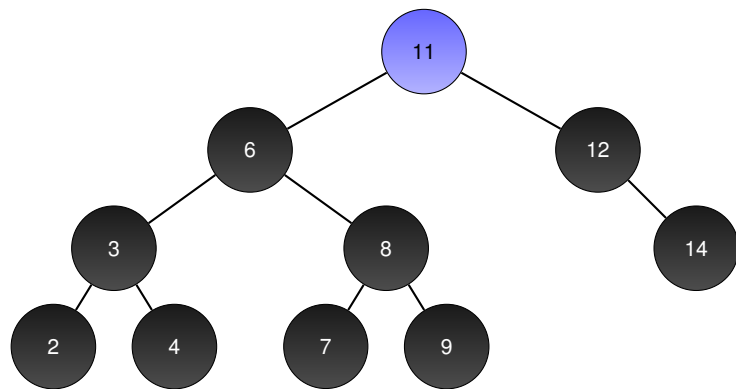
Visite di alberi — in-order



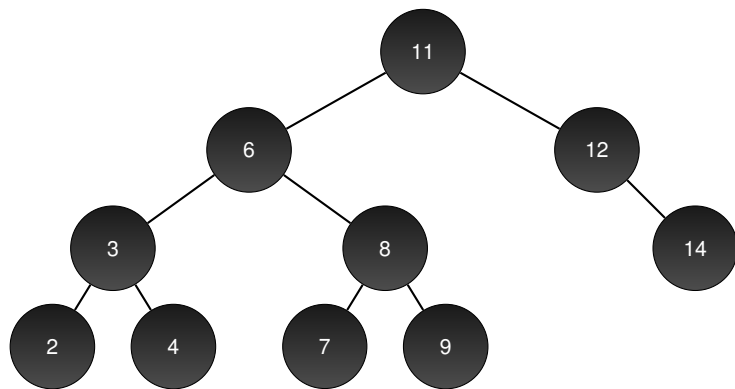
Visite di alberi — in-order



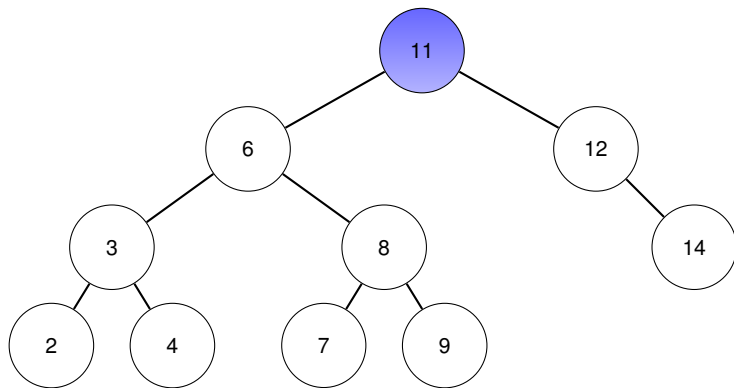
Visite di alberi — in-order



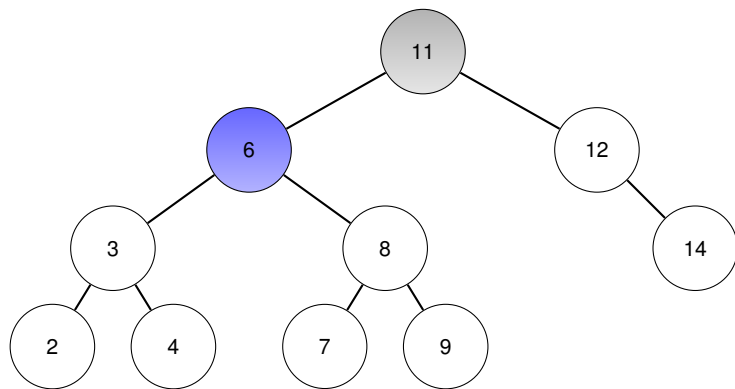
Visite di alberi — in-order



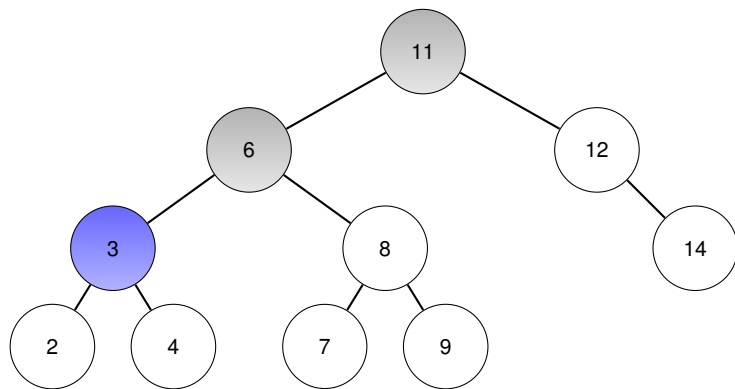
Visite di alberi — post-order



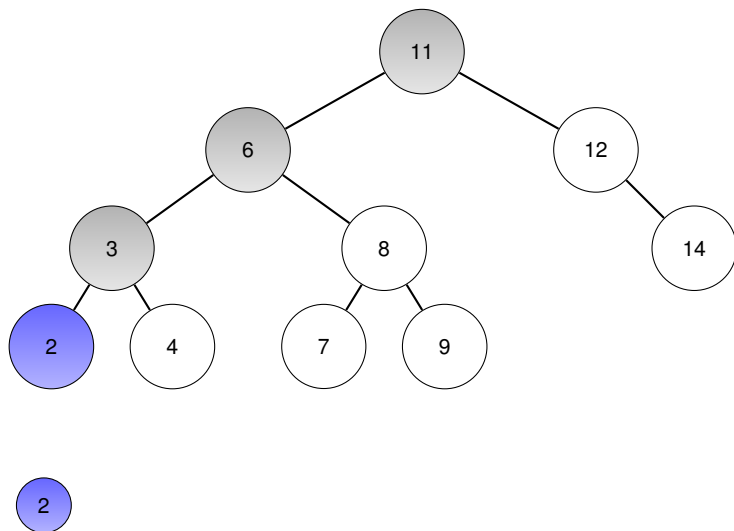
Visite di alberi — post-order



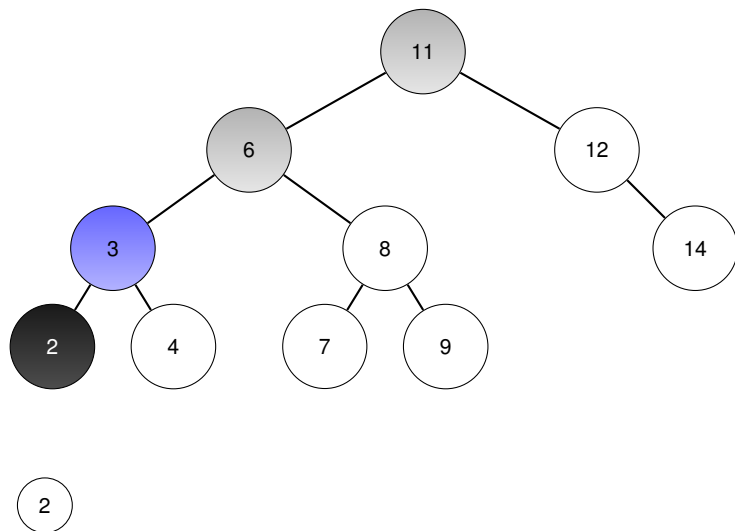
Visite di alberi — post-order



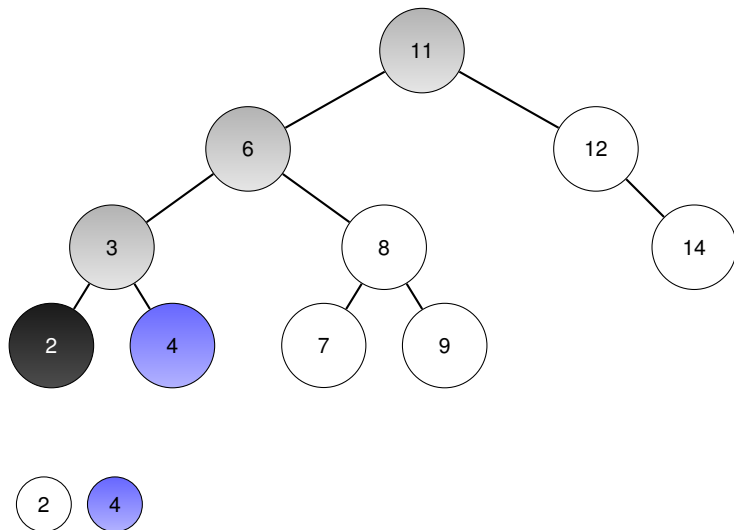
Visite di alberi — post-order



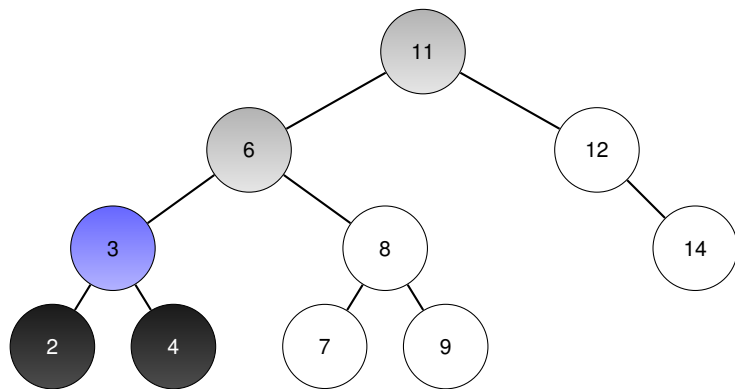
Visite di alberi — post-order



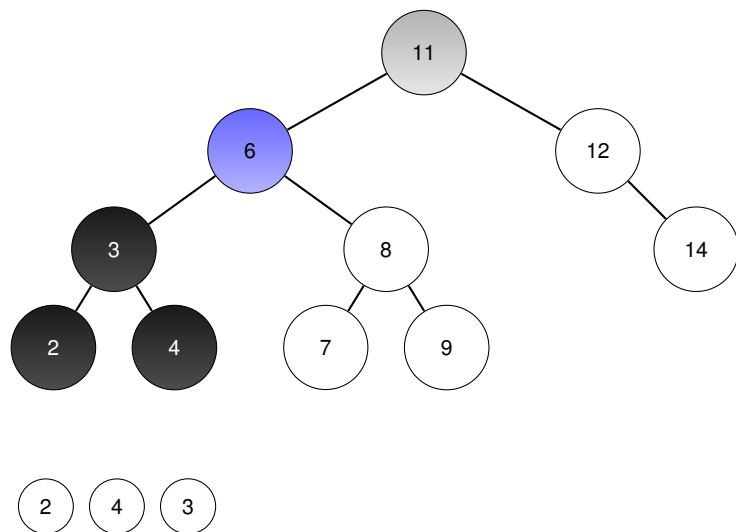
Visite di alberi — post-order



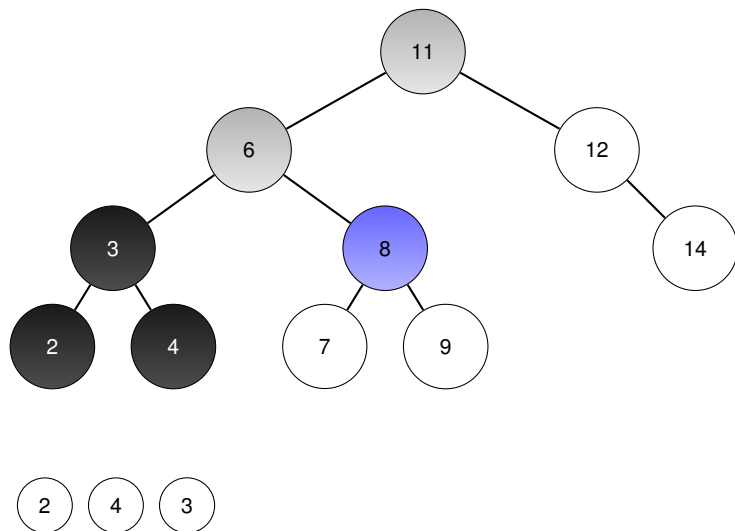
Visite di alberi — post-order



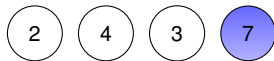
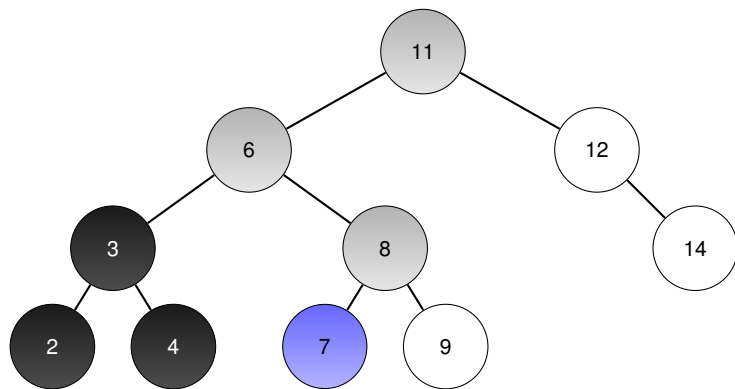
Visite di alberi — post-order



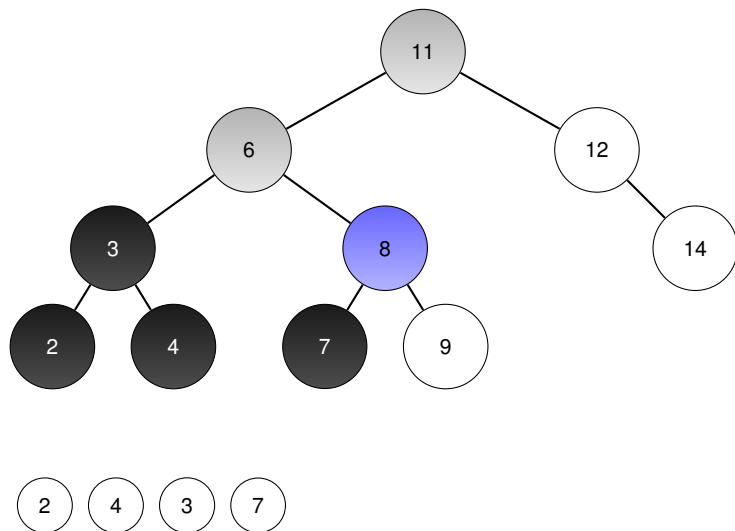
Visite di alberi — post-order



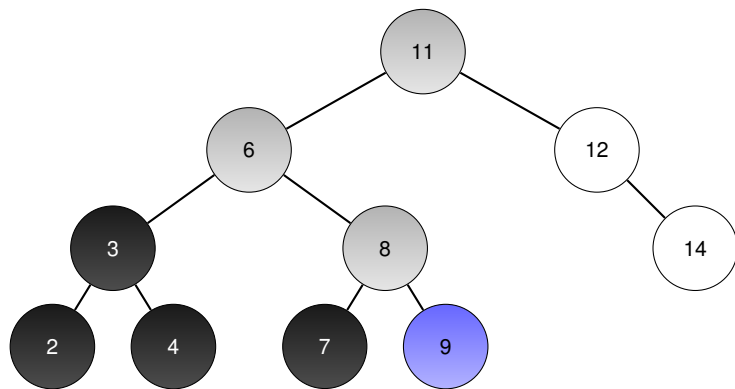
Visite di alberi — post-order



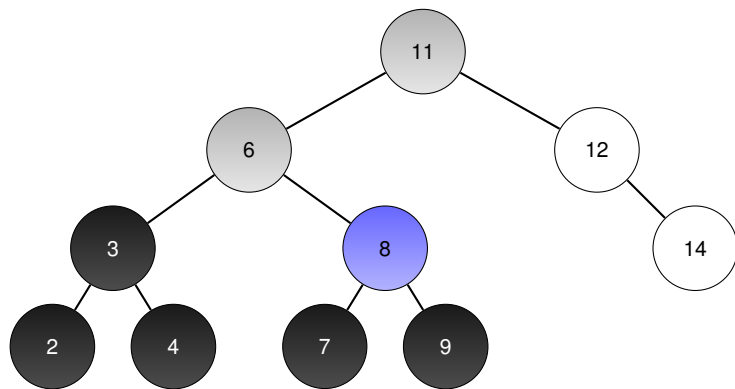
Visite di alberi — post-order



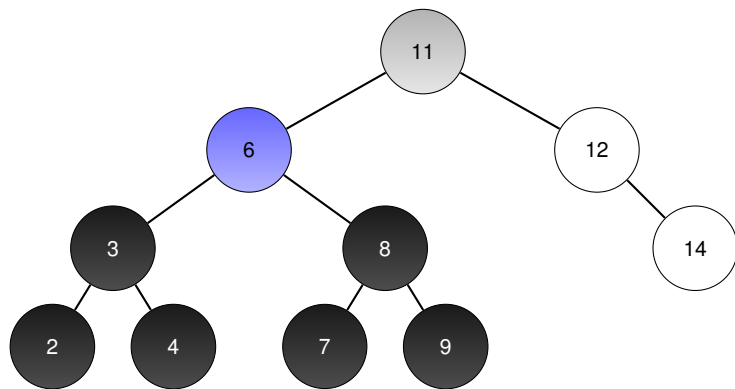
Visite di alberi — post-order



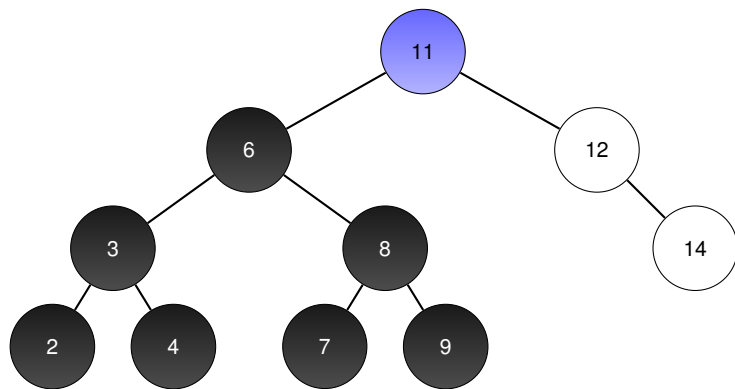
Visite di alberi — post-order



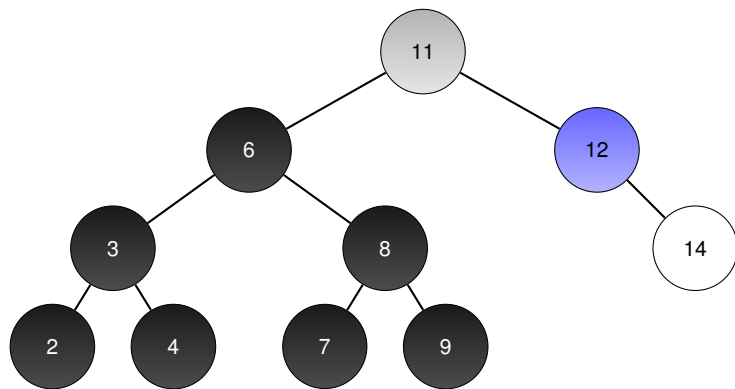
Visite di alberi — post-order



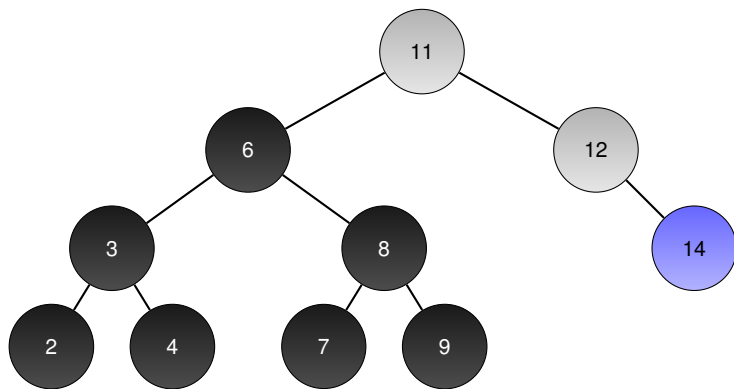
Visite di alberi — post-order



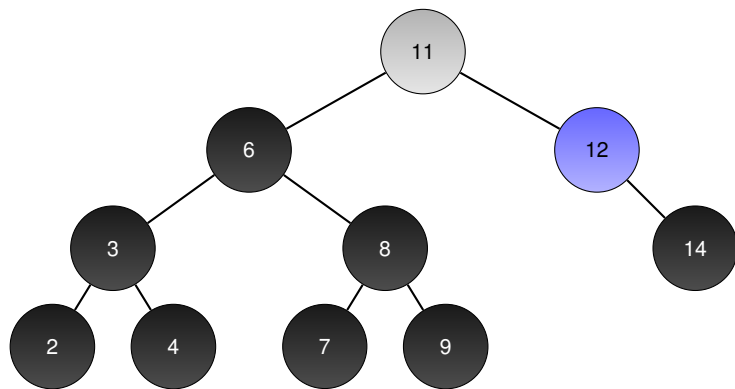
Visite di alberi — post-order



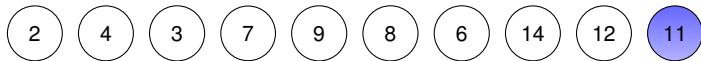
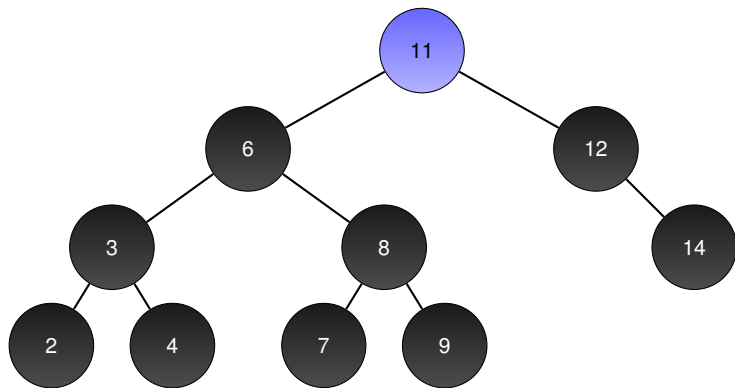
Visite di alberi — post-order



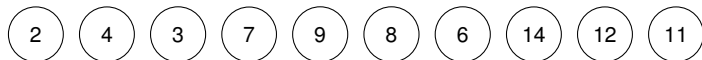
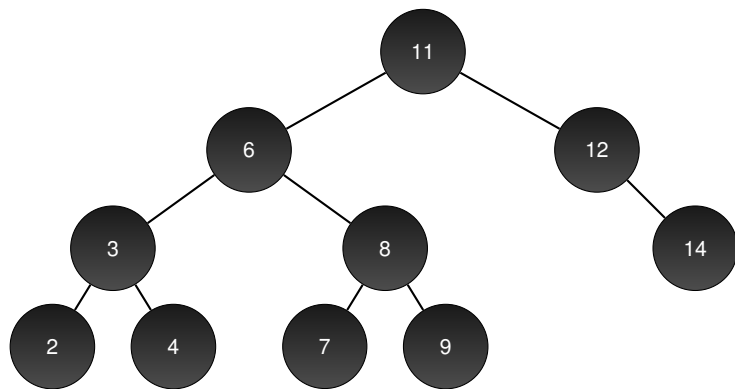
Visite di alberi — post-order



Visite di alberi — post-order



Visite di alberi — post-order



Visite di alberi - ricostruzione dell'albero

Ricostruzione albero da visite pre-order e in-order

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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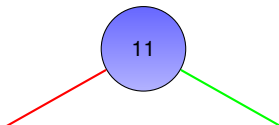
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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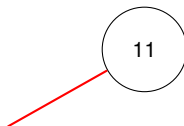
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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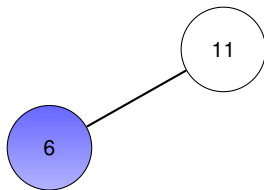
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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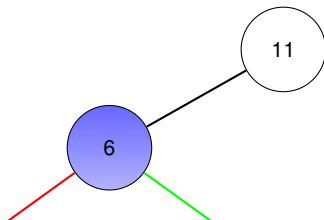
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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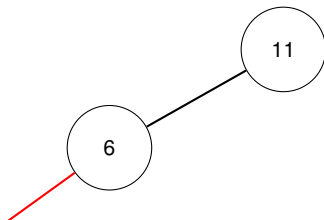
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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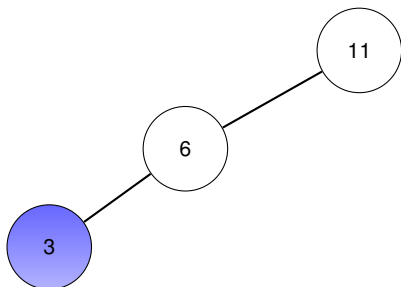
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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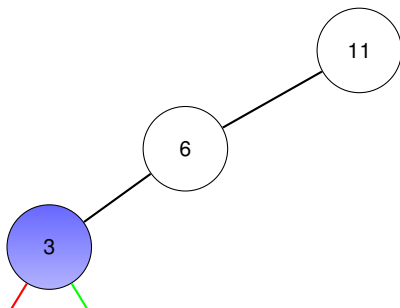
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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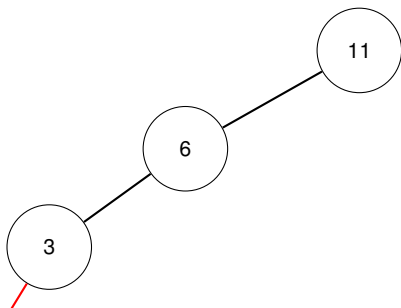
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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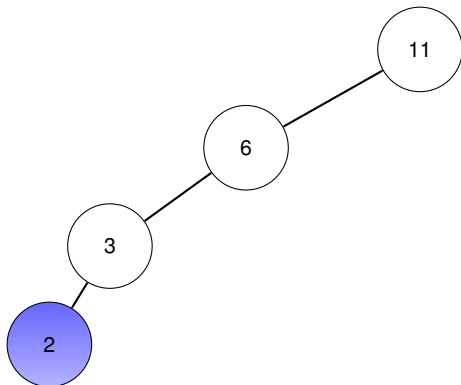
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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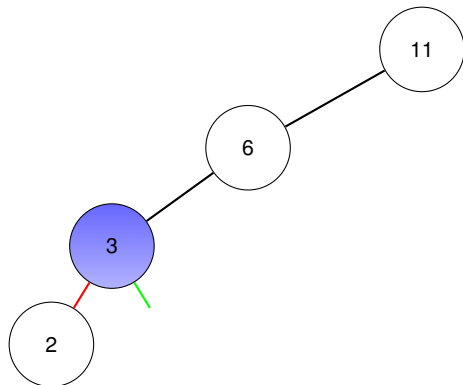
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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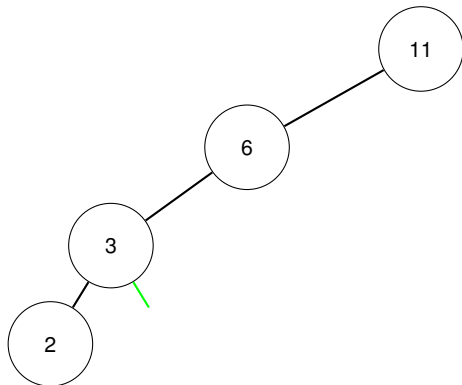
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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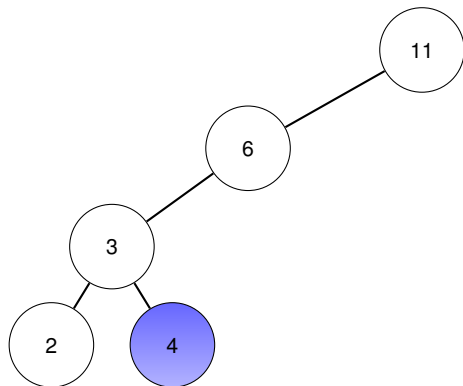
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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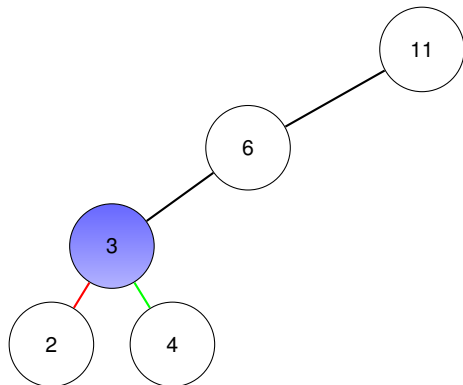
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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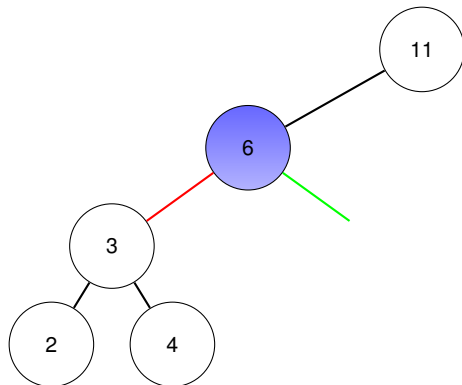
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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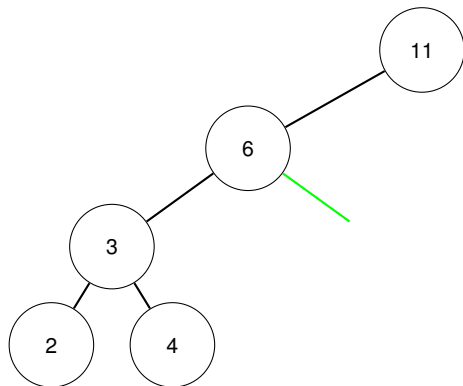
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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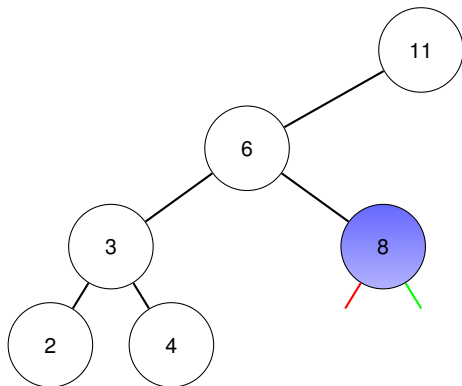
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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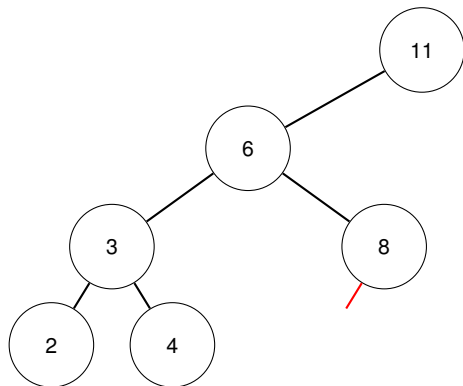
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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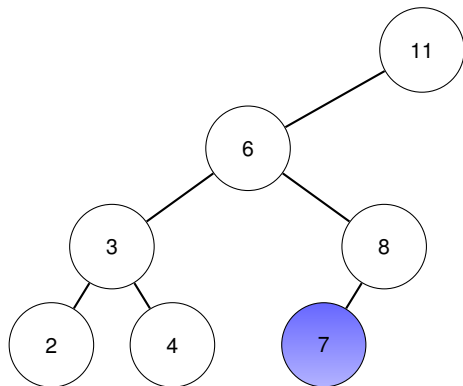
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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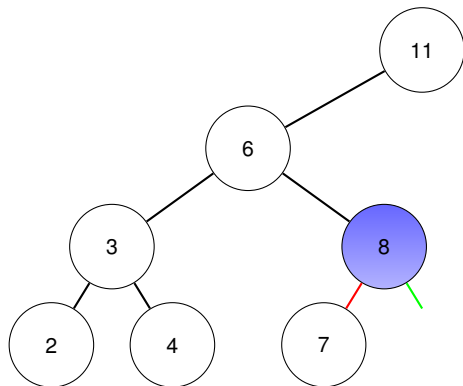
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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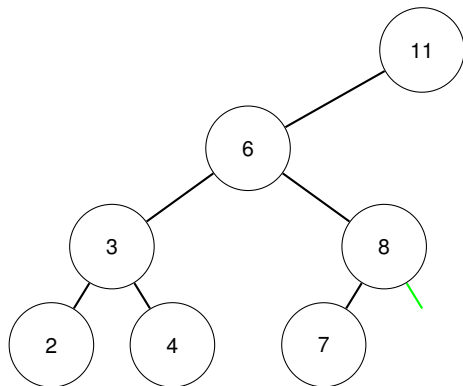
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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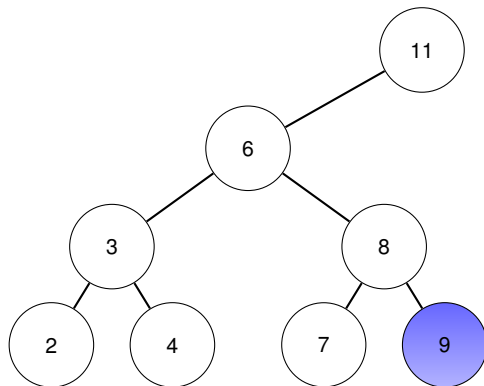
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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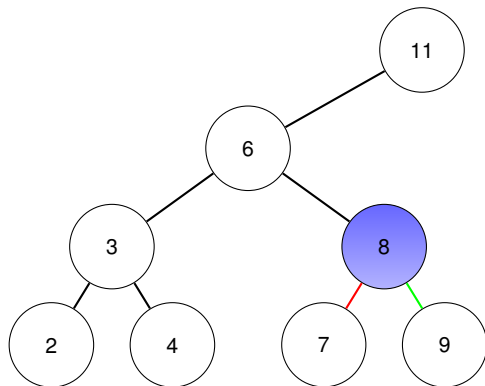
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
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In-order:

2	3	4	6	7	8	9	11	12	14
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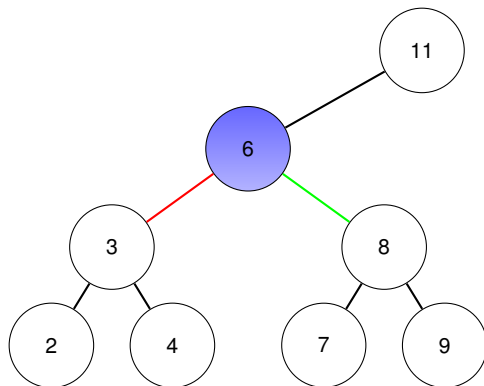
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
----	---	---	---	---	---	---	---	----	----

In-order:

2	3	4	6	7	8	9	11	12	14
---	---	---	---	---	---	---	----	----	----



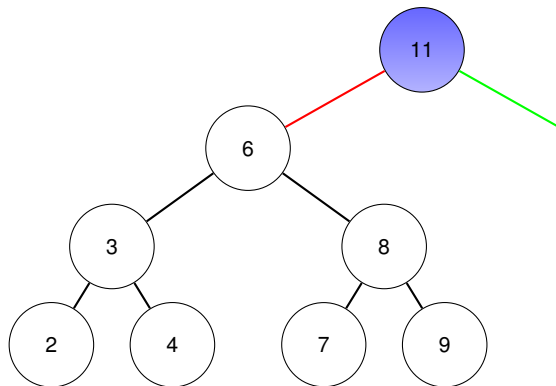
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
----	---	---	---	---	---	---	---	----	----

In-order:

2	3	4	6	7	8	9	11	12	14
---	---	---	---	---	---	---	----	----	----



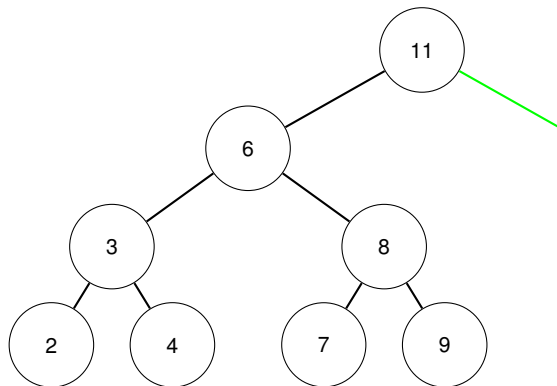
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
----	---	---	---	---	---	---	---	----	----

In-order:

2	3	4	6	7	8	9	11	12	14
---	---	---	---	---	---	---	----	----	----



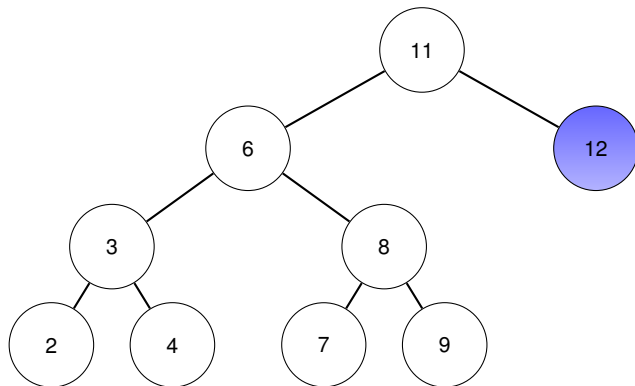
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
----	---	---	---	---	---	---	---	----	----

In-order:

2	3	4	6	7	8	9	11	12	14
---	---	---	---	---	---	---	----	----	----



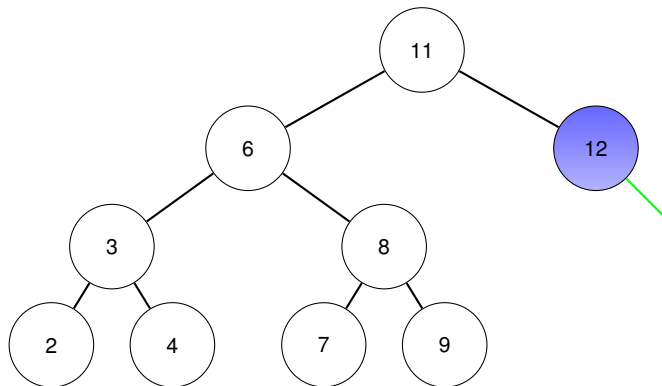
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
----	---	---	---	---	---	---	---	----	----

In-order:

2	3	4	6	7	8	9	11	12	14
---	---	---	---	---	---	---	----	----	----



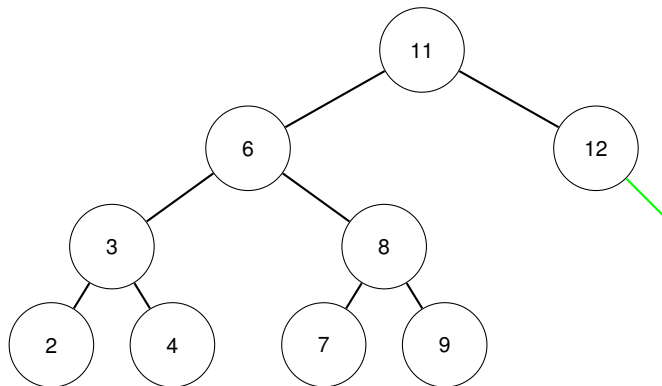
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
----	---	---	---	---	---	---	---	----	----

In-order:

2	3	4	6	7	8	9	11	12	14
---	---	---	---	---	---	---	----	----	----



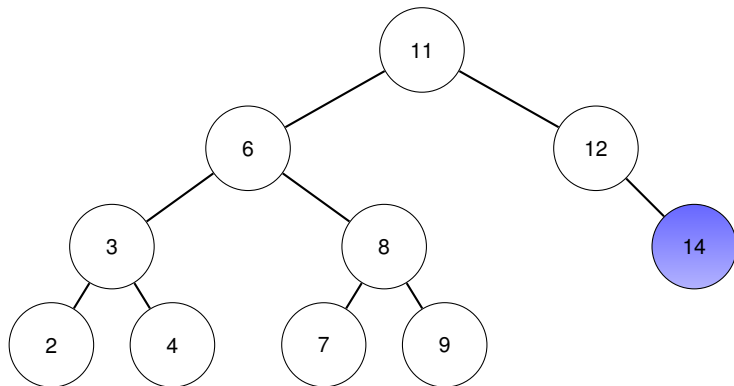
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
----	---	---	---	---	---	---	---	----	----

In-order:

2	3	4	6	7	8	9	11	12	14
---	---	---	---	---	---	---	----	----	----



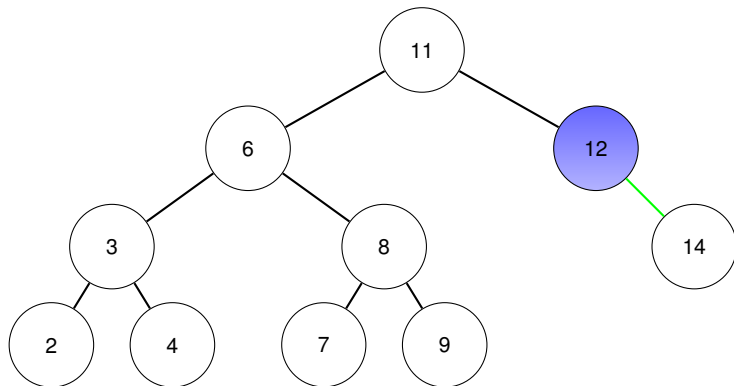
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
----	---	---	---	---	---	---	---	----	----

In-order:

2	3	4	6	7	8	9	11	12	14
---	---	---	---	---	---	---	----	----	----



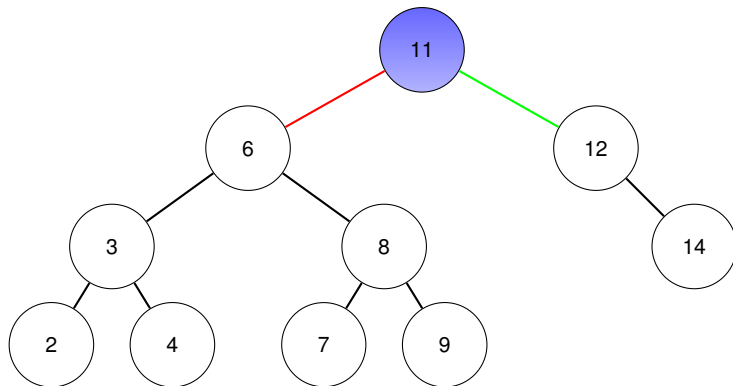
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
----	---	---	---	---	---	---	---	----	----

In-order:

2	3	4	6	7	8	9	11	12	14
---	---	---	---	---	---	---	----	----	----



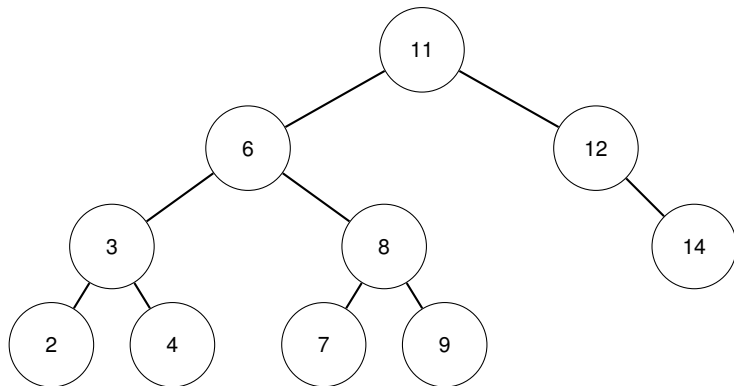
Visite di alberi - ricostruzione dell'albero

Pre-order:

11	6	3	2	4	8	7	9	12	14
----	---	---	---	---	---	---	---	----	----

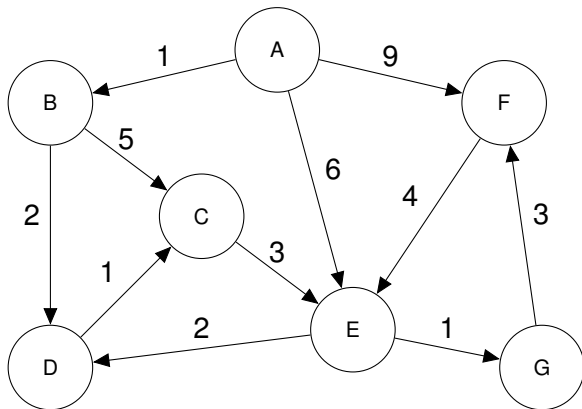
In-order:

2	3	4	6	7	8	9	11	12	14
---	---	---	---	---	---	---	----	----	----

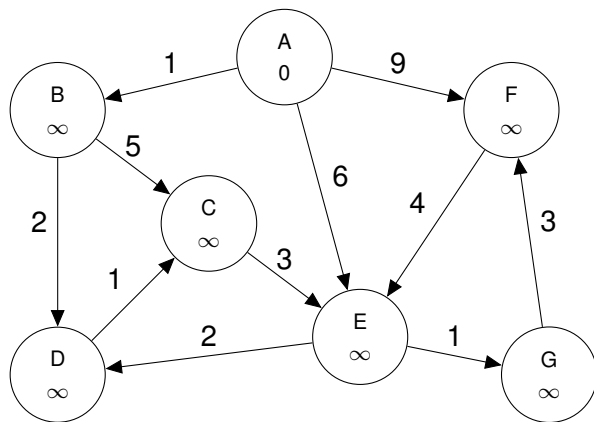


Camminimi Minimi - Algoritmo di Dijkstra

Calcolare i cammini minimi a partire dal nodo sorgente A

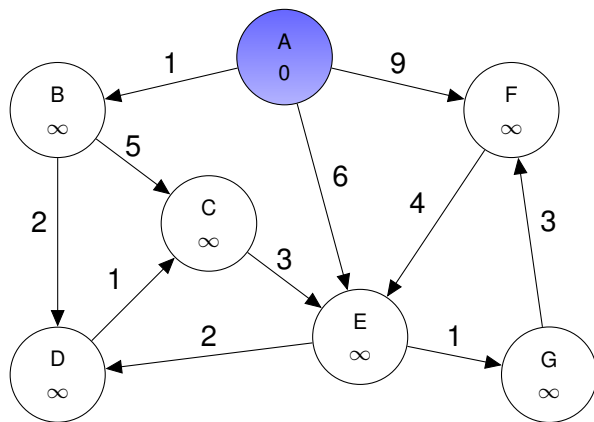


Cammini Minimi - Algoritmo di Dijkstra



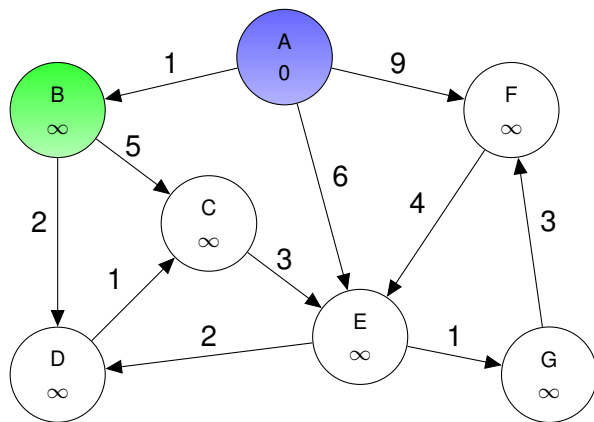
A: 0
B: ∞
C: ∞
D: ∞
E: ∞
F: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra



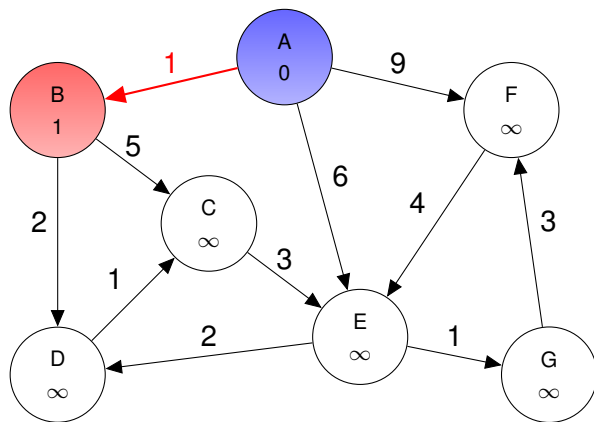
A: 0
B: ∞
C: ∞
D: ∞
E: ∞
F: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra



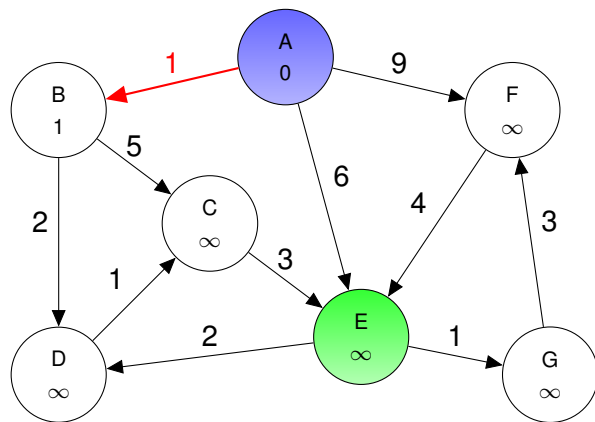
A: 0
B: ∞
C: ∞
D: ∞
E: ∞
F: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra



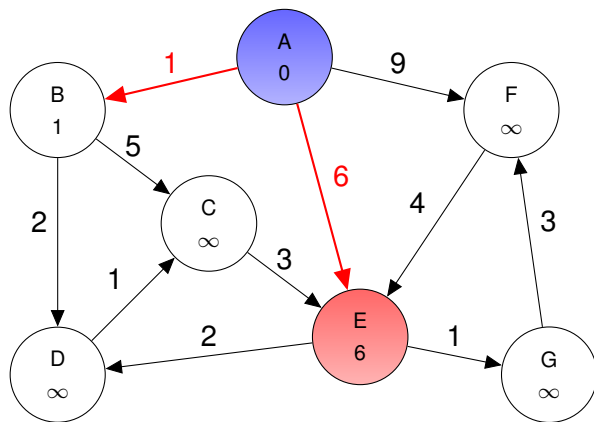
A: 0
B: 1
C: ∞
D: ∞
E: ∞
F: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra



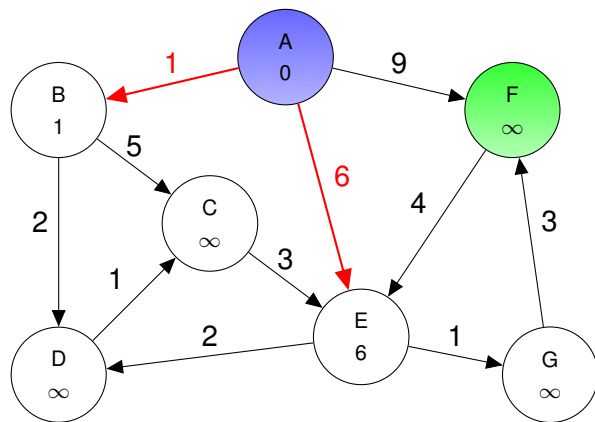
A: 0
B: 1
C: ∞
D: ∞
E: ∞
F: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra



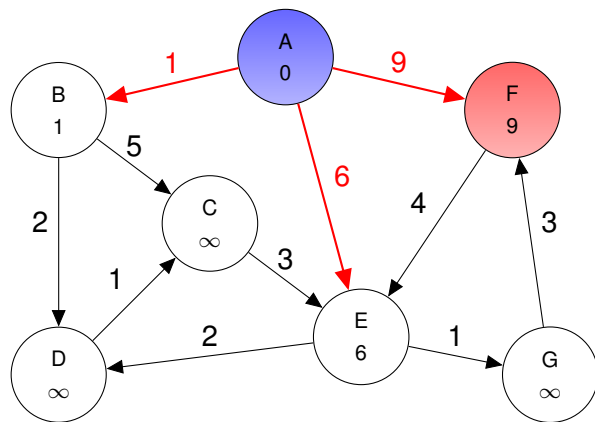
A: 0
B: 1
E: 6
C: ∞
D: ∞
F: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra



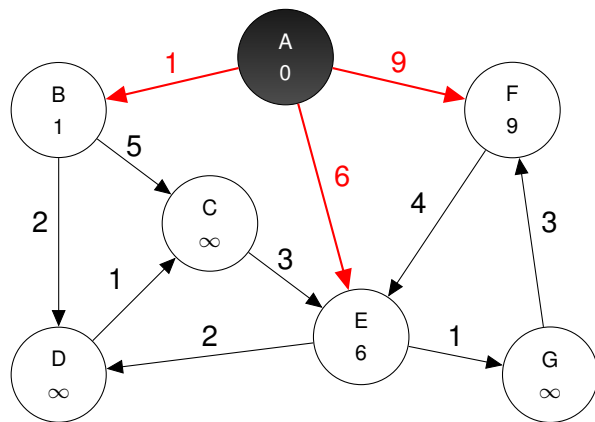
A: 0
B: 1
E: 6
C: ∞
D: ∞
F: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra



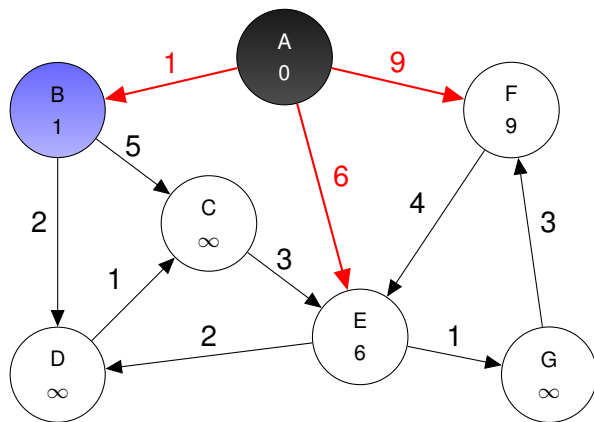
A: 0
B: 1
E: 6
F: 9
C: ∞
D: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra



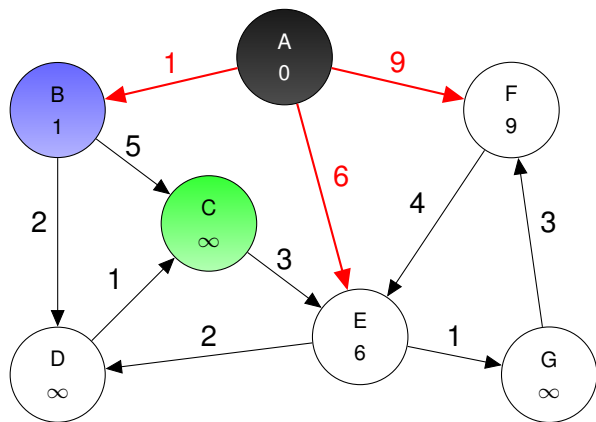
B: 1
E: 6
F: 9
C: ∞
D: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra



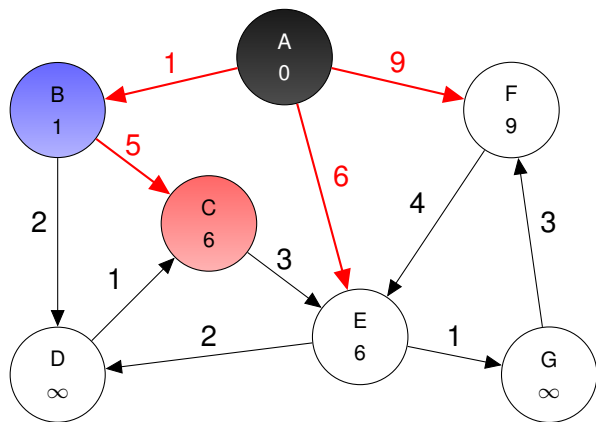
B: 1
E: 6
F: 9
C: ∞
D: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra



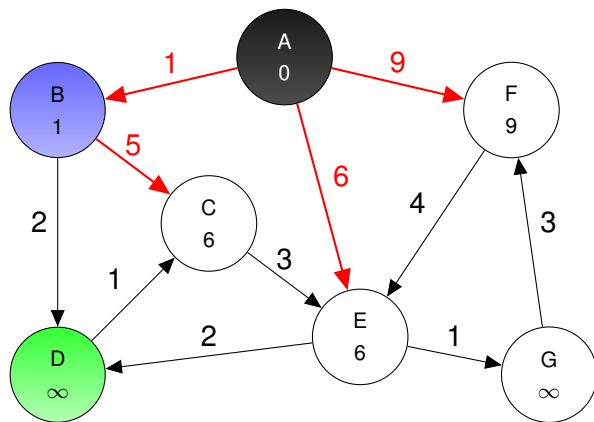
B: 1
E: 6
F: 9
C: ∞
D: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra



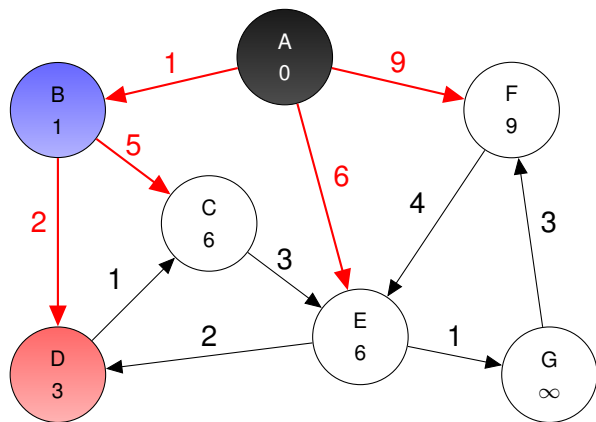
B: 1
C: 6
E: 6
F: 9
D: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra



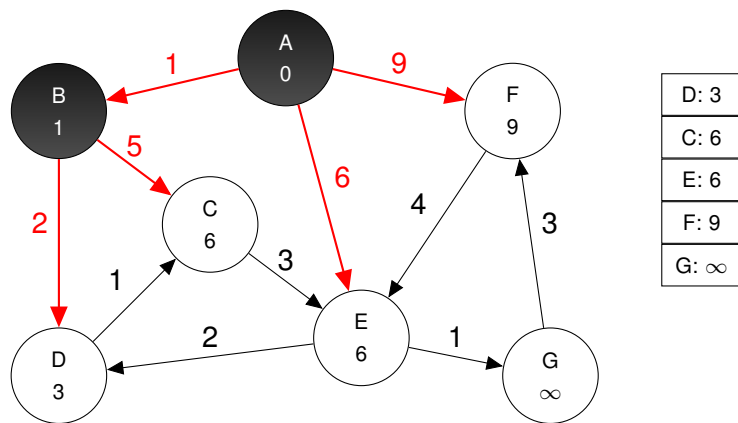
B: 1
C: 6
E: 6
F: 9
D: ∞
G: ∞

Cammini Minimi - Algoritmo di Dijkstra

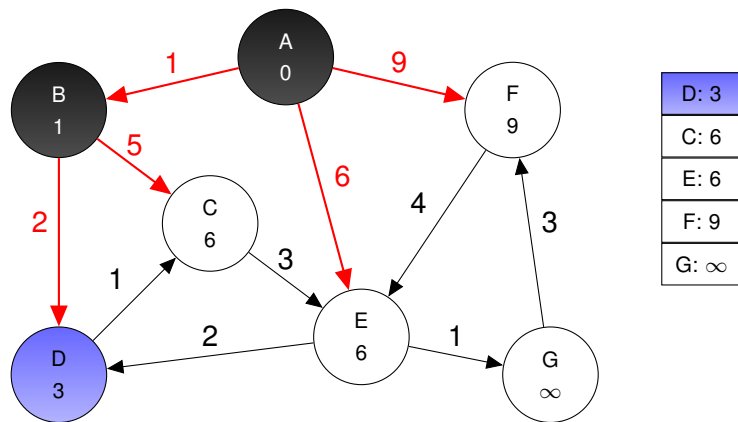


B: 1
D: 3
C: 6
E: 6
F: 9
G: ∞

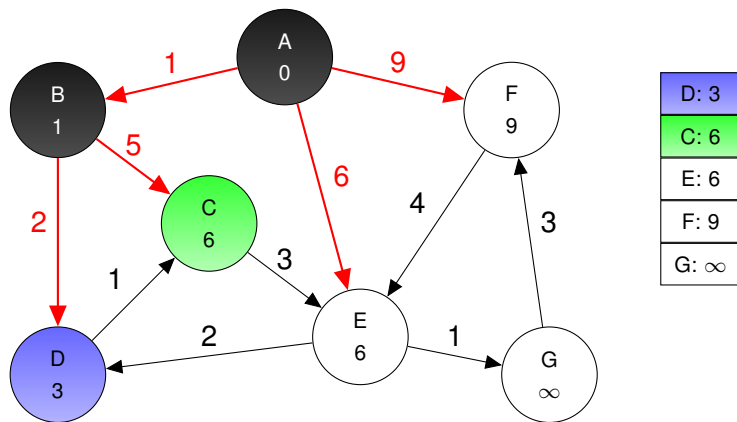
Cammini Minimi - Algoritmo di Dijkstra



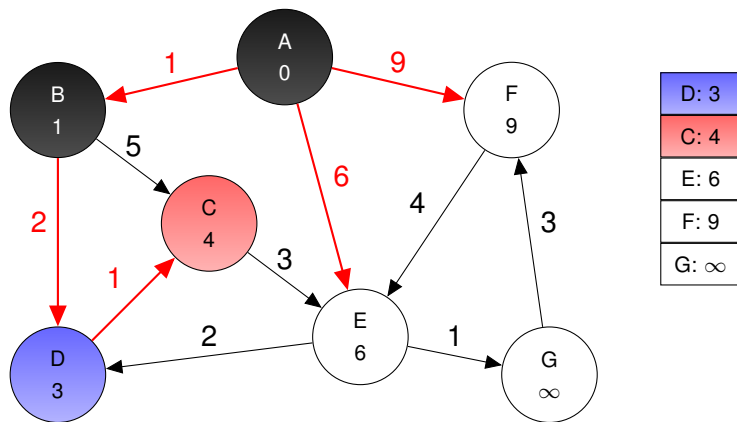
Cammini Minimi - Algoritmo di Dijkstra



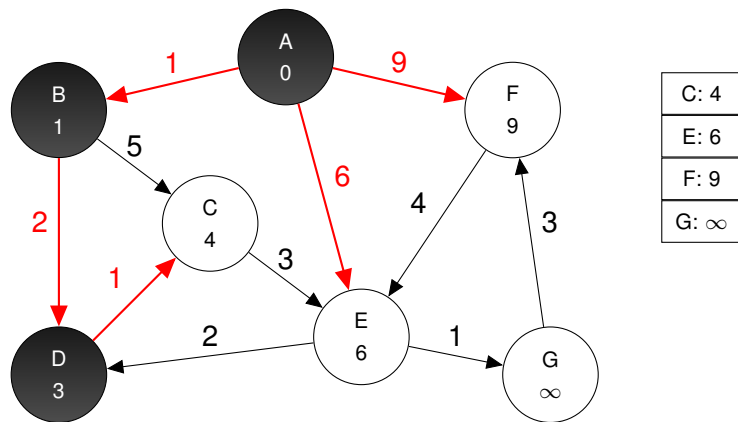
Cammini Minimi - Algoritmo di Dijkstra



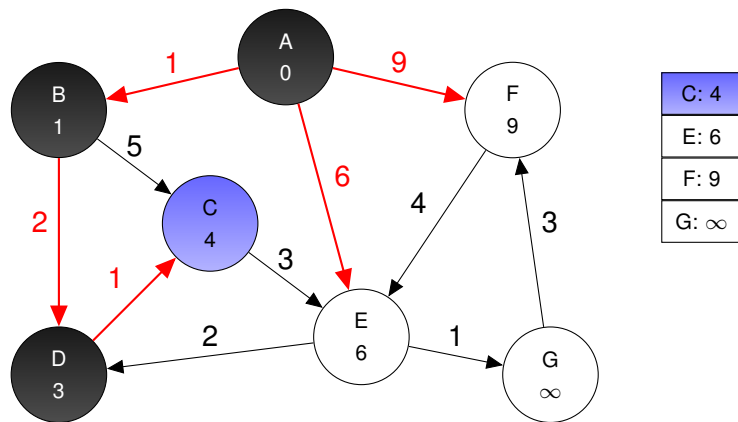
Cammini Minimi - Algoritmo di Dijkstra



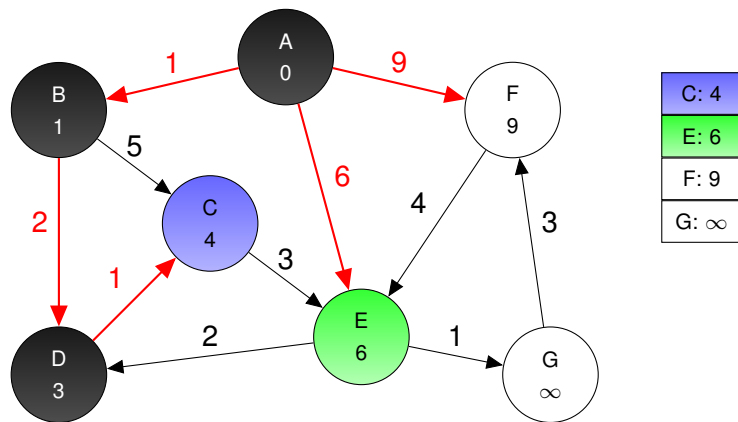
Cammini Minimi - Algoritmo di Dijkstra



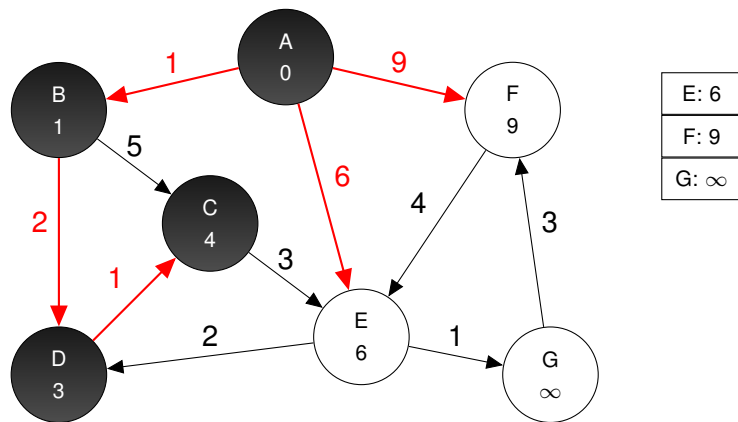
Cammini Minimi - Algoritmo di Dijkstra



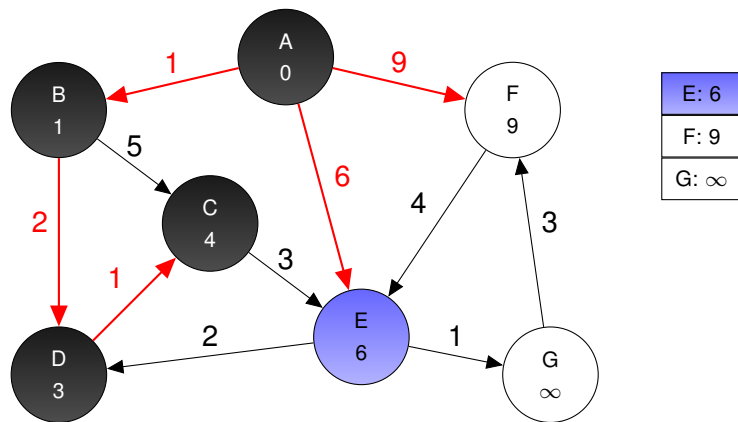
Cammini Minimi - Algoritmo di Dijkstra



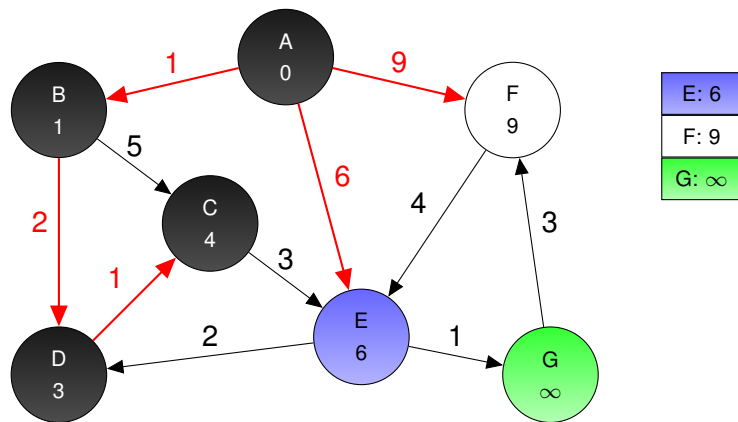
Cammini Minimi - Algoritmo di Dijkstra



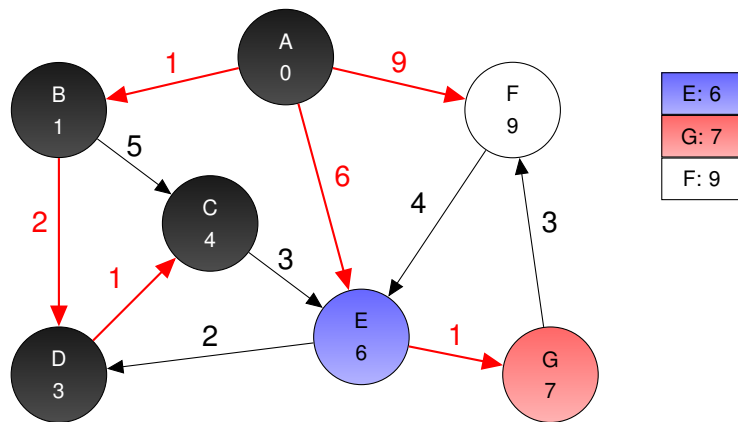
Cammini Minimi - Algoritmo di Dijkstra



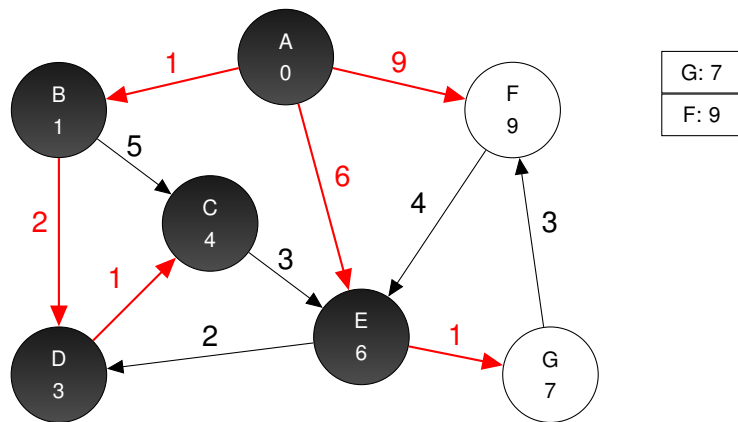
Cammini Minimi - Algoritmo di Dijkstra



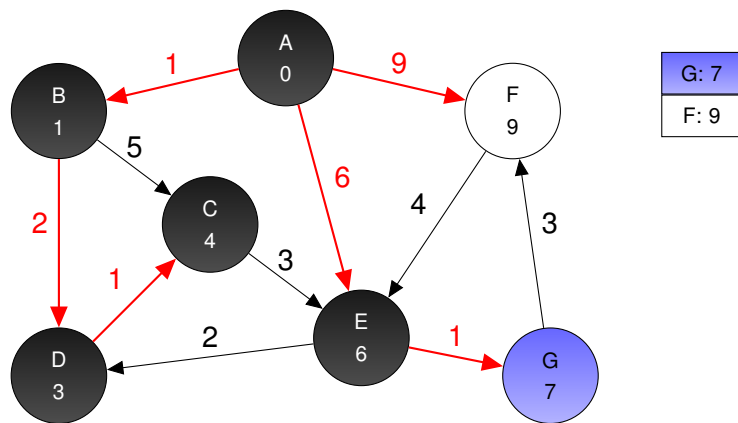
Cammini Minimi - Algoritmo di Dijkstra



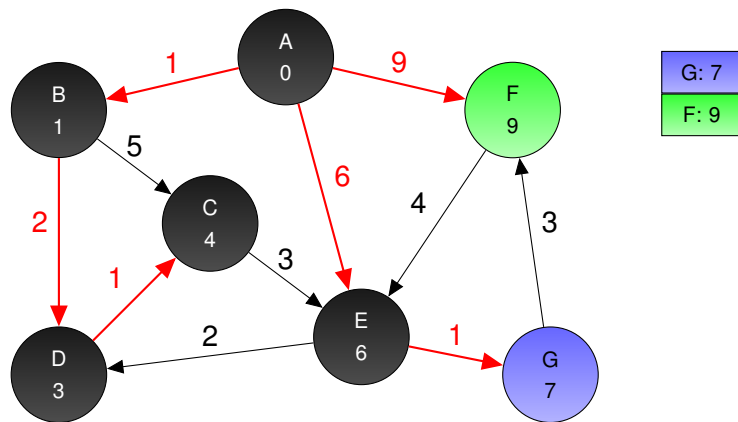
Cammini Minimi - Algoritmo di Dijkstra



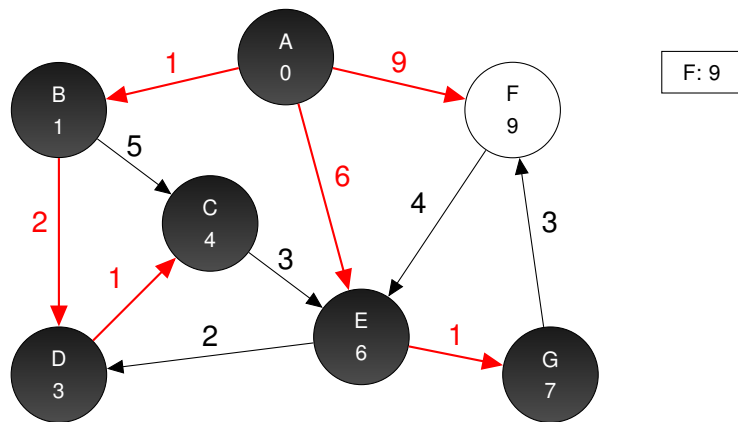
Cammini Minimi - Algoritmo di Dijkstra



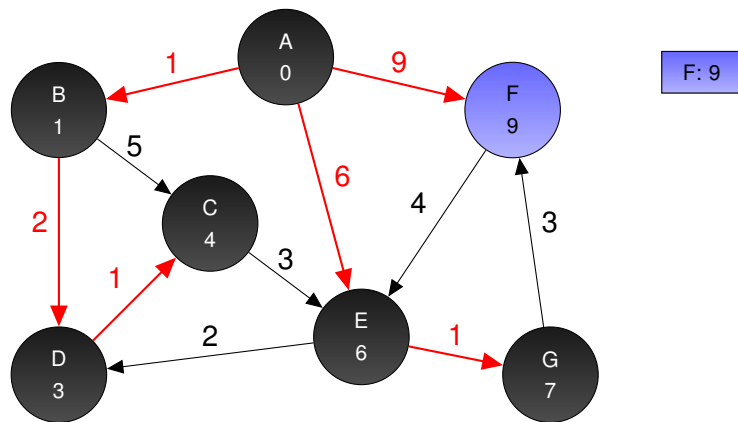
Cammini Minimi - Algoritmo di Dijkstra



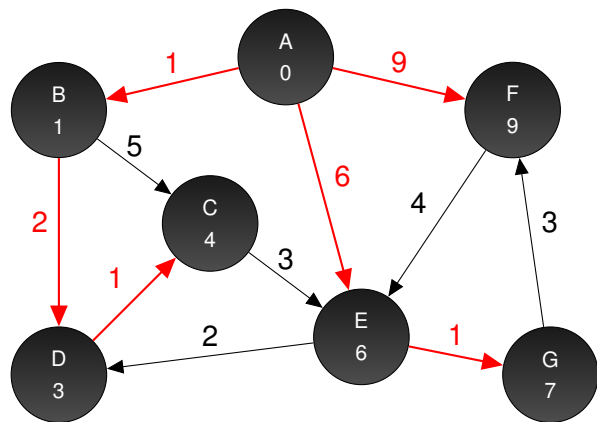
Cammini Minimi - Algoritmo di Dijkstra



Cammini Minimi - Algoritmo di Dijkstra

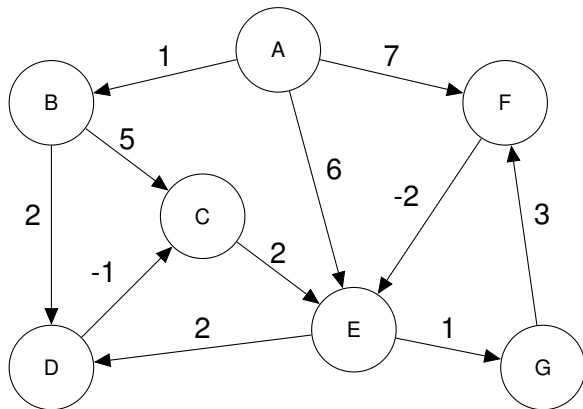


Cammini Minimi - Algoritmo di Dijkstra



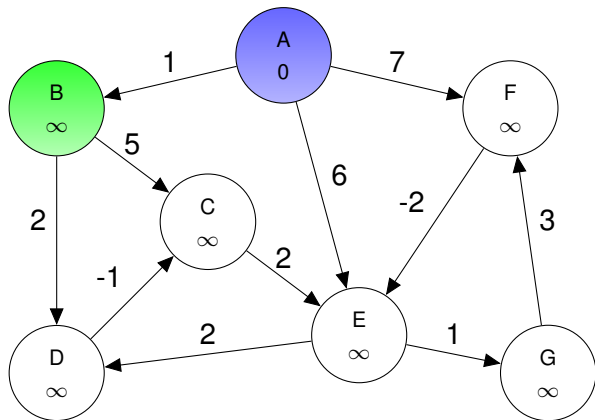
Camminimi Minimi - Algoritmo di Bellman-Ford

Calcolare i cammini minimi a partire dal nodo sorgente A



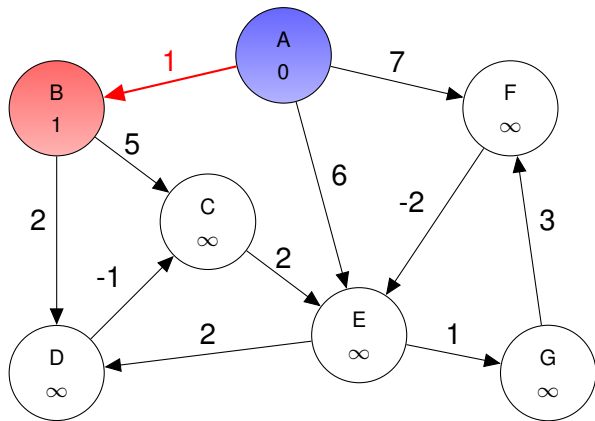
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



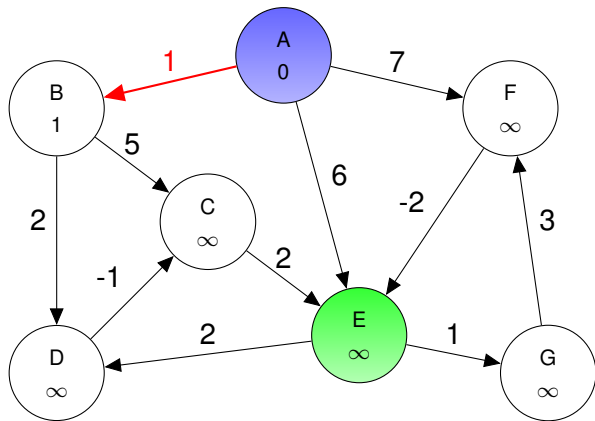
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



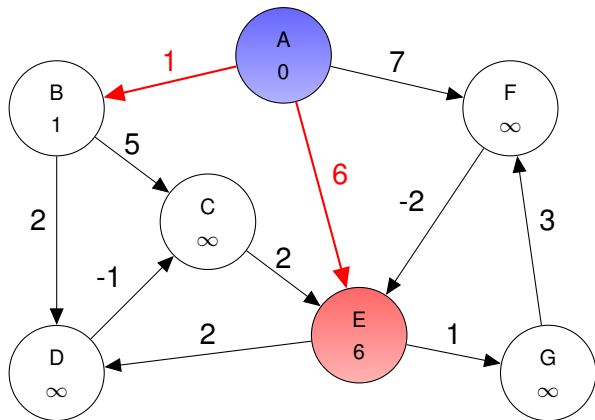
Camminimi Minimi - Algoritmo di Bellman-Ford

Iterazione 1



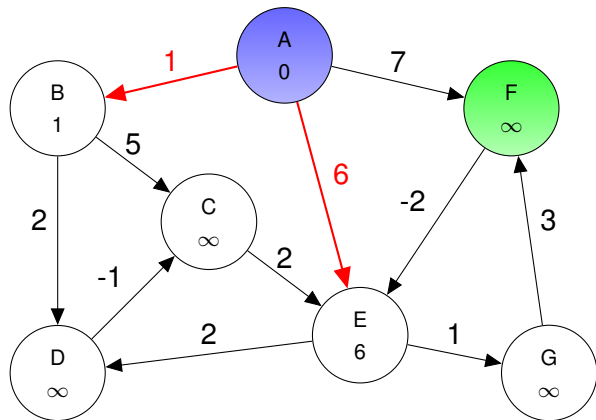
Camminimi Minimi - Algoritmo di Bellman-Ford

Iterazione 1



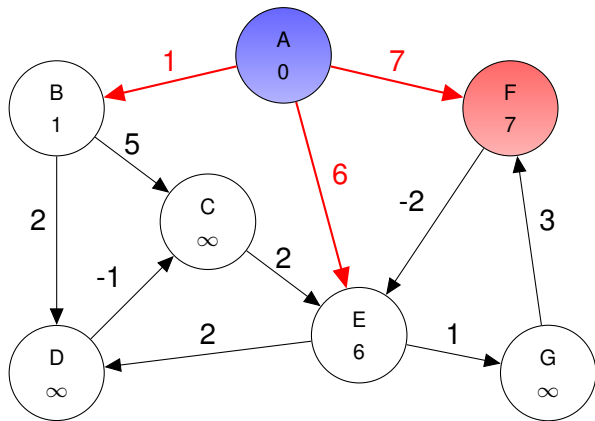
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



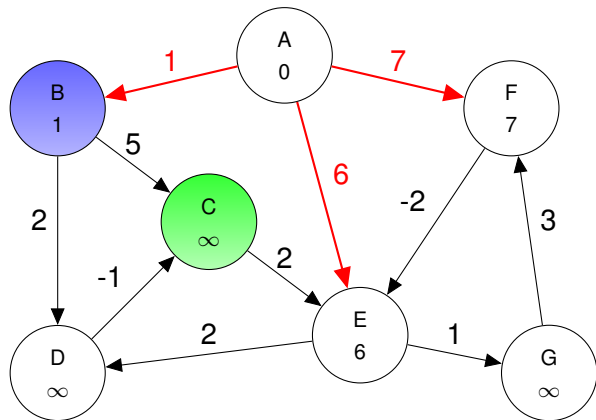
Camminimi Minimi - Algoritmo di Bellman-Ford

Iterazione 1



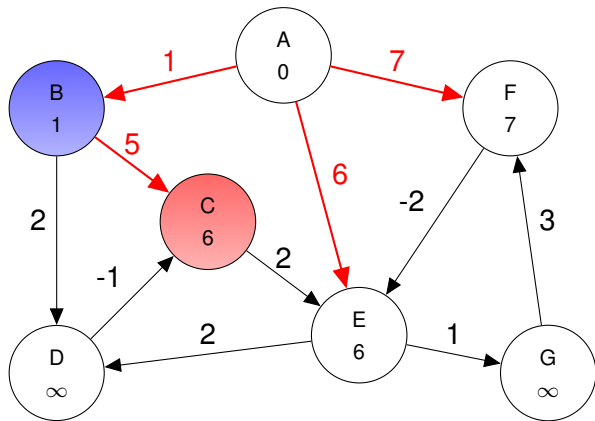
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



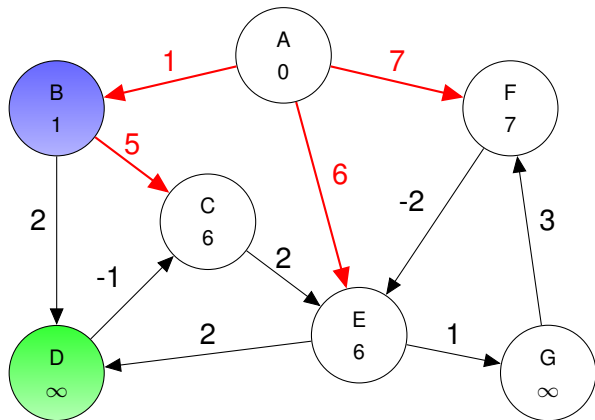
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



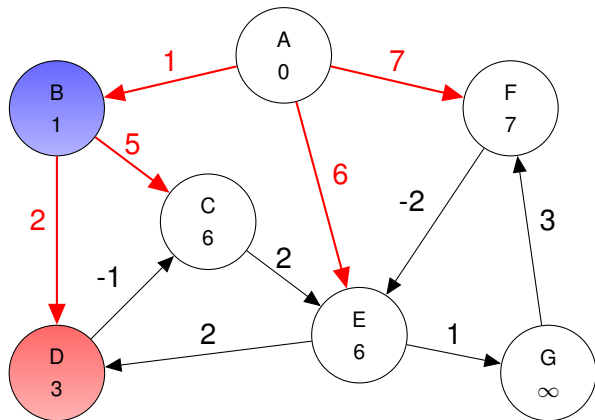
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



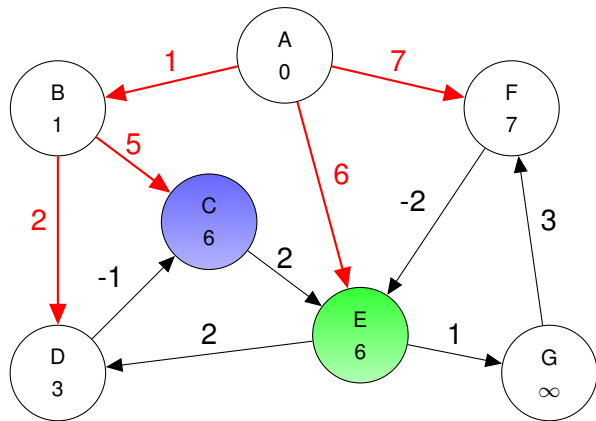
Camminimi Minimi - Algoritmo di Bellman-Ford

Iterazione 1



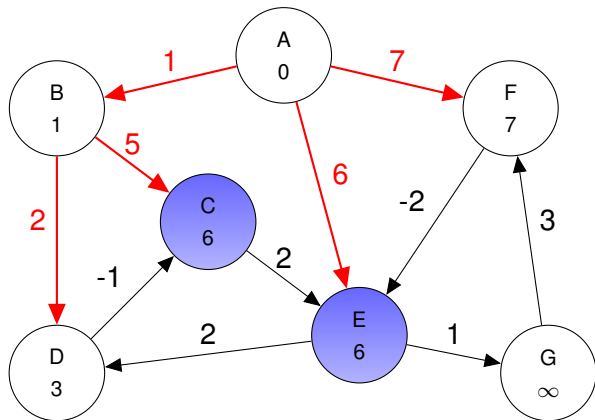
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



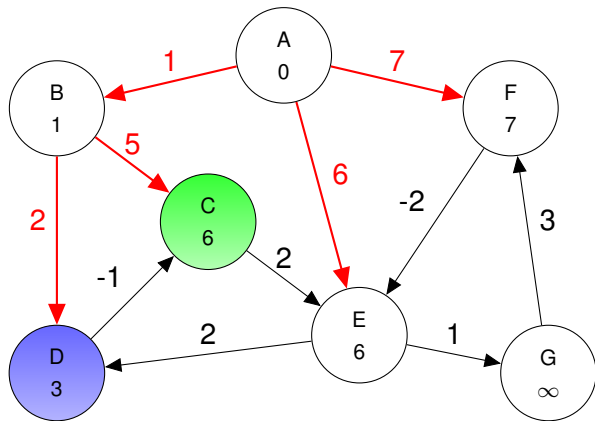
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



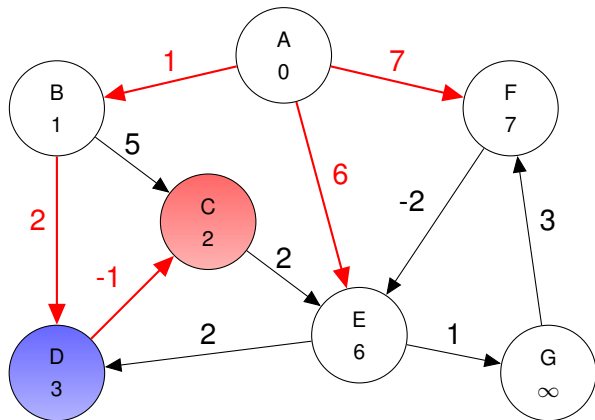
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



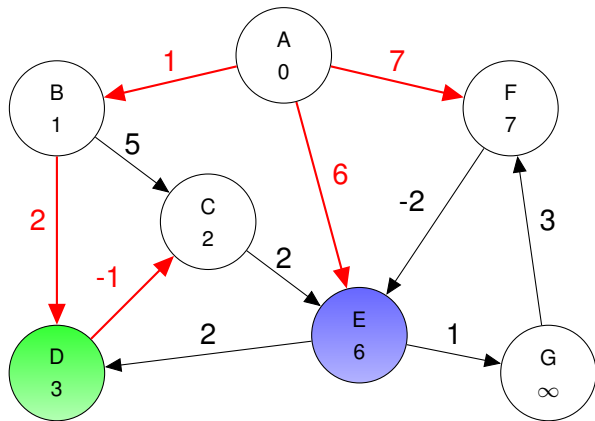
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



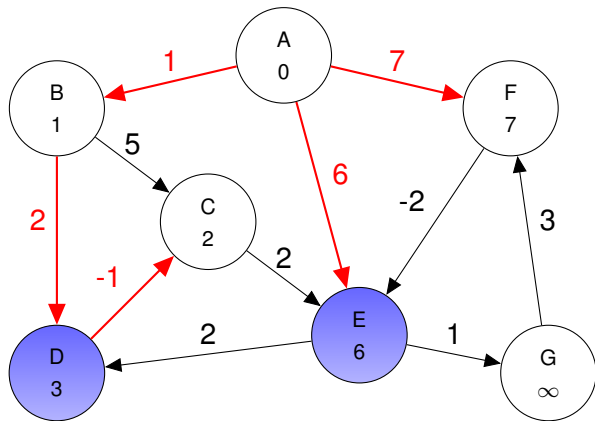
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



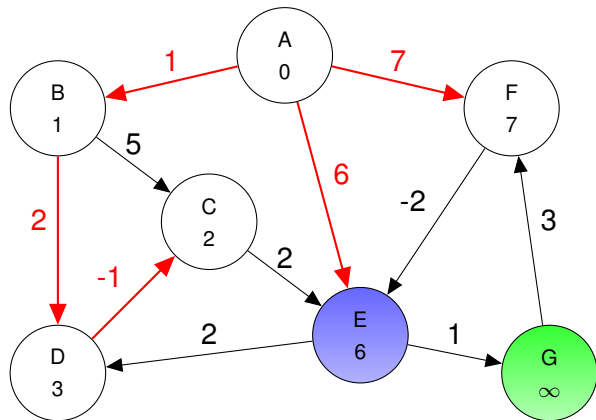
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



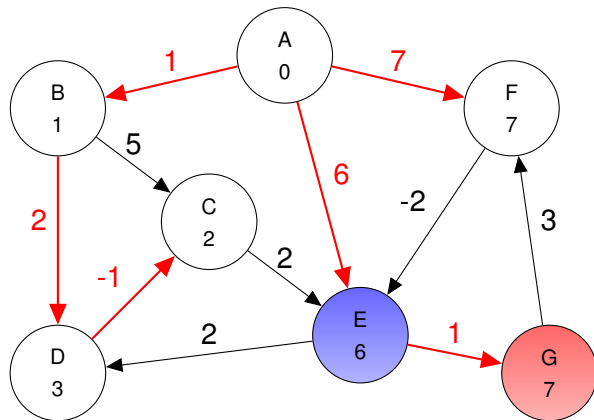
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



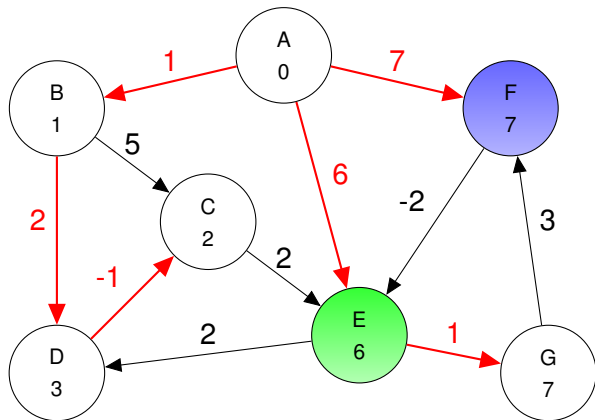
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



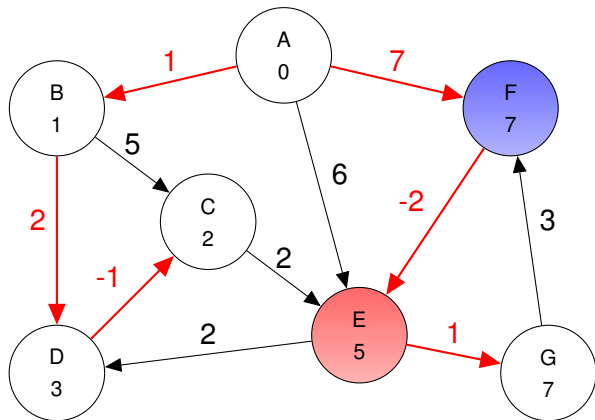
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



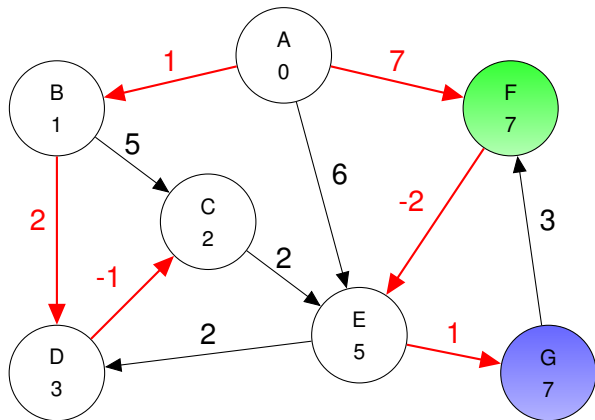
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



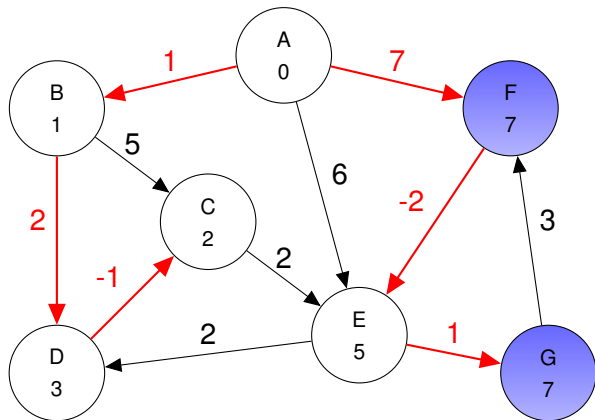
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



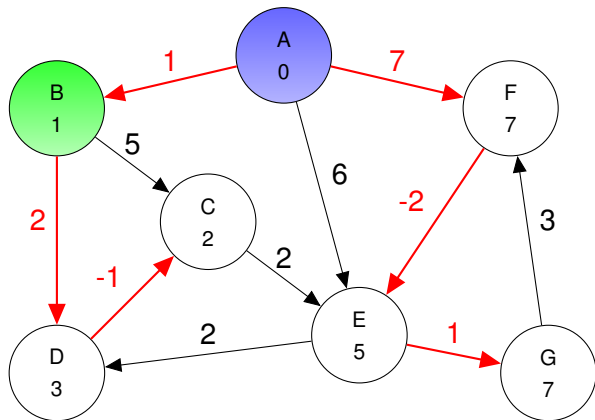
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 1



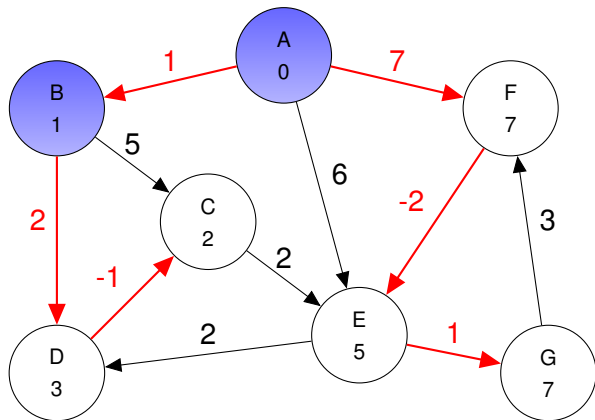
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2

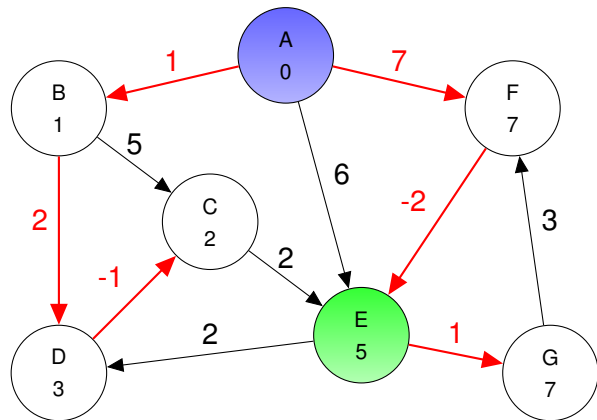


Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2

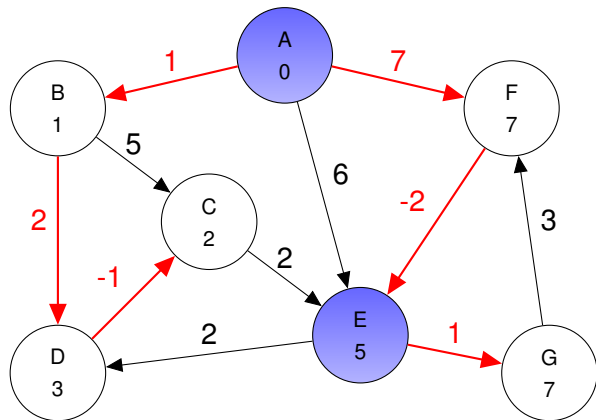


Iterazione 2



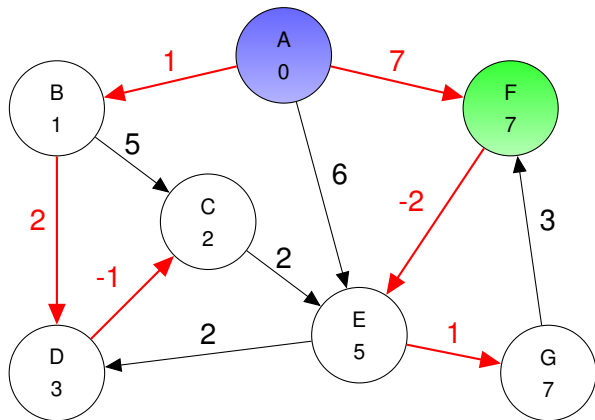
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



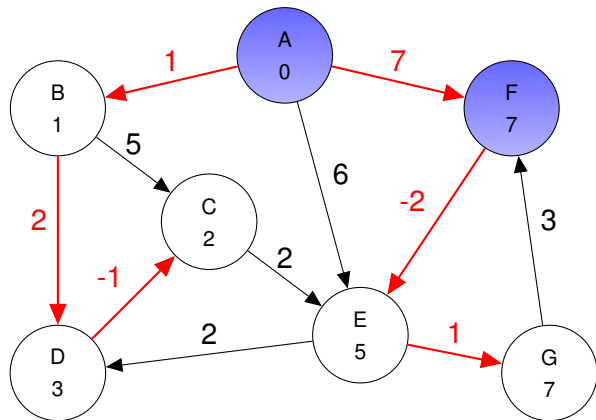
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



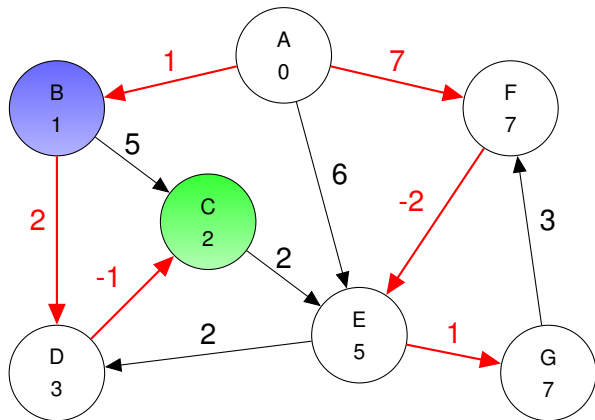
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



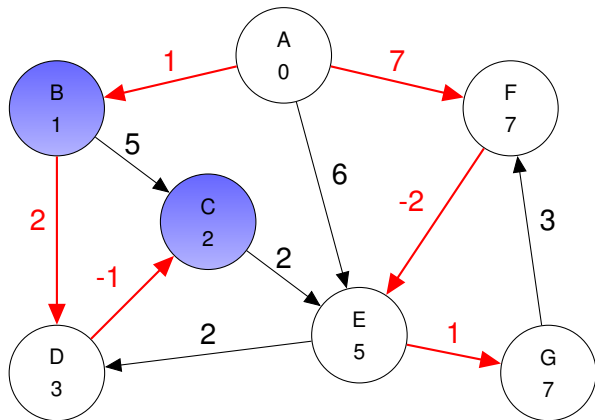
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



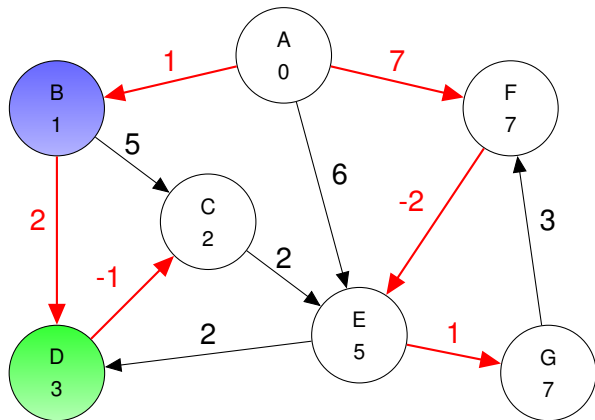
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



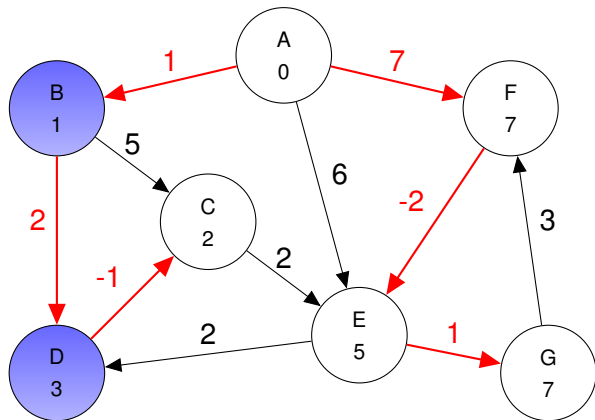
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



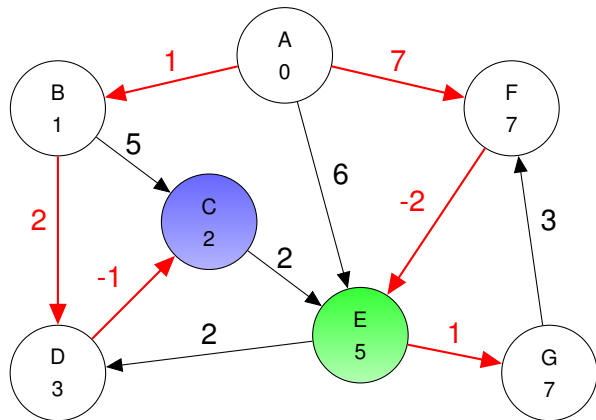
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



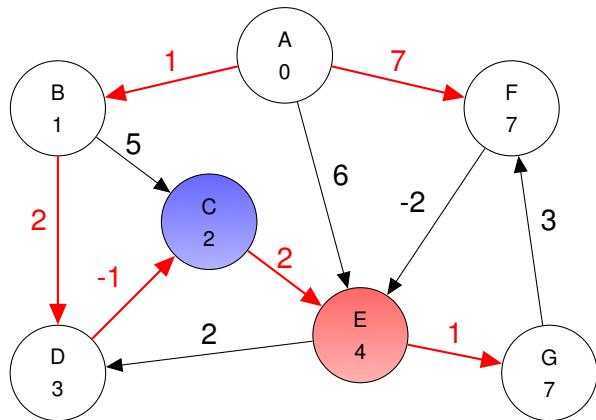
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



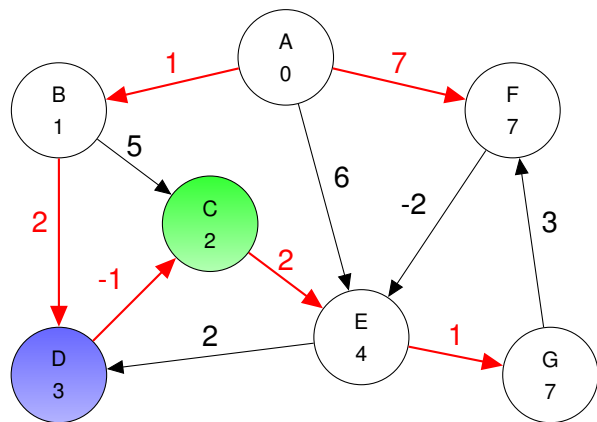
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



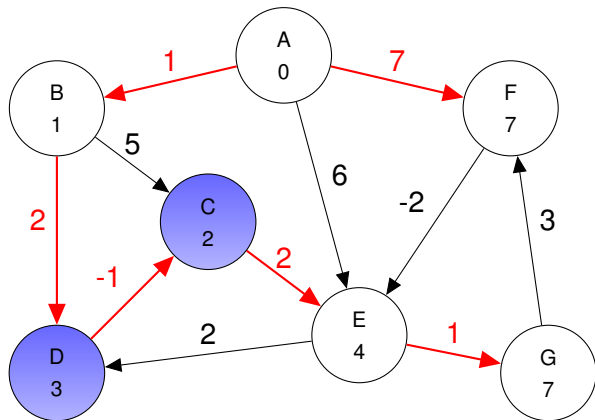
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



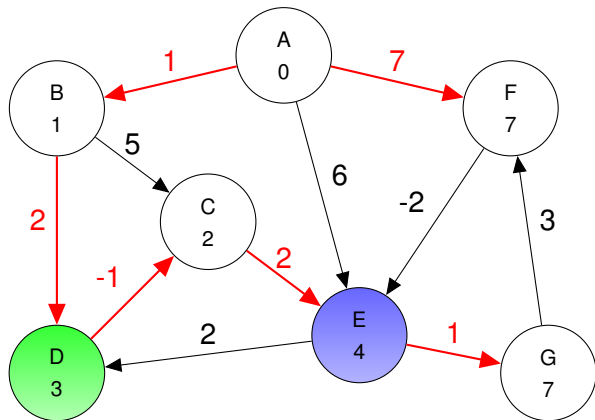
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



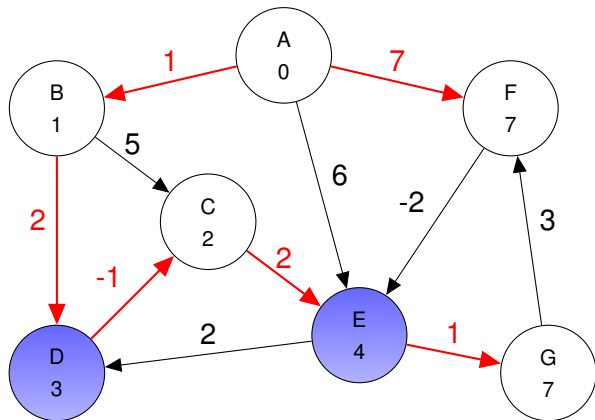
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



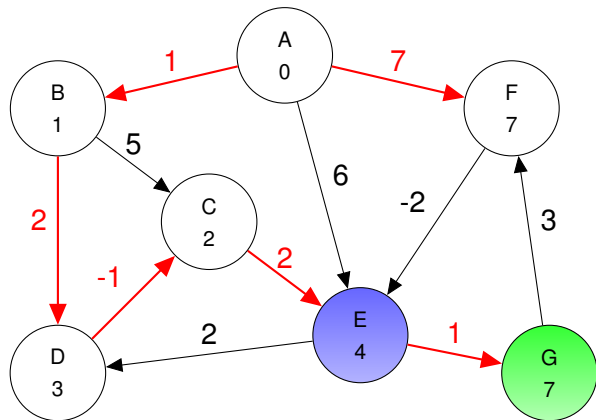
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



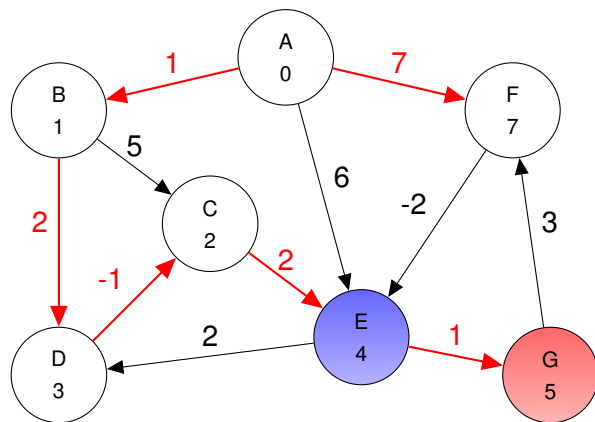
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



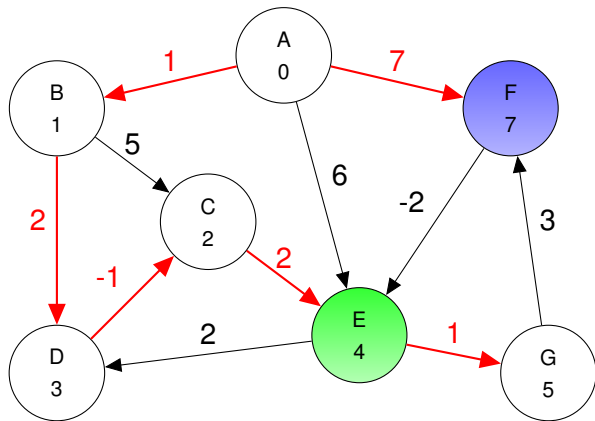
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



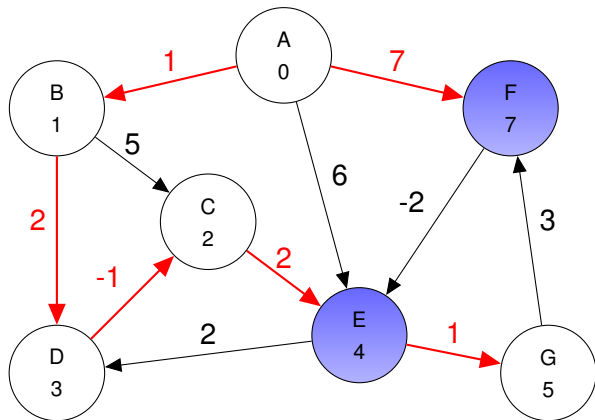
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



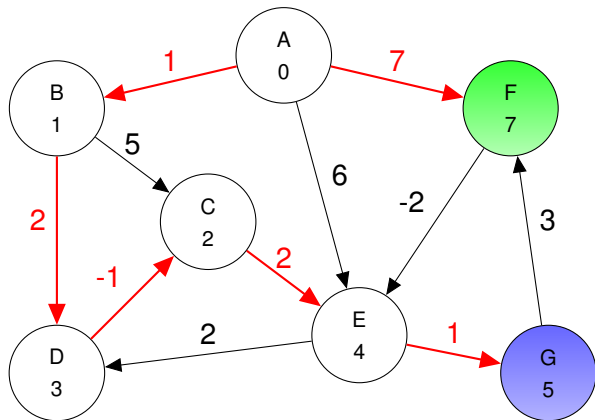
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2



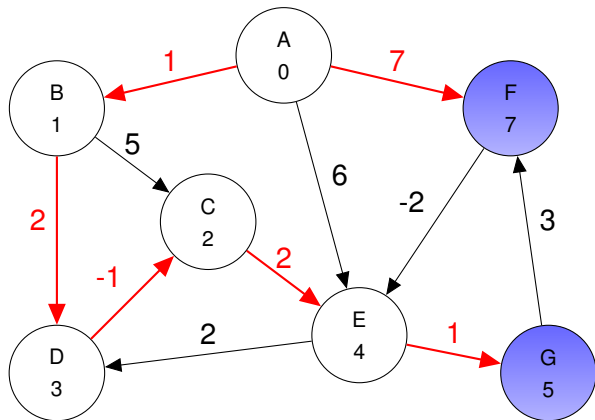
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2

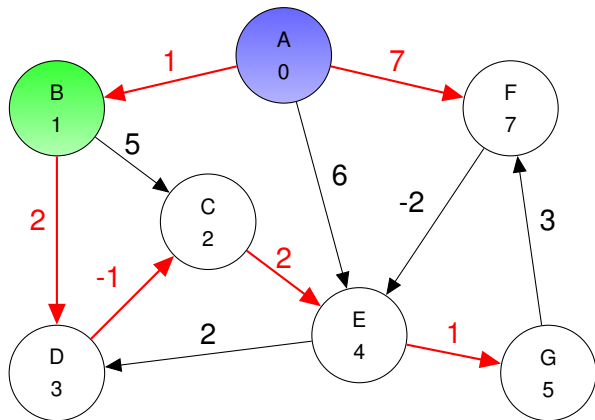


Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 2

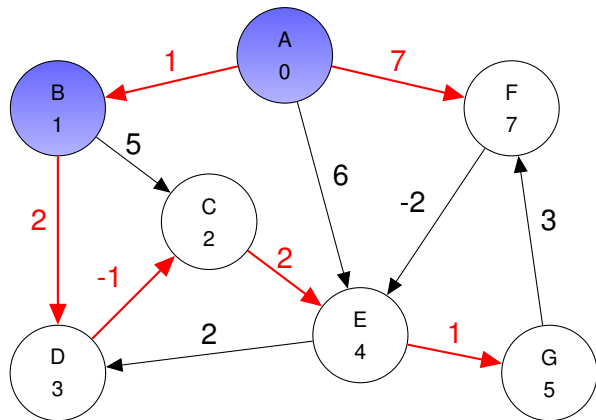


Iterazione 3

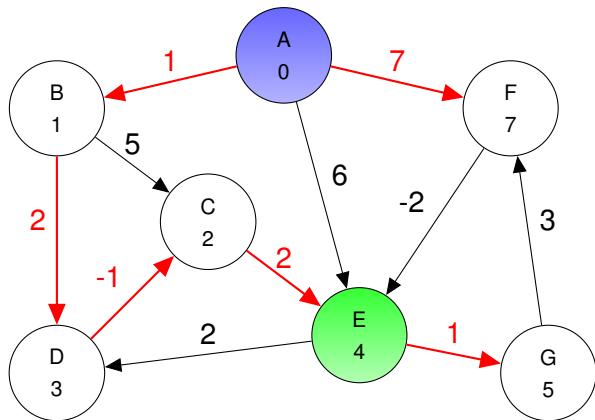


Cammini Minimi - Algoritmo di Bellman-Ford

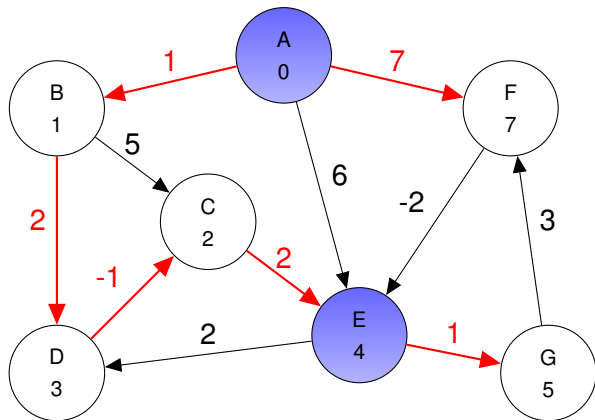
Iterazione 3



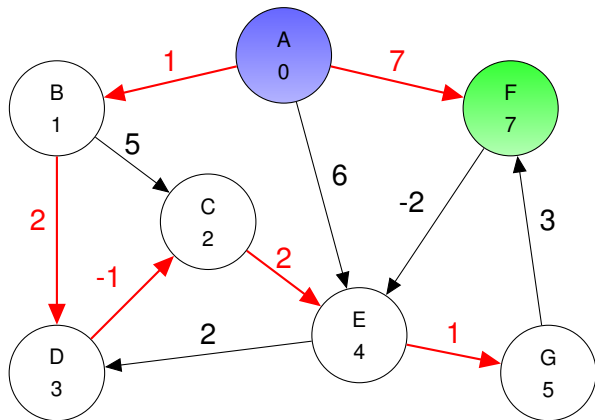
Iterazione 3



Iterazione 3

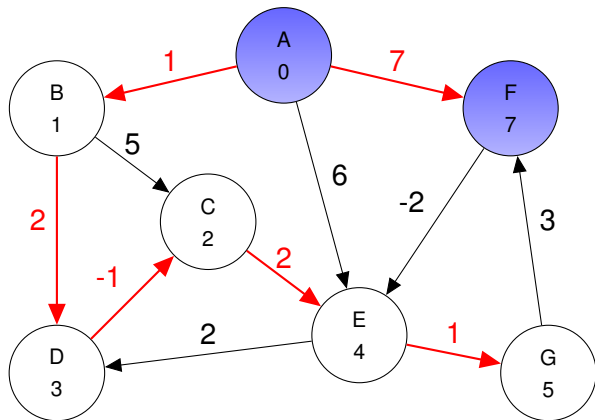


Iterazione 3



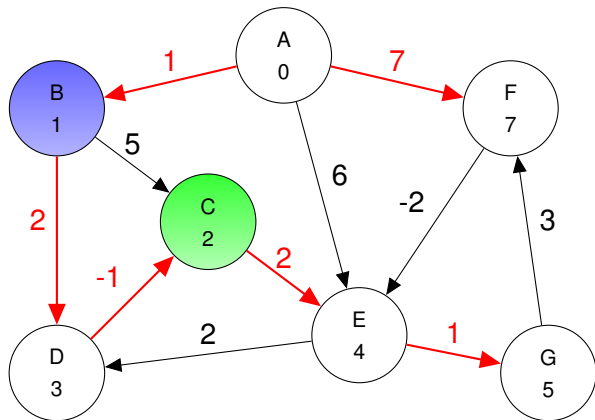
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 3



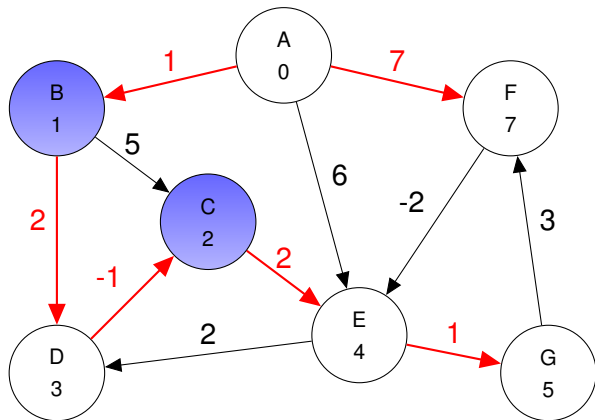
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 3



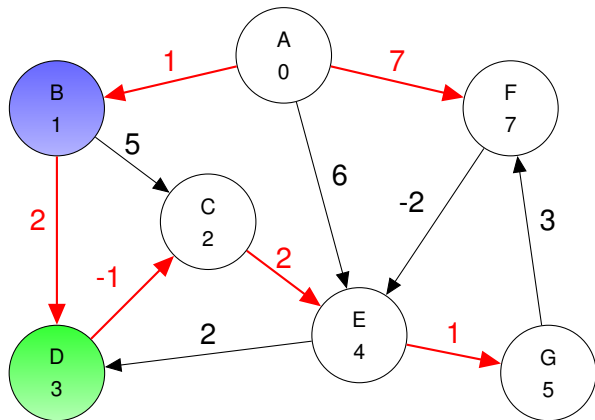
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 3



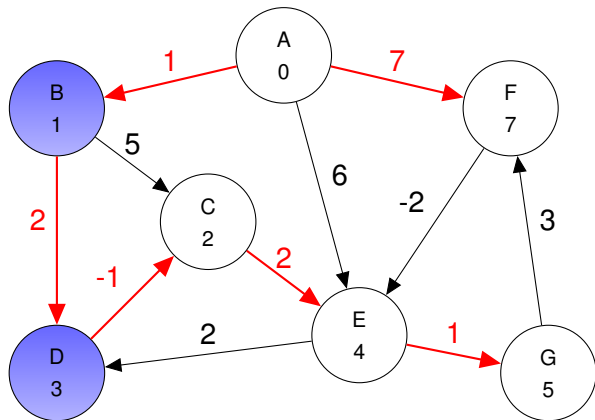
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 3

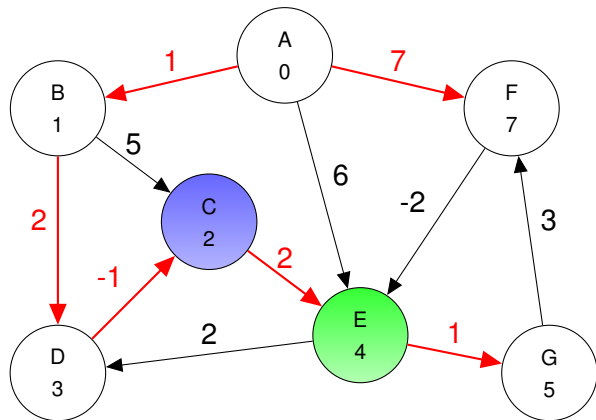


Cammini Minimi - Algoritmo di Bellman-Ford

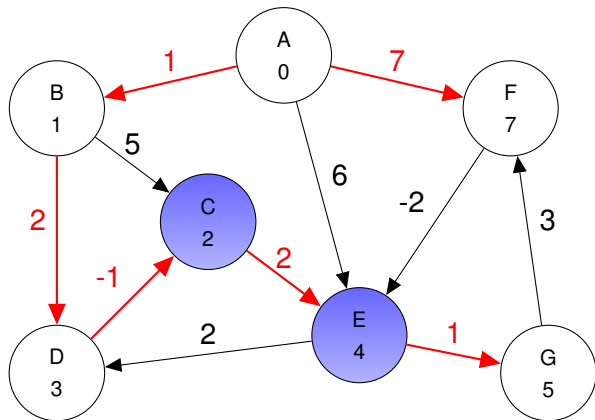
Iterazione 3



Iterazione 3

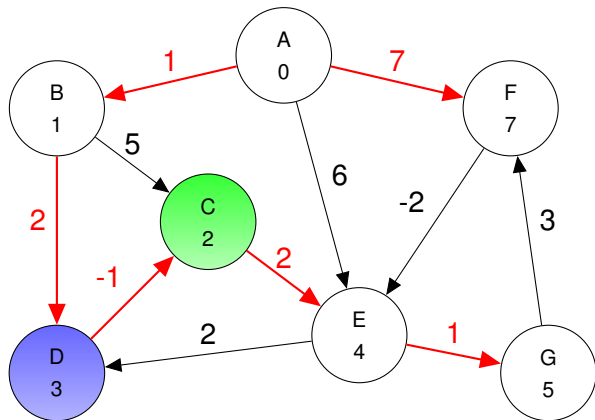


Iterazione 3



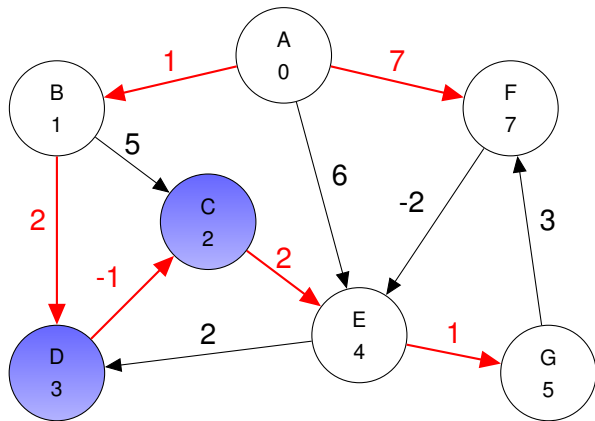
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 3



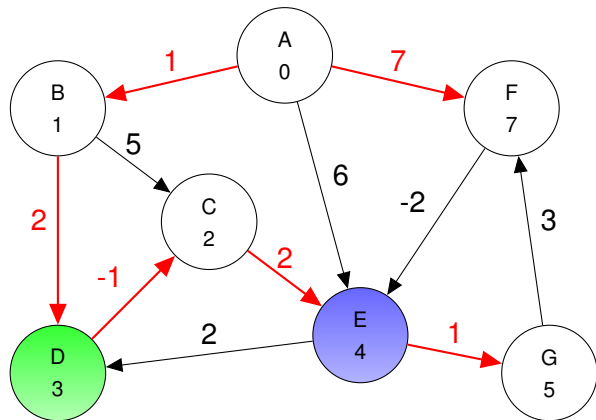
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 3



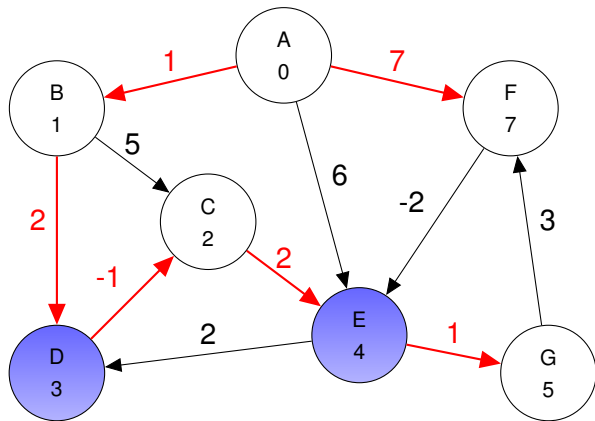
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 3



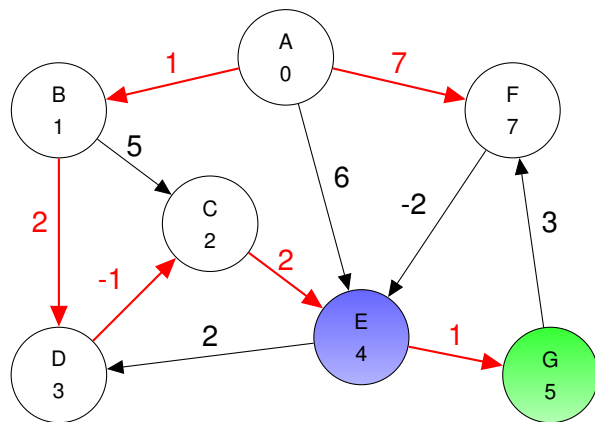
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 3



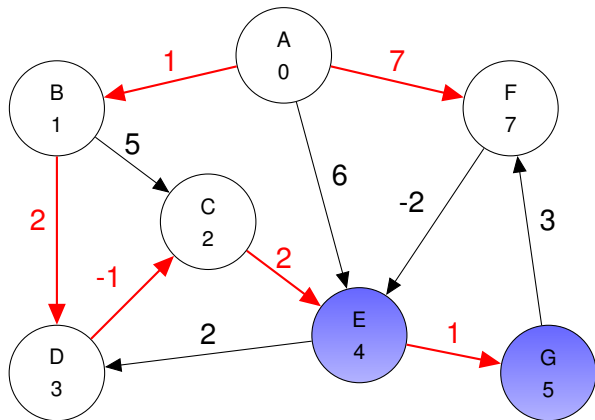
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 3

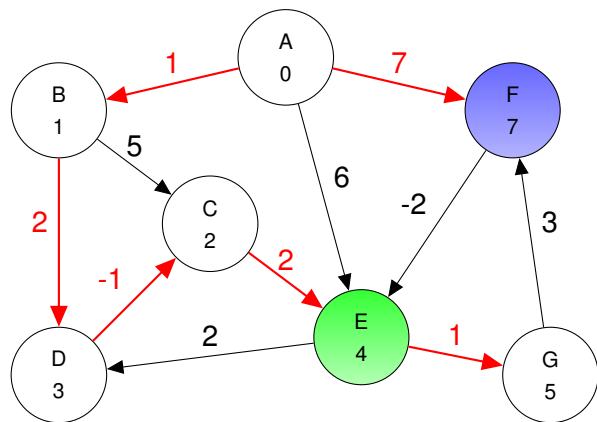


Cammini Minimi - Algoritmo di Bellman-Ford

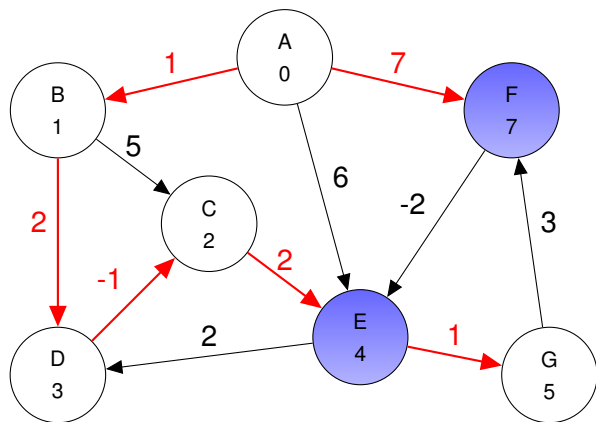
Iterazione 3



Iterazione 3

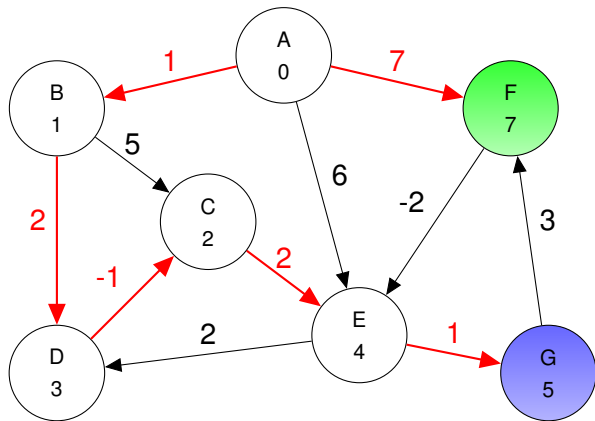


Iterazione 3



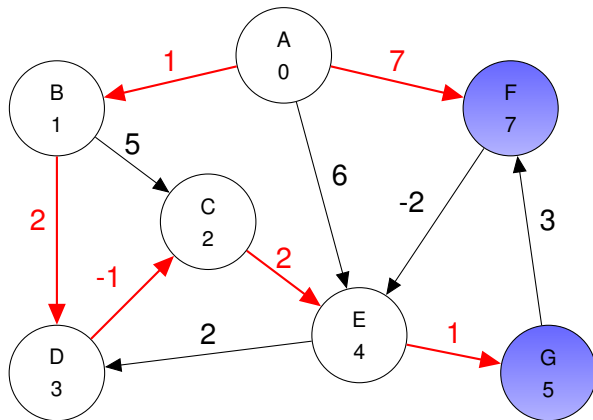
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 3



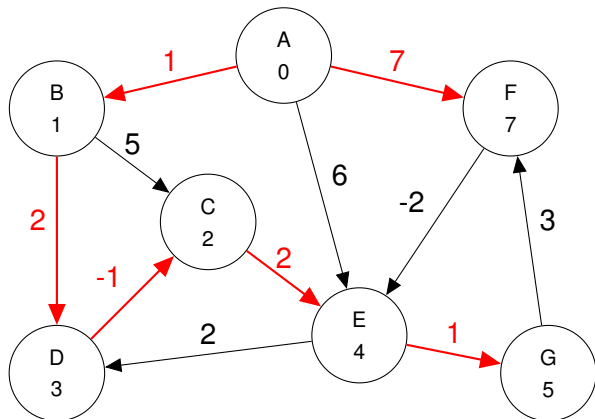
Cammini Minimi - Algoritmo di Bellman-Ford

Iterazione 3



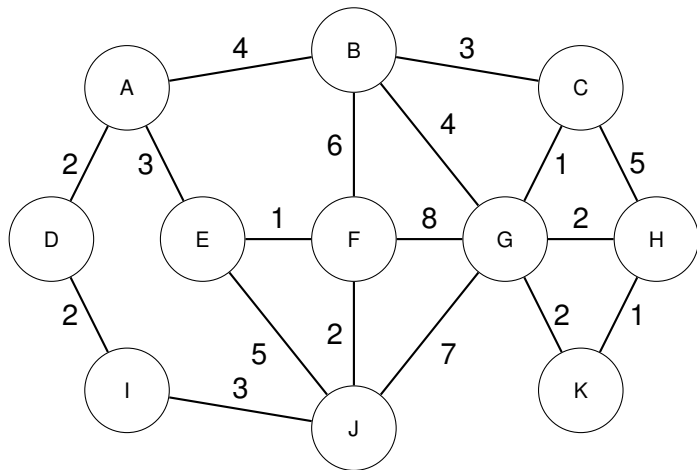
Camminimi Minimi - Algoritmo di Bellman-Ford

Alla terza iterazione non sono avvenute modifiche — non ci sono cicli negativi e possiamo terminare l'algoritmo

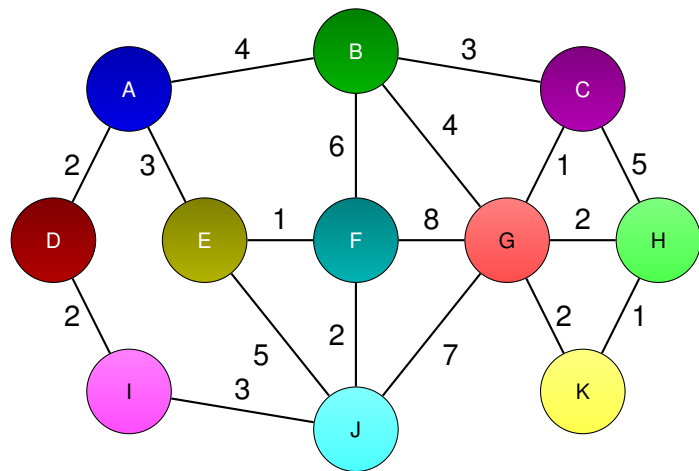


Minimum spanning tree

Calcolare un Minimum spanning tree per il seguente grafo pesato:

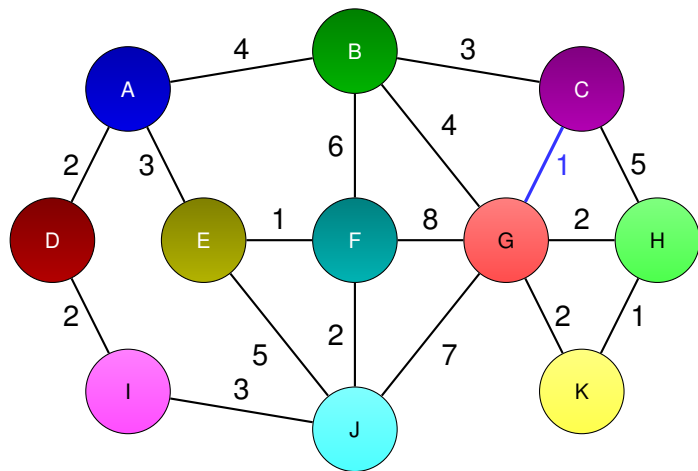


Minimum spanning tree - Algoritmo di Kruskal



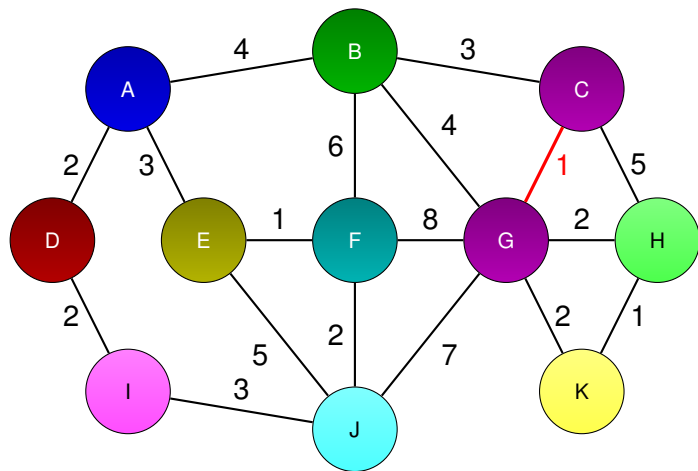
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



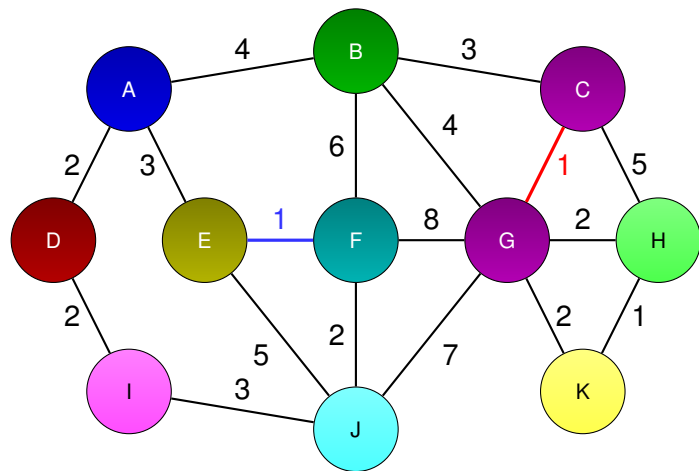
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



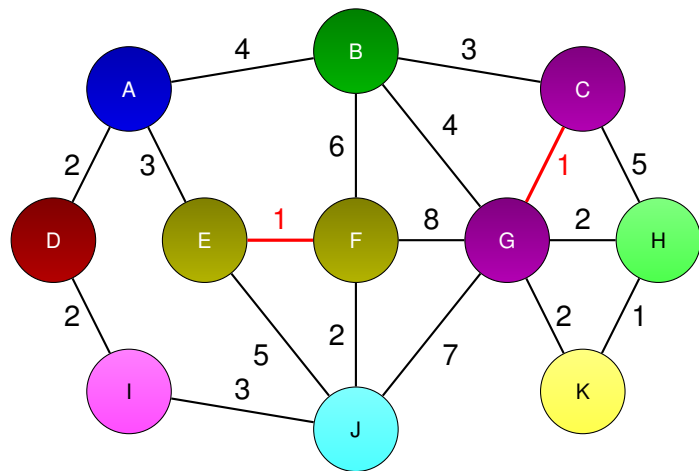
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



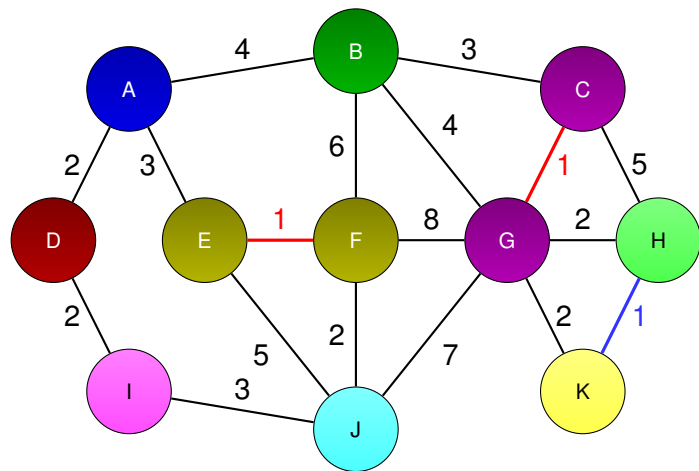
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



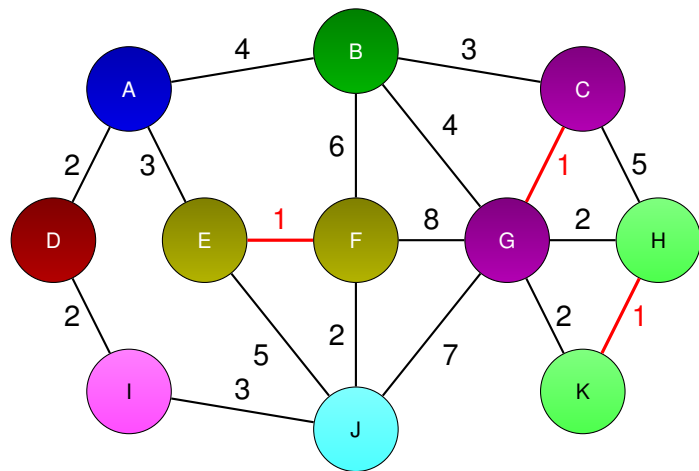
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



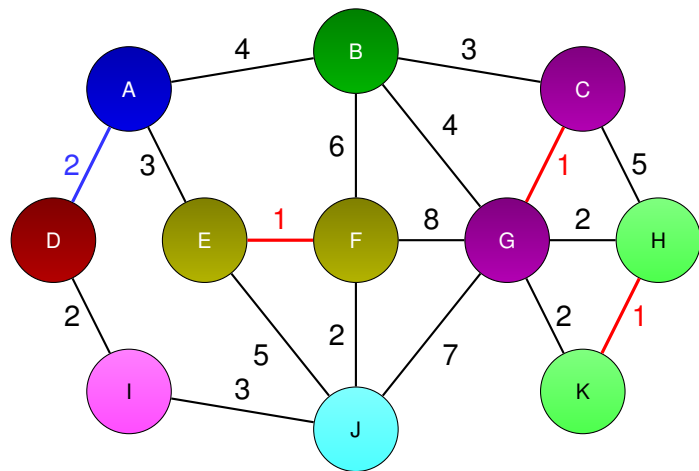
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



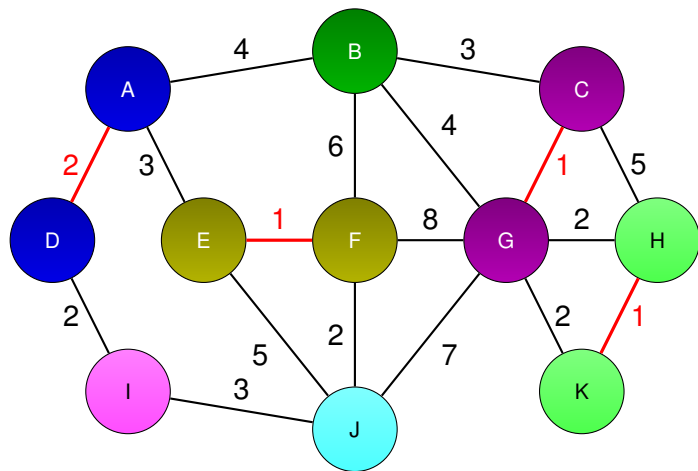
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



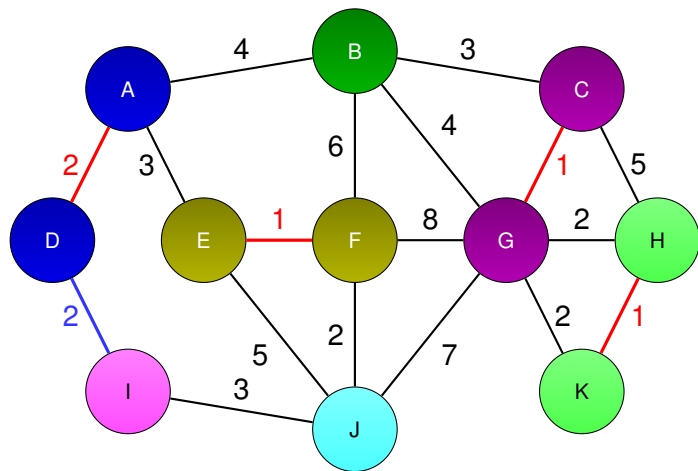
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



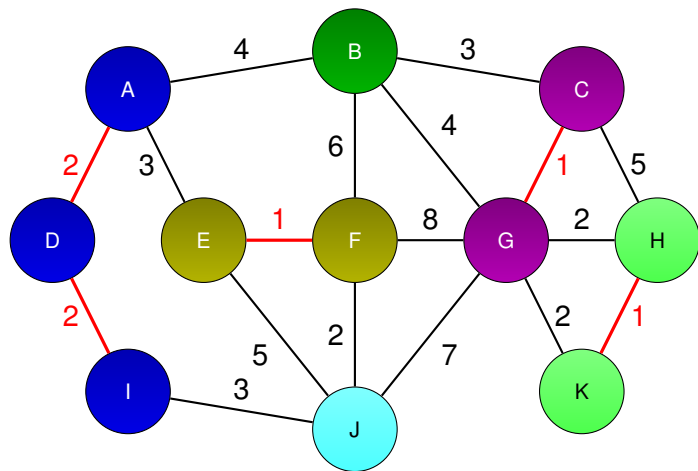
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



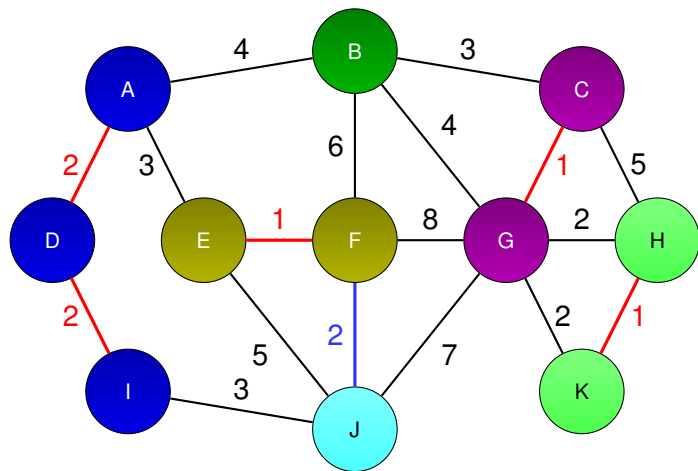
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – I: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



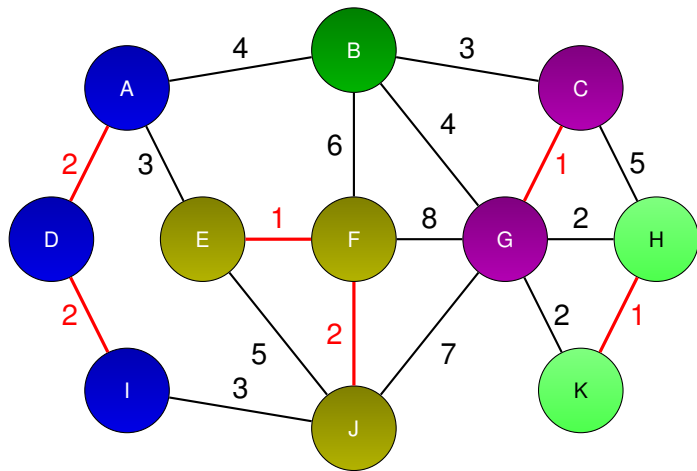
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



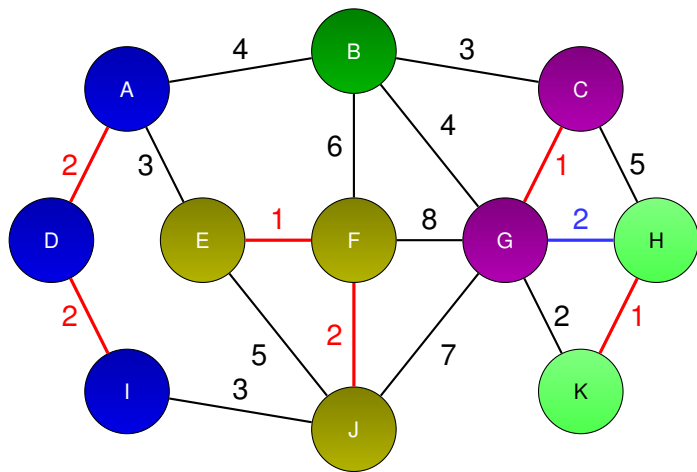
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



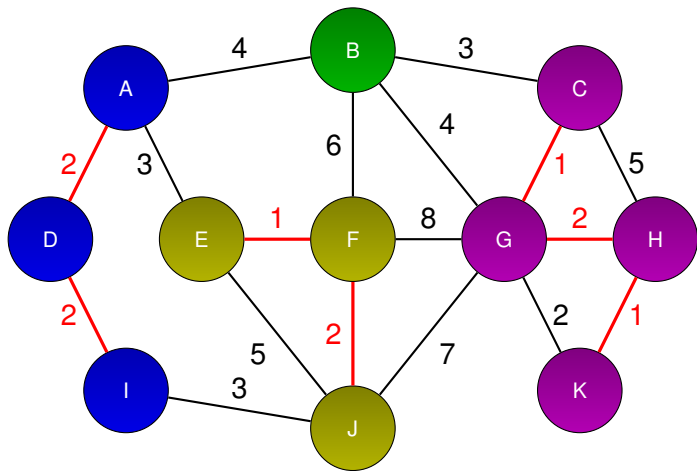
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



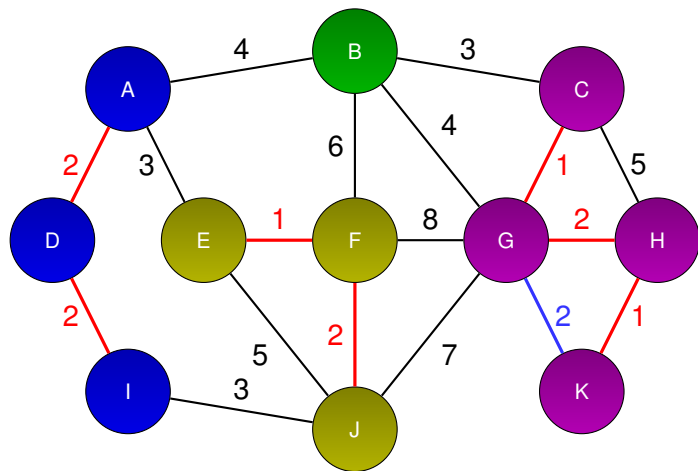
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



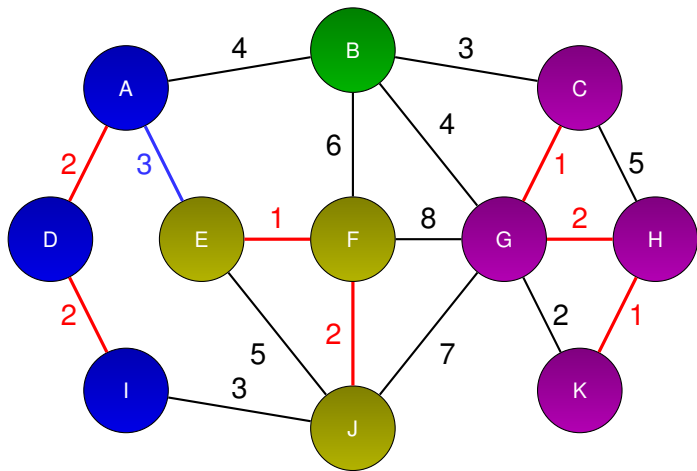
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



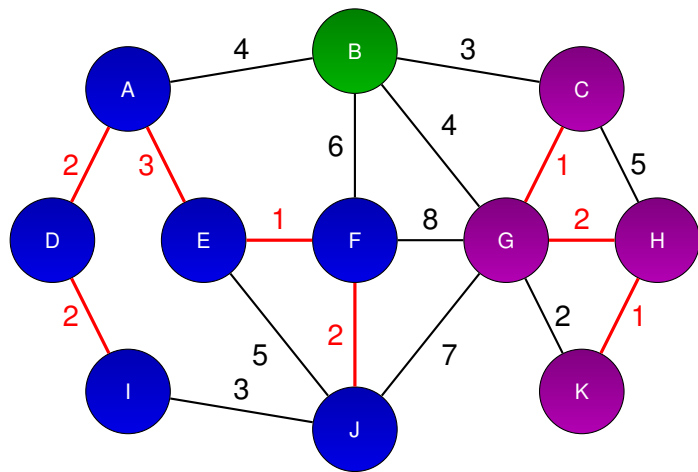
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



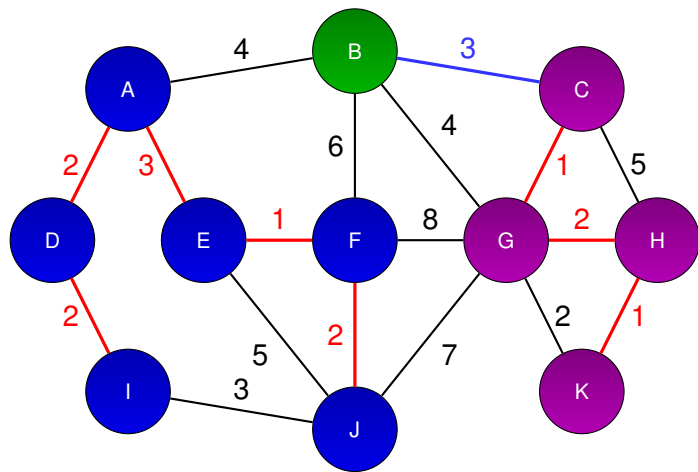
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



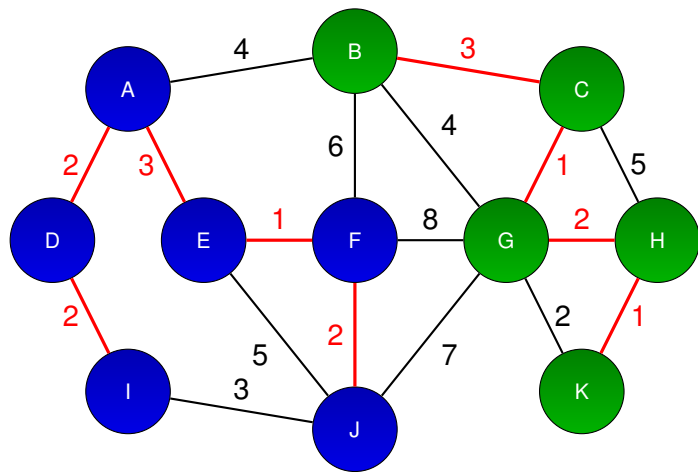
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



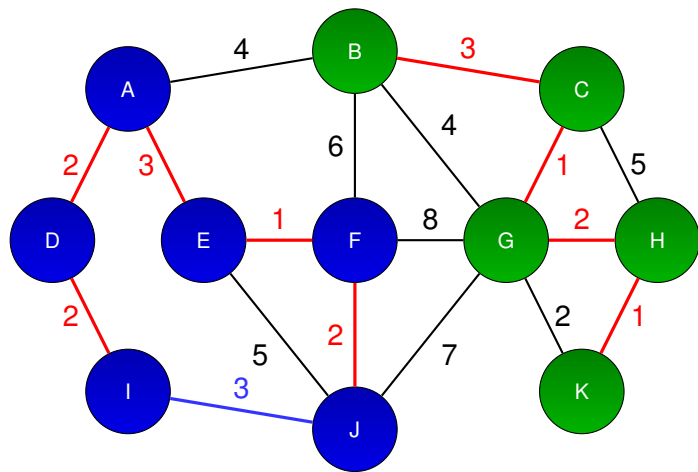
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



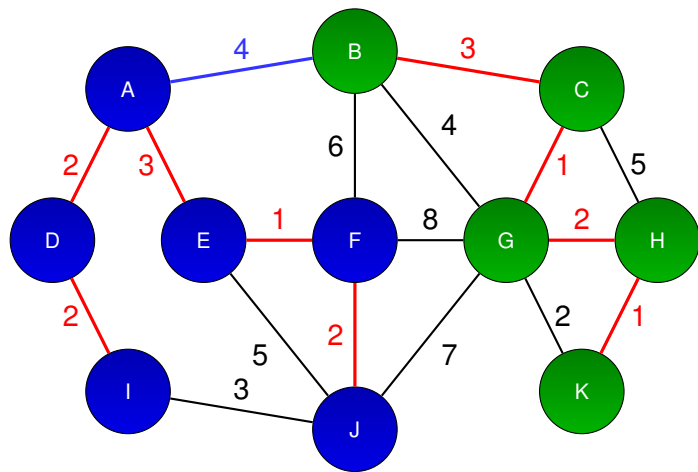
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



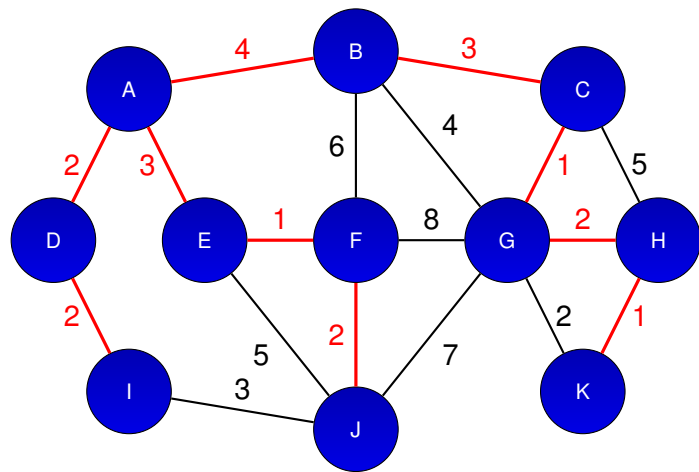
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



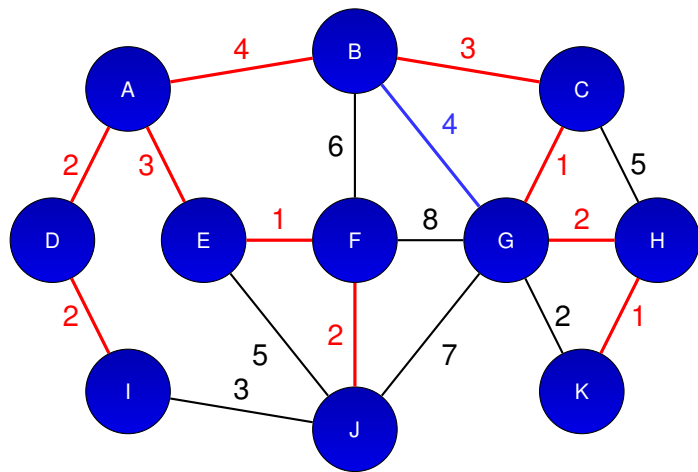
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



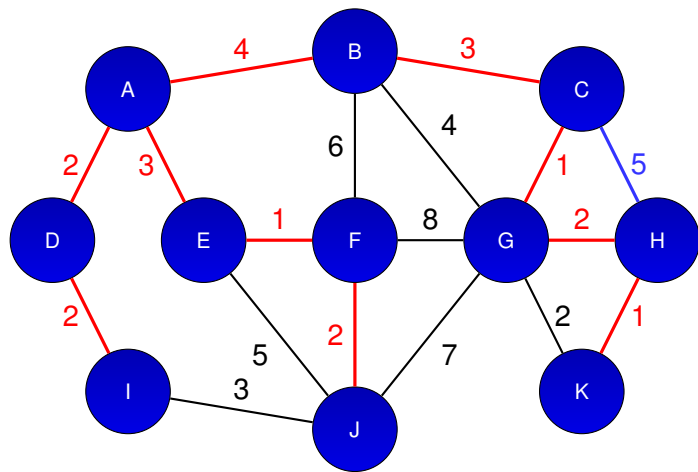
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



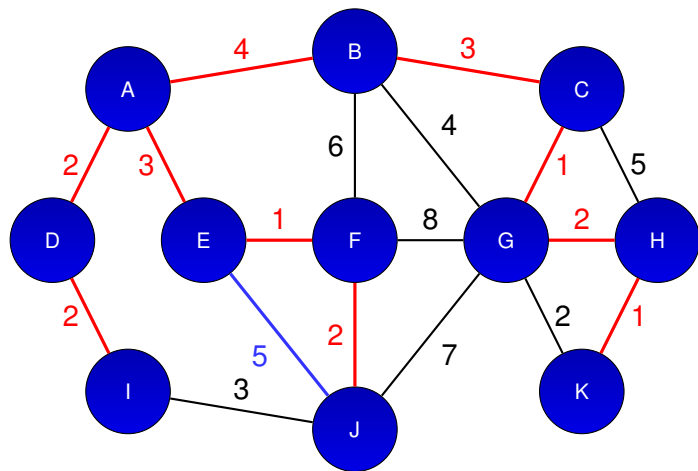
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



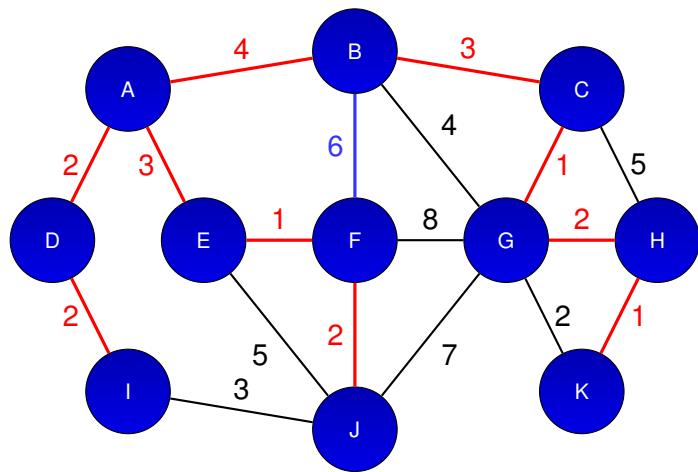
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



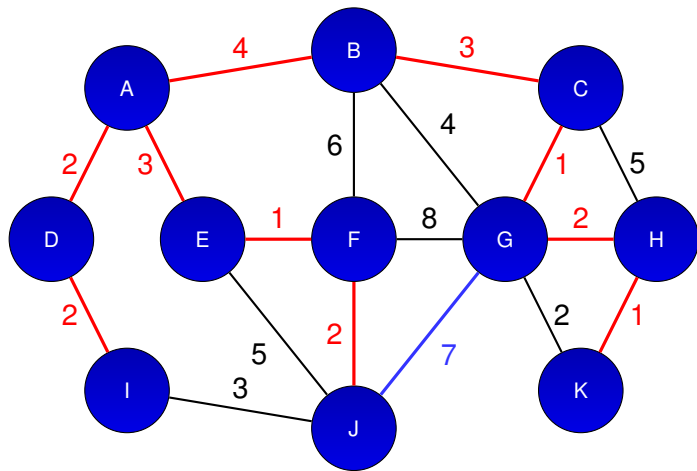
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



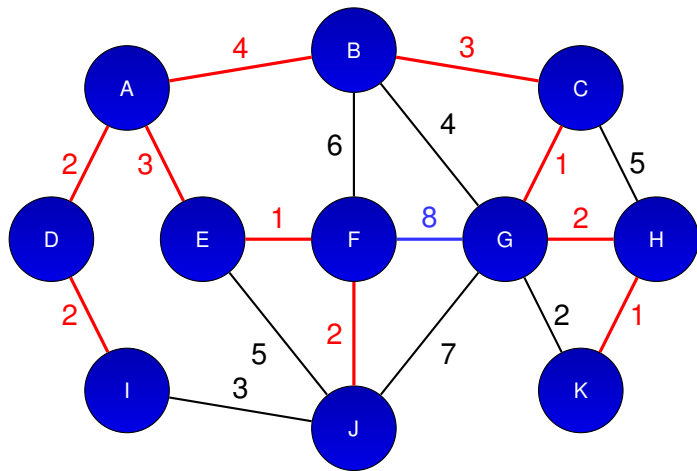
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



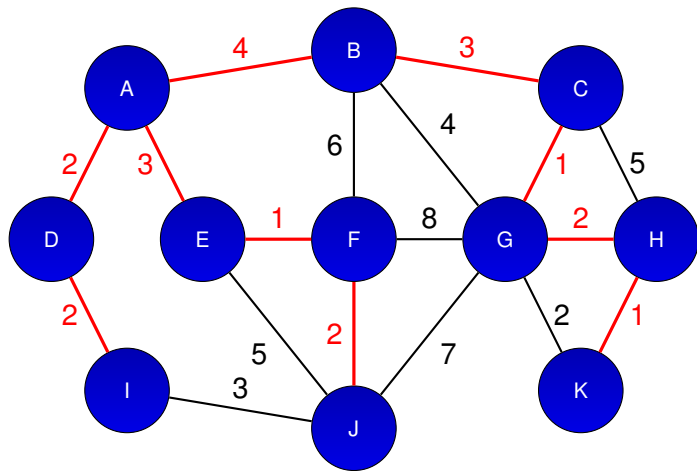
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal



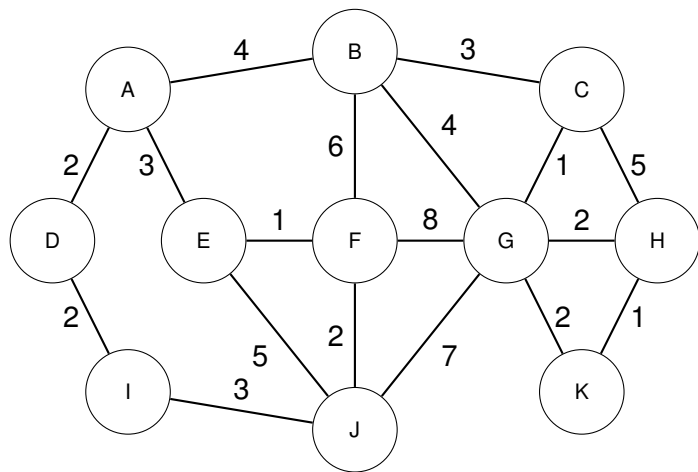
C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

Minimum spanning tree - Algoritmo di Kruskal

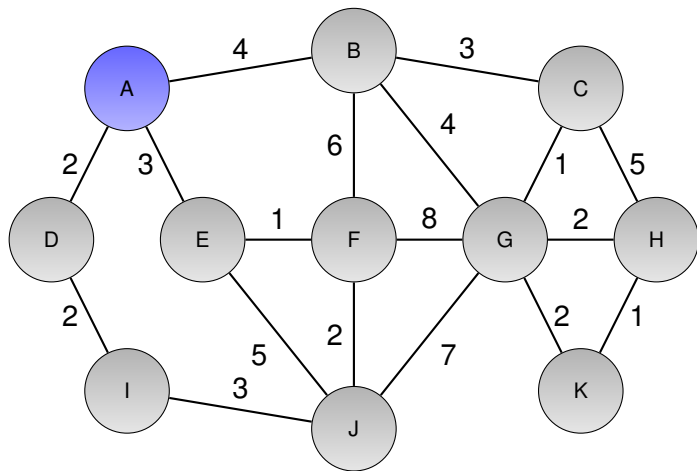


C – G: 1
E – F: 1
H – K: 1
A – D: 2
D – I: 2
F – J: 2
G – H: 2
G – K: 2
A – E: 3
B – C: 3
I – J: 3
A – B: 4
B – G: 4
C – H: 5
E – J: 5
B – F: 6
G – J: 7
F – G: 8

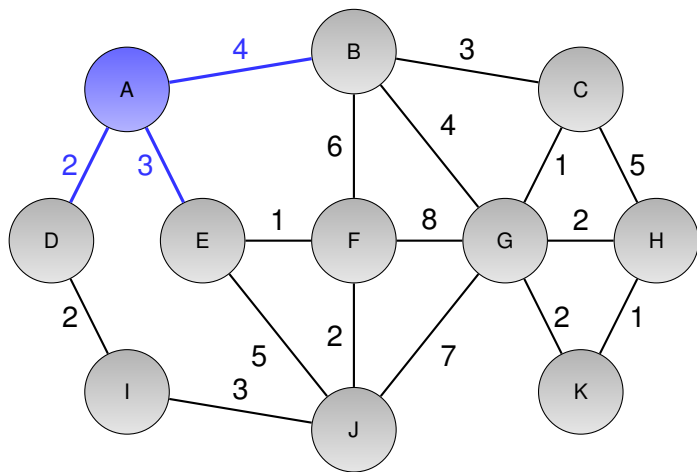
Minimum spanning tree - Algoritmo di Prim



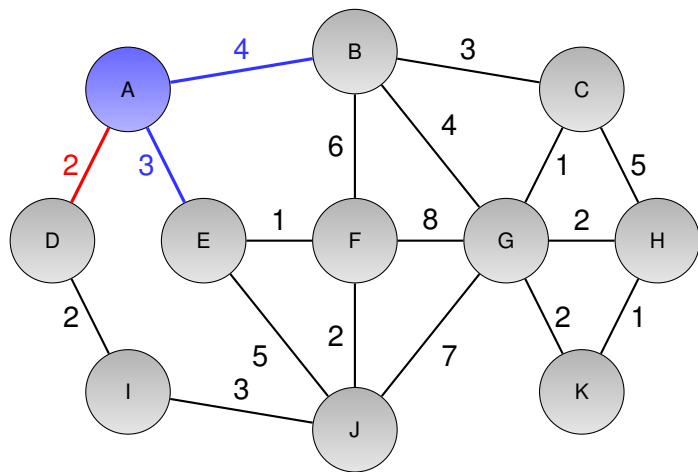
Minimum spanning tree - Algoritmo di Prim



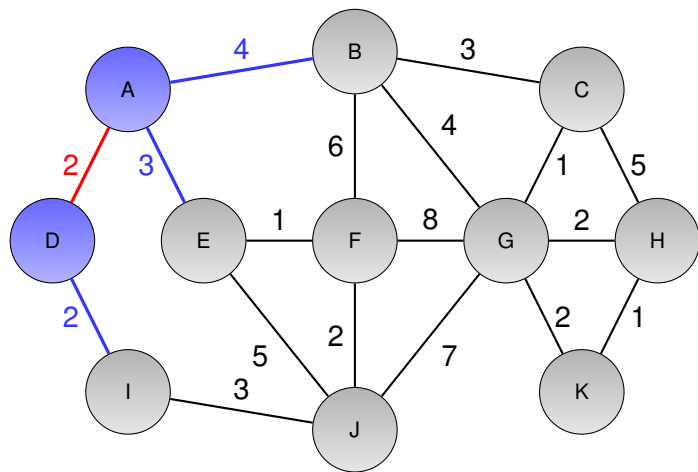
Minimum spanning tree - Algoritmo di Prim



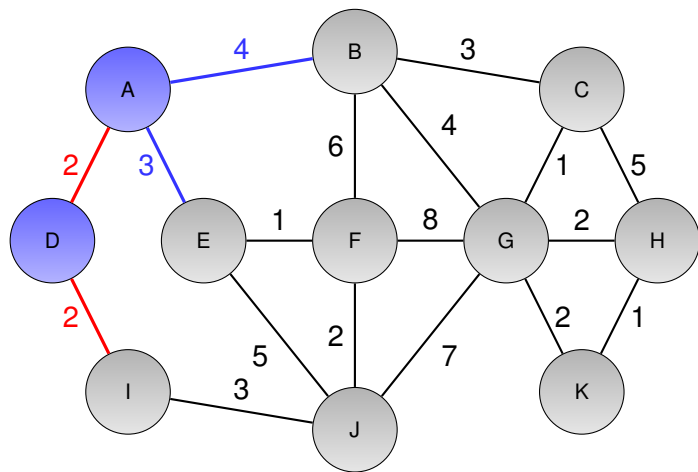
Minimum spanning tree - Algoritmo di Prim



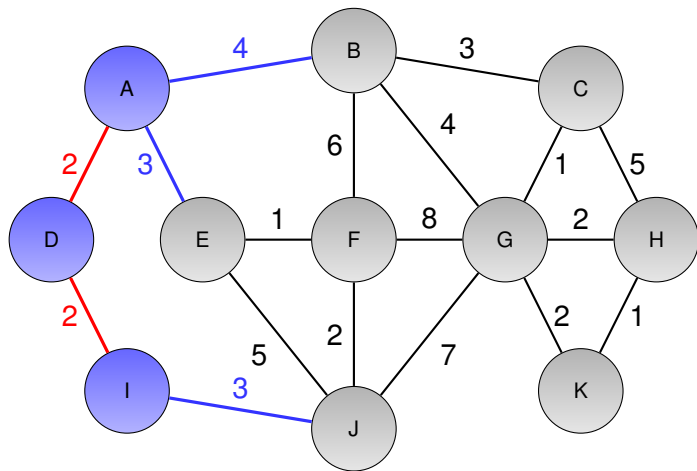
Minimum spanning tree - Algoritmo di Prim



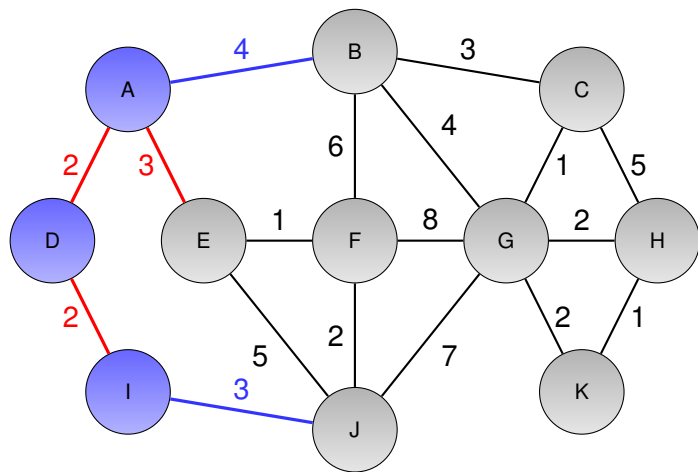
Minimum spanning tree - Algoritmo di Prim



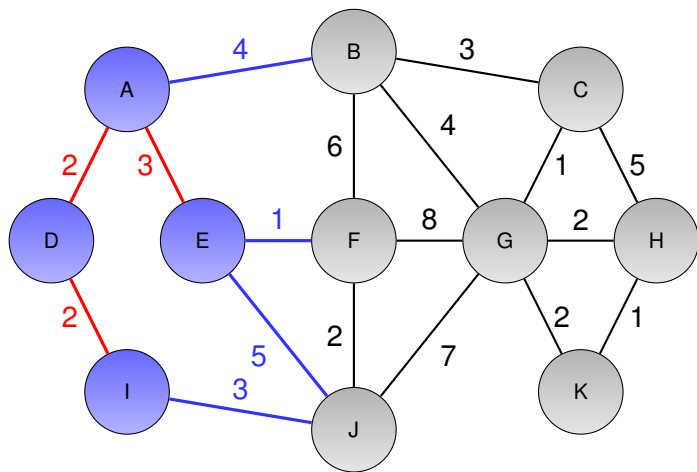
Minimum spanning tree - Algoritmo di Prim



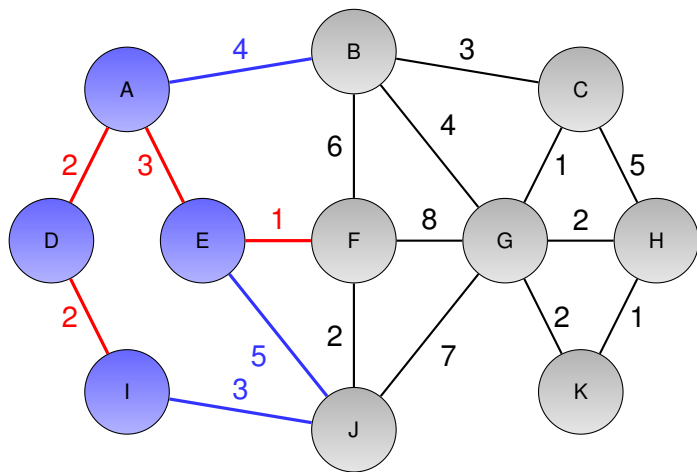
Minimum spanning tree - Algoritmo di Prim



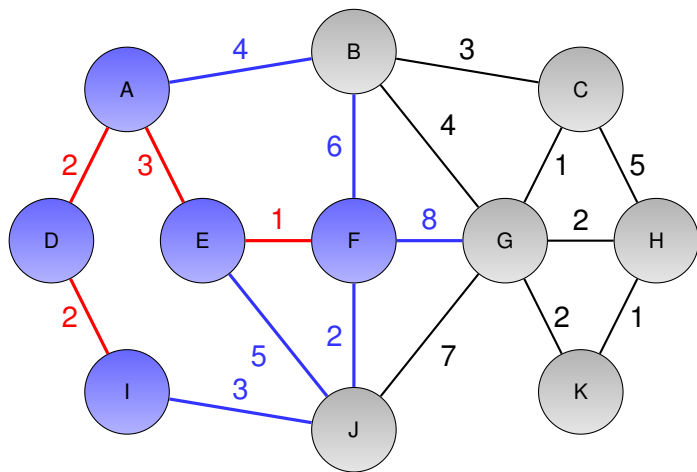
Minimum spanning tree - Algoritmo di Prim



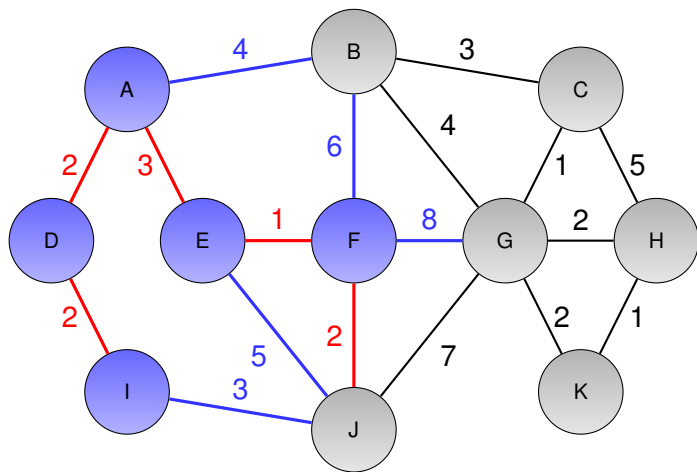
Minimum spanning tree - Algoritmo di Prim



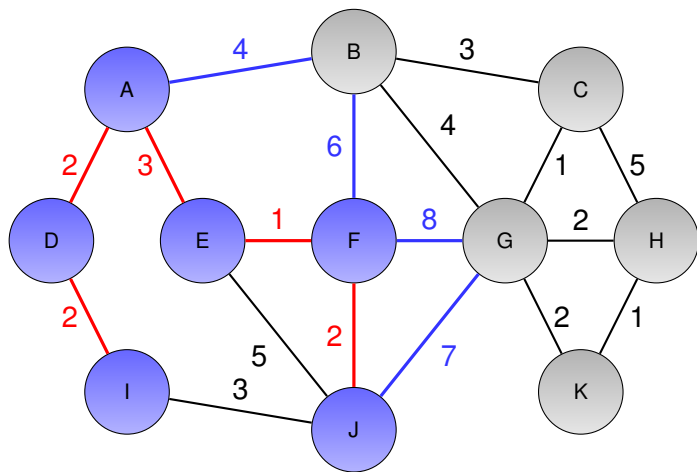
Minimum spanning tree - Algoritmo di Prim



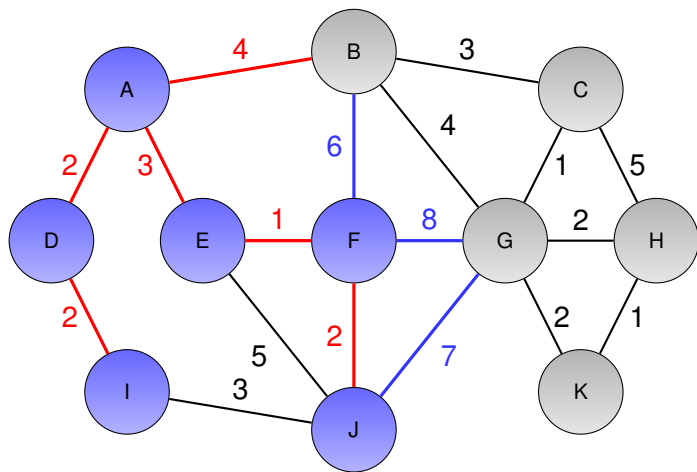
Minimum spanning tree - Algoritmo di Prim



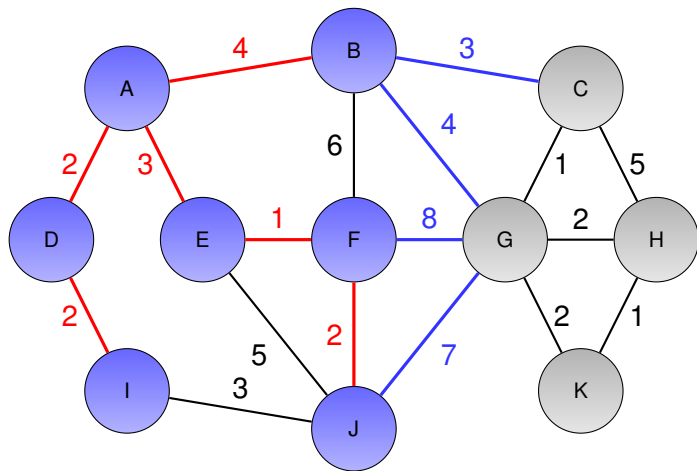
Minimum spanning tree - Algoritmo di Prim



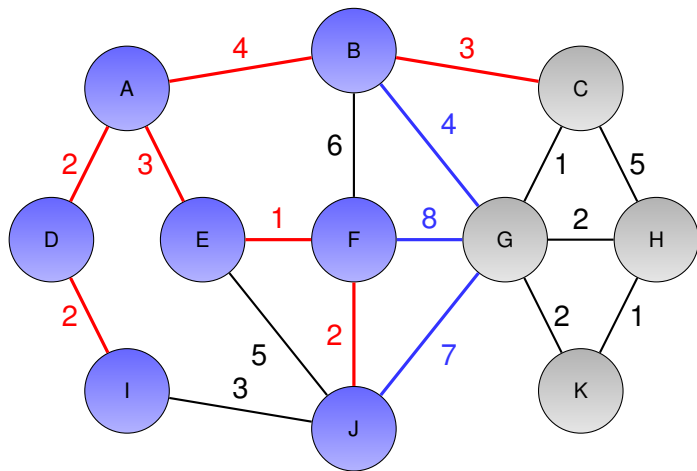
Minimum spanning tree - Algoritmo di Prim



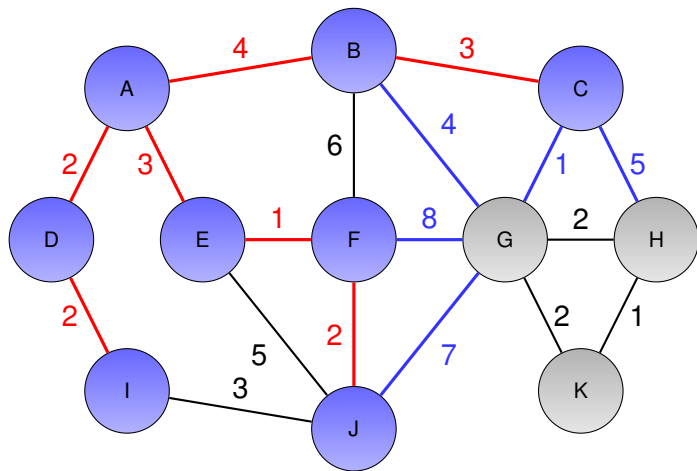
Minimum spanning tree - Algoritmo di Prim



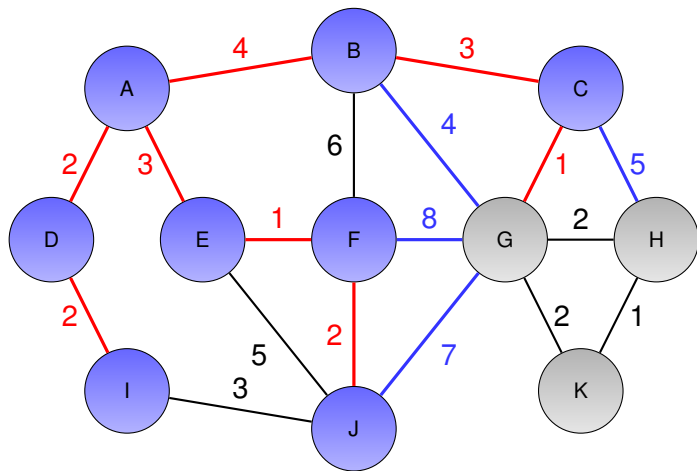
Minimum spanning tree - Algoritmo di Prim



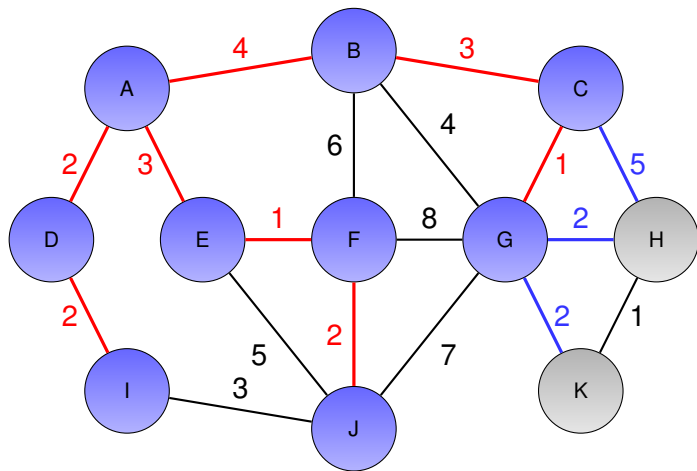
Minimum spanning tree - Algoritmo di Prim



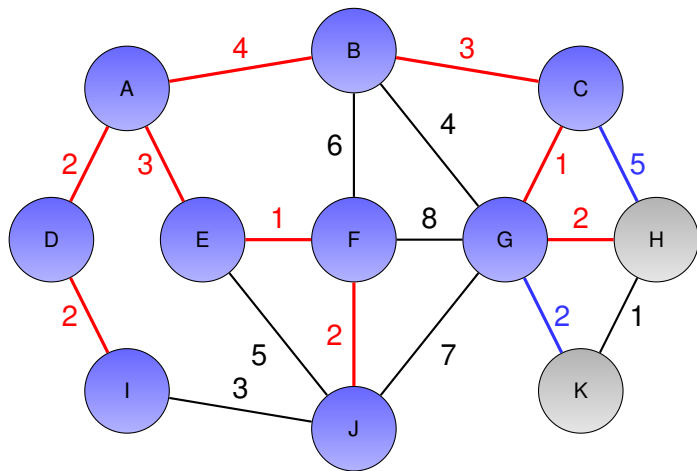
Minimum spanning tree - Algoritmo di Prim



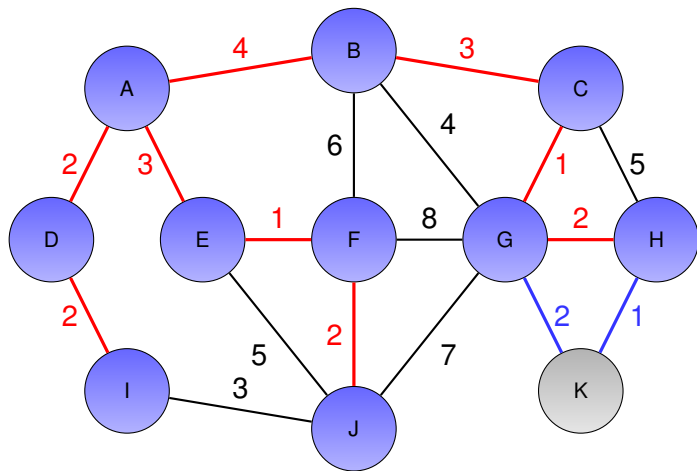
Minimum spanning tree - Algoritmo di Prim



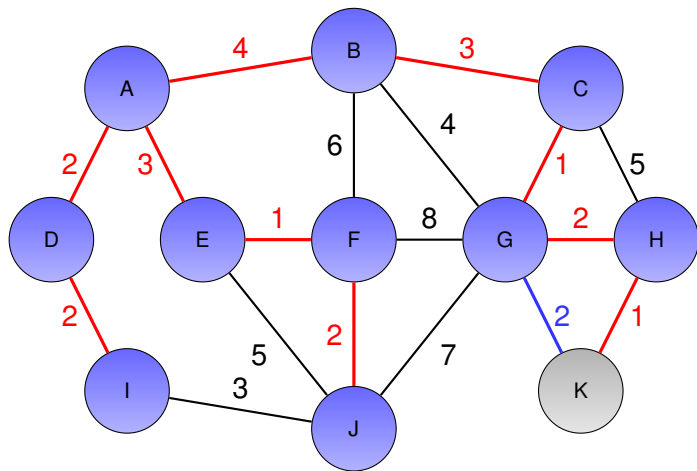
Minimum spanning tree - Algoritmo di Prim



Minimum spanning tree - Algoritmo di Prim



Minimum spanning tree - Algoritmo di Prim



Minimum spanning tree - Algoritmo di Prim

